w/ attached ORIGINAL copy of RFA

Republic of the Philippines
OFFICE OF THE PRESIDENT

COMMISSION ON HIGHER EDUCATION

30 June 2022

ATTY. DANILO L. CONCEPCION
President
University of the Philippines
Office of the UP President
2nd Floor Quezon Hall, UP Campus, Diliman
Quezon City
Tel. No. (632) 928-3014, (632) 928-0110
Email Address: op@up.edu.ph

THROUGH: **CARMENCITA C. PADILLA, M.D**
Chancellor, University of the Philippines Manila

NOTICE TO PROCEED

Project Title:	**Phytochemical characterization of macroalgae for food and high value products (PhycoPRO)**
Project Leader: Lead Institution:	**Michael Y. Roleda, Dr. rer. nat.** University of the Philippines
Co-Project Leader: Lead Institution:	**Irene B. Rodriguez, Dr. rer. nat.** University of the Philippines
Project Collaborator: Institution:	**Rey Y. Capangpangan, Ph.D.** Mindanao State University - Naawan
Project Collaborator: Institution:	**Michael A. Clores, Ph.D.** Partido State University
Project Collaborator: Institution:	**Richard V. Dumilag, Ph.D.** Sorsogon State University
Project Collaborator: Institution:	**Ronelie C. Salvador, Ph.D.** University of Eastern Philippines
Project Duration: Total Budget:	Two (2) years **PhP 183,054,284.54**

Republic of the Philippines
OFFICE OF THE PRESIDENT

COMMISSION ON HIGHER EDUCATION

Dear **President Concepcion:**

Pursuant to the attached duly-executed Research Funding Agreement (RFA), implementation of this project commences on July 1, 2022.

Releases of the approved amounts shall be in accordance with the terms and conditions stipulated in the RFA, more specifically, Annex "B", and shall be subject to applicable government accounting and auditing rules and regulations.

As the lead Higher Education Institution (HEI), you shall be responsible for informing your collaborating institutions of the abovementioned arrangement and ensuring the completion of the project in keeping with the approved "Research Project Proposal, Work Plan and Budget".

Kindly acknowledge receipt and acceptance of this NOTICE TO PROCEED by signing these two (2) copies in the space provided below. Keep one copy and return immediately the other to the Commission on Higher Education (CHED), Attention: Leading the Advancement of Knowledge in Agriculture and Sciences Program Management and Coordination Office (LAKAS-PMCO), Higher Education Development Center Building, Ground Floor, C.P. Garcia Avenue, UP Campus, Diliman, Quezon City, Philippines.

Thank you and we look forward to a mutually rewarding research partnership. We take pleasure in working together with your institution in the pursuit of high science research aimed to contribute to the accelerated development of Philippine Higher Education Institutions (HEIs) to become world-class research institutions.

Truly yours,

J. PROSPERO E. DE VERA III, DPA
Chairman
Commission on Higher Education

Copy furnished:

MARIA CYNTHIA ROSE BANZON-BAUTISTA, Ph.D.
Vice President for Academic Affairs
University of the Philippines
3rd Floor Quezon Hall, UP Campus, Diliman

ELNOR C. ROA, Ph.D.
Chancellor, Mindanao State University - Naawan Campus

RAUL G. BRADECINA, Ph.D
President, Partido State University

Republic of the Philippines
OFFICE OF THE PRESIDENT

COMMISSION ON HIGHER EDUCATION

CHERRY I. ULTRA, Ph.D.
President, University of Eastern Philippines

ATTY. RYAN L. ESTEVEZ
OIC President, Sorsogon State University

MICHAEL Y. ROLEDA, Dr. rer. nat.
Project Leader, University of the Philippines

IRENE B. RODRIGUEZ, Dr. rer. nat.
Co-Project Leader, University of the Philippines

REY Y. CAPANGPANGAN, Ph.D.
Project Collaborator, Mindanao State University - Naawan

MICHAEL A. CLORES, Ph.D.
Project Collaborator, Partido State University

RICHARD V. DUMILAG, Ph.D.
Project Collaborator, Sorsogon State University

RONELIE C. SALVADOR, Ph.D.
Project Collaborator, roneliesalvador@yahoo.com

Received by:

(Lead HEI President or authorized representative)

Date: _____

RESEARCH FUNDING AGREEMENT

Between and among

THE COMMISSION ON HIGHER EDUCATION,

THE UNIVERSITY OF THE PHILIPPINES,

THE MINDANAO STATE UNIVERSITY - NAAWAN,

THE PARTIDO STATE UNIVERSITY,

THE SORSOGON STATE UNIVERSITY,

-and-

THE UNIVERSITY OF EASTERN PHILIPPINES

KNOW ALL MEN BY THESE PRESENTS:

This Research Funding Agreement (hereinafter referred to as "RFA") entered into this JUL 11 2022 QUEZON CITY by:

The **PHILIPPINE COMMISSION ON HIGHER EDUCATION**, a government agency created pursuant to Republic Act No. 7722 with principal address at Higher Education Development Center, C.P. Garcia Avenue, UP Diliman, Quezon City represented herein by its Chairman, **J. PROSPERO E. DE VERA III, DPA,** hereinafter referred to as the "**CHED**";

The **UNIVERSITY OF THE PHILIPPINES**, the national university, created by virtue of Act No. 1870, as amended and strengthened by Republic Act 9500, with principal office at Quezon Hall, UP Diliman, Quezon City, Philippines, represented herein by UP Manila Chancellor, **CARMENCITA D. PADILLA, M.D.**, pursuant to the approval of its Board of Regents, hereinafter referred to as the "**UP**";

The **MINDANAO STATE UNIVERSITY-NAAWAN CAMPUS,** with office address at Pedro Pagalan St., Poblacion, Naawan, Misamis Oriental, 9023, represented herein by its Chancellor, **ELNOR C. ROA, Ph.D.**, hereinafter referred to as **"MSU-Naawan"**;

The **PARTIDO STATE UNIVERSITY,** with office address at San Juan Bautista St. Goa Camarines Sur 4422, represented herein by its President, **RAUL G. BRADECINA, Ph.D.**, hereinafter referred to as **"ParSU"**;

The **SORSOGON STATE UNIVERSITY,** with office address at Magsaysay St, Sorsogon City, represented herein by its _____, _____, hereinafter referred to as **"SorSU"**;

and

The **UNIVERSITY OF EASTERN PHILIPPINES**, with office address at University Town, Catarman N. Samar, 6400, represented herein by its President, **CHERRY I. ULTRA, Ph.D.**, hereinafter referred to as "**UEP**";

Hereinafter collectively referred to as "**UNIVERSITIES**".

CHED and **UNIVERSITIES** are collectively referred to as the "Parties" or individually as "Party".

RECITALS:

WHEREAS, the Government of the Republic of the Philippines, pursuant to Republic Act No. _____, also known as the "General Appropriations Act (GAA) for Fiscal Year (FY) _____," has appropriated funds to implement, through the **CHED**, the special collaborative project for research and Higher Education Institution (HEI) capacity building known as **"Leading the Advancement of Knowledge in Agriculture and Sciences (LAKAS) Program"** (hereinafter referred to as "LAKAS Program");

WHEREAS, the LAKAS Program focuses on research and development efforts in four (4) strategic technology areas of (1) Agriculture, Environment, Forestry, and Fisheries, (2) Health and Nutrition, (3) Science and Engineering, and (4) National Quality Infrastructure to contribute to the accelerated development of Philippine HEIs to become world-class institutions with the aim of contributing to and spurring innovation, economic development, and global competitiveness of the Philippines;

WHEREAS, the **UNIVERSITIES** as collaborating research institutions are interested in participating in the LAKAS Program, and submitted a project proposal, titled **"Phytochemical Characterization of Macroalgae for Food and High-Value Products (PhycoPRO)",** which project has been approved by the **CHED** for funding and implementation per CEB Resolution No. 294-2022 dated 07 June 2022, under the terms and conditions of this RFA;

WHEREAS, the **CHED** and the **UNIVERSITIES** believe that by entering into this RFA they will be able to achieve the objectives of the LAKAS Program to their respective and mutual beneficial interest;

WHEREAS, the representatives of the **UNIVERSITIES** have been properly authorized to enter into this agreement by their respective Governing Boards;

WHEREAS, this RFA together with attachments or annexes, which are incorporated by reference, constitutes the complete agreement and understanding between the contracting parties with respect to this LAKAS project titled **"Phytochemical Characterization of Macroalgae for Food and High-Value Products (PhycoPRO)"**;

NOW, THEREFORE, for and in consideration of the foregoing premises and of the terms and conditions stipulated, the Parties hereby agree, as follows:

ARTICLE 1. The PROJECT

1.1 With the **CHED** as the Philippine Government's funding agency of the LAKAS Program and the **UNIVERSITIES** as Philippine collaborating HEIs, for purposes of implementing the **LAKAS 2021-035 "Phytochemical Characterization of Macroalgae**

for Food and High-Value Products (PhycoPRO)", the Parties expressly agree that this RFA, which sets the terms and conditions shall govern their legal and contractual relationship in the implementation of the herein project as described below:

PROJECT DESCRIPTION & SCOPE OF WORK OF COLLABORATING/PARTNER HEIs:

Project Title:	Phytochemical Characterization of Macroalgae for Food and High Value Products (PhycoPRO)
Project Code Number:	LAKAS 2021-035
Name and Address of Lead HEI:	**University of the Philippines** 2nd Floor, Quezon Hall, UP Diliman Quezon City, Metro Manila, Philippines
Name of Project Leader/ Position Title/Contact Details:	**Project Leader:** **Michael Y. Roleda, Dr. rer. nat.** Professor / Dr. rer. nat 09998486567/ (632) 8922-3962 myroleda@up.edu.ph Marine Science Institute, University of the Philippines Diliman, Velasquez Str., Quezon City **Co-Project Leader:** **Irene B. Rodriguez, Dr. rer. nat.** Associate Professor / Dr. rer. nat 09260641309/(632) 8922-3962 ibrodriguez@msi.upd.edu.ph Marine Science Institute, University of the Philippines Diliman, Velasquez Str., Quezon City
Names and Addresses of Collaborating HEIs:	**Mindanao State University-Naawan Campus,** Pedro Pagalan St., Poblacion, Naawan, Misamis Oriental, 9023 **Partido State University,** San Juan Bautista St. Goa Camarines Sur 4422 **Sorsogon State University,** Aguada Norte, Magallanes, Sorsogon, 4705, Philippines **University of Eastern Philippines,** University Town, Catarman N. Samar, 6400
Names of Collaborators per HEI/SUC/Position Titles/ Contact Details:	**Project Collaborator:** **Rey Y. Capangpangan, Ph.D.** Professor VI/ Director for International Affairs and Linkages +63 977 820 3227 rey.capangpangan@msunaawan.edu.ph Mindanao State University at Naawan, Poblacion, Naawan, Misamis Oriental, 9023, Philippines

	Project Collaborator: **Michael A. Clores, Ph.D.** Associate Professor II +63 9175215772 michael.clores@parsu.edu.ph Partido State University, Caramoan Campus, Barangay Cadong, Caramoan, Camarines Sur, 4429, Philippines **Project Collaborator:** **Richard V. Dumilag, Ph.D.** Associate Professor V +63 9361585522 richard.dumilag@sorsu.edu.ph Sorsogon State University, Magallanes Campus, Aguada Norte, Magallanes, Sorsogon, 4705, Philippines **Project Collaborator:** **Ronelie C. Salvador, Ph.D.** Professor VI/Dean, College of Agriculture, Fisheries and Natural Resources 09084303550 roneliesalvador@yahoo.com University of Eastern Philippines, University Town, Catarman N. Samar, 6400
Brief Project Rationale/ Description:	The Philippines is host to 1079 identified seaweed species and still has many unidentified species to be catalogued. Among the taxonomically identified species, only a small fraction (>5%) is substantially used for food and industrial applications. This can be attributed to a poor understanding of its biochemical composition, health benefit and safety for consumption. Among the marketed and currently farmed species, the Philippines has an economic advantage in the global seaweed industry with an available farmable area of ~200,000 hectares along the coastlines. Currently, roughly 12% of this area is utilized for farming of seaweed species and strains from 4 genera namely: *Kappaphycus, Eucheuma, Gracilaria* and *Caulerpa*. Seaweed farming sustains the livelihood of more than 200,000 fishing families and generates about 250 million US dollars in export earnings. The diversity of seaweed species available in the Philippines and its available farmable area represent a massive potential for seaweed farming in the country. However, there are several challenges that must be addressed to maximize the potential of the seaweed industry in the country. These include multi-disciplinary basic research focused on the comprehensive phytochemical characterization of seaweeds and research into associated epiphytes of seaweeds to understand their role in the quality of seaweed-derived products.

	This project is envisioned to enhance the status of the seaweed industry in the Philippines by providing critical knowledge through a comprehensive phytochemical characterization of seaweeds to identify species that have higher economic potential and pinpoint locations where their growth will be maximal. The major output of this project is a handbook of seaweeds outlining their phytochemical properties, which will be made available to the Bureau of Fisheries and Aquatic Resources (BFAR), seaweed farmers, and other stakeholders involved in the seaweed industry as a guide for mariculture activities. The publication of a handbook of seaweeds outlining its phytochemical properties will have societal impacts on long-, medium-, and short-term scales. For long-term impact, the knowledge gained from this project will serve as the starting point of future efforts into product development for various applications including food and food additives and high-value compounds such as pigments, nutraceuticals, and pharmaceuticals. For medium-term impact, the identification of seaweed species that are more profitable for a specific location will guide seaweed farmers on what species they will prioritize for farming and improvement of mariculture methods, which may result in poverty alleviation in coastal communities. For short-term impact, the identification of seaweed species with high nutritional content and safe for direct consumption will provide additional or alternative food sources for the general population, who are clamoring for nutritious dietary sources. The project is envisioned to contribute to knowledge generation about potential uses of seaweeds that will have societal and economic impacts on coastal communities and the Philippine population in general. Millions of marginalized small-scale fisherfolk are dependent upon seaweed farming for their livelihoods. However, they are faced with problems of declining productivity and associated incomes due to a dependence on old and outdated seaweed farming methods that limit people to very shallow (<2m) coastal waters, old varieties of seaweed that have been vegetatively propagated for the last five decades, and the impacts of climate change. These factors interact to result in low growth rates and high mortality from epiphytism and ice-ice disease. As a result, seaweed farmers are increasingly returning to fishing, resulting in unsustainable levels of exploitation that diminish natural resources and trap people in poverty. This project proposal will build upon our expertise in the production of new and high-performing Kappaphycus cultivars generated from spores of wild local stocks through sexual reproduction and growing them up in local seaweed nurseries for distribution to seaweed farmers. Promising new strains will be tested for carrageenan content to preselect those with the highest quality and

<table>
<tr><td></td><td>yield. Good quality biomass will command a higher buying price that will effectively increase household incomes of small-scale fisherfolk/ seaweed farmers to at least P15-20,000 per month.</td></tr>
<tr><td>Total Budget:</td><td>Grand Total: PhP 183,054,284.54

First Tranche (Month 1 to 6): PhP63,493,097.98
Second Tranche (Month 7 to 18): PhP102,550,042.27
*Last Tranche (Month 19 to 24): PhP17,011,144.30

For UP:
<table>
<tr><th>Schedule</th><th>Amount, PhP</th></tr>
<tr><td>First Tranche (Month 1 to 6)</td><td>PhP58,188,809.98</td></tr>
<tr><td>Second Tranche (Month 7 to 18)</td><td>PhP66,591,466.27</td></tr>
<tr><td>*Last Tranche (Month 19 to 24)</td><td>PhP12,906,856.30</td></tr>
<tr><td>Total</td><td>PhP137,687,132.54</td></tr>
</table>
For MSU-Naawan:
<table>
<tr><th>Schedule</th><th>Amount, PhP</th></tr>
<tr><td>First Tranche (Month 1 to 6)</td><td>PhP1,071,072.00</td></tr>
<tr><td>Second Tranche (Month 7 to 18)</td><td>PhP9, 512,144.00</td></tr>
<tr><td>*Last Tranche (Month 19 to 24)</td><td>PhP1,071,072.00</td></tr>
<tr><td>Total</td><td>PhP11,654,288.00</td></tr>
</table>
For SorSU:
<table>
<tr><th>Schedule</th><th>Amount, PhP</th></tr>
<tr><td>First Tranche (Month 1 to 6)</td><td>PhP1,731,072.00</td></tr>
<tr><td>Second Tranche (Month 7 to 18)</td><td>PhP9,812,144.00</td></tr>
<tr><td>*Last Tranche (Month 19 to 24)</td><td>PhP1,131,072.00</td></tr>
<tr><td>Total</td><td>PhP12,674,288.00</td></tr>
</table></td></tr>
</table>

For ParSU:

Schedule	Amount, PhP
First Tranche (Month 1 to 6)	PhP1,011,072.00
Second Tranche (Month 7 to 18)	PhP10,032,144.00
*Last Tranche (Month 19 to 24)	PhP1,011,072.00
Total	PhP12,054,288.00

For UEP:

Schedule	Amount, PhP
First Tranche (Month 1 to 6)	PhP1,491,072.00
Second Tranche (Month 7 to 18)	PhP6,602,144.00
*Last Tranche (Month 19 to 24)	PhP891,072.00
Total	PhP8,984,288.00

** subject to availability of funds*

1.2 PROJECT WORK PLAN AND BUDGET

1.2.1 Attached as integral part and annexes of this RFA are the following documents:

a) Annex "A" - Approved Collaborative Research Project;

b) Annex "B1" - Schedule of Research Project Fund Releases for the **UNIVERSITIES;**

1.2.2. A budget allocation in the amount of **One Hundred Eighty-Three Million Fifty-Four Thousand Two Hundred Eighty-Four Pesos and Fifty-Four Centavos (PhP 183,054,284.54)** for the whole project shall be distributed according to the approved funding requirements of the implementing institutions by the **CHED** subject to the applicable Philippine Government accounting and auditing laws, rules and regulations. Funds will be released to the **UNIVERSITIES** based on Annex "B" mentioned above which takes into consideration the approved Project Work Plan and Budget, as well as the Schedules embodied in the aforementioned approved project proposal.

ARTICLE 2. DEFINITION OF TERMS

2.1 **Administrative Overhead Cost (Indirect Cost)** — refers to those facilities and administration costs that are incurred for common or joint activities and, therefore, may not be identified readily and specifically with a particular sponsored project.

2.2 **Capacity Building –** A systematic process through which the Philippine HEIs could strengthen the professional knowledge and skills in science and technology; develop and train faculty, students and staff and thereby enhance their ability to identify and meet development challenges.

In the global context, capacity building refers to the ability of individuals and institutions to make and implement decisions and perform functions in an effective, efficient and sustainable manner. At the individual level, capacity building refers to the process of changing attitudes and behaviors imparting knowledge and developing skills while maximizing the benefits of participation, knowledge exchange and ownership. At the institutional level, it focuses on the overall organizational performance and functioning capabilities, as well as the ability of an organization to adapt to change.

2.3 **Deliverables -** A tangible object or completed task produced as a result of a research project that is intended to be delivered to the **CHED**.

2.4 **Financial Report -** An official record prepared by the **UNIVERSITIES** for endorsement of the Project Leader to the LAKAS Program Management and Coordination Office (PMCO). This includes the actual costs and disbursements of the reporting period.

2.5 **Lead Higher Education Institution (HEI) -** Any entity accountable for and responsible for the full implementation of the collaborative activities of the project.

2.6 **Milestones and Milestone Report - Milestones** are high-level descriptions of activities or tasks with corresponding outputs or deliverables. Milestones shall also be quantified in terms of percentage of the deliverable for the year, e.g. a 25% milestone corresponds to activities, tasks and outputs that comprise 25% of the total project deliverables for the year. A corresponding payment of 25% shall be released upon accomplishment of this milestone as verified by the **CHED**.

The **milestone report** shall include a description of tasks completed, corresponding outputs and evidence thereof. In recognition of the unpredictability of research activities and outputs, milestones to be reflected in the RFA may be worded in general terms while still being sufficiently descriptive of activities to allow for reasonable verification and evaluation by the **CHED**.

2.7 **Project Budget -** A detailed breakdown of the financial support requested from the funding agency (**CHED**), reflecting the best estimate of the costs requested to conduct the work outlined in the project proposal.

2.8 **Project Equipment -** Any apparatus used primarily for research valued at PhP50,000.00 or more.

2.9 **Project Leader** - refers to the individual identified in the Research Funding Agreement who has primary responsibility for the scientific and technical direction of the project, its overall implementation, and the submission of necessary reports.

2.10 **Co-Project Leader or Equivalent** - refers to the individual as may be identified by the Project Leader who shares responsibility with the participating HEI/s for the scientific and technical direction of the project.

2.11 **Project Work Plan** - A description of the project to be accomplished with an outline of how it will be done over the project duration including the staffing and resources needed in the conduct of the project.

2.12 **Technical Report** - A document prepared for each project by the Project Leader for submission to the **CHED** through the LAKAS PMCO, describing the progress of a research project in terms of detailed technical specifications such as materials, functions, features, operations, market potential, and similar information.

ARTICLE 3. OBLIGATIONS OF THE CHED

A. TO THE UNIVERSITIES

3.1 To provide a total funding assistance to the **UNIVERSITIES** in the amount of **One Hundred Eighty-Three Million Fifty-Four Thousand Two Hundred Eighty-Four Pesos and Fifty-Four Centavos (PhP 183,054,284.54).** Fund releases shall be in accordance with the approved Schedule of Research Project Fund Releases (Annex "B"). It is understood that the **UNIVERSITIES** shall shoulder any and all expenses in excess of the approved project budget. Excess funds, if any, shall be turned over to the **CHED**. Such releases shall be subject to the submission of financial statements as provided under applicable Philippine Government accounting and auditing laws, rules and regulations.

3.1.1 In the case of the University of the Philippines, the budget shall be channeled through UP-Manila.

B. MONITORING MECHANISM

3.2 The **CHED** shall develop and implement a project monitoring mechanism which identifies and considers gaps and problem areas which must be immediately addressed, in the event that the project incurs delays or deviations from its approved Milestones Table or Project Work Plan and Budget, as applicable;

3.3 The **UP** as the lead implementing HEI and represented by the Project Leader shall assist **CHED** in monitoring the project implementation and shall compile and submit to **CHED** all required reports and submission.

C. PAYMENTS/FUND RELEASES

3.4 All payments/funds released to the **UNIVERSITIES** shall be in accordance with the approved Schedules mentioned in Subsections 3.1 above.

3.5 The release and remittance of research funds to the **UNIVERSITIES** shall be in the name of the following **UNIVERSITIES,** in such amounts as may be indicated in Annex "B":

a. University of the Philippines

b. Mindanao State University – Naawan

c. Sorsogon State University

d. Partido State University

e. University of Eastern Philippines

D. Dissemination of LAKAS Project Research Results

3.6 The **CHED** or the **UNIVERSITIES** shall facilitate and provide avenue/s where the research results could effectively and adequately be disseminated and utilized with the end in view of achieving the Philippine capacity building objective of the LAKAS Program.

ARTICLE 4. OBLIGATIONS OF THE COLLABORATING RESEARCH INSTITUTIONS

The UNIVERSITIES, individually and collectively, shall:

4.1 Undertake the Project strictly in accordance with the ATTACHED approved Collaborative Research Project (Annex "A").

4.2 Execute a Memorandum of Agreement (MOA) among themselves, as needed, to set the terms and conditions to govern the legal and contractual relationship of the concerned Parties in the implementation of the herein project.

4.3 Utilize the research funds in accordance with the approved Research Project Work Plan and Budget.

4.4 Obtain all relevant permits, licenses or clearances that may be required for the project.

4.5 Issue Official Receipts or such other documents that will prove receipt of every amount received from the **CHED**.

4.6 Monitor the accomplishment vis-à-vis the approved Project Work Plan and Budget for the **UNIVERSITIES**, and submit to the **CHED** such Technical and Financial Reports as may reasonably be necessary, during the project duration.

4.7 To the extent it is legally able to do so, grant the **CHED** a sublicensable, non-exclusive and royalty-free license to use, copy, produce, reproduce, modify, adapt, implement, advance and otherwise exploit the research results and any intellectual property rights for educational and research purposes for the life of the patent rights/copyrights to: (i) the patent rights that are conceived and reduced to practice during the performance of research under this RFA; and (ii) in the case of copyrights, the copyrights that are first created in the course of and under this particular RFA.

4.8 Provide such support services and facilities as may be necessary to fulfill the requirement of the project including but not limited to the Project staff, their salaries/fees/wages or such other remunerations, office space, research facilities/equipment during the conduct of the research project.

4.9 Ensure that there are at all times a designated Project Leader and the project Teams to undertake the implementation of the project and ensure that they will carry out the project strictly in accordance with the approved Work Plan and Budget.

4.10 Execute an Inter-Institutional Agreement (IIA) with the **CHED** as a signatory that will define the specific terms and conditions in the event that a disclosure of invention is made by either party and both decide to jointly develop/exploit/commercialize the same.

4.11 Present the project results in a forum for critiquing.

4.12 Submit a Final Terminal Report (in hard and soft copies) within sixty (60) calendar days from the date of completion of the research project. Such a report shall contain complete technical and financial details including the disclosure of a verified list of IP/IPRs generated by or any project income derived from the research project as of the date of project termination. The Financial Report at the end of the project shall reflect the final adjustments of costs and payments. The Project Leader shall be responsible in collecting, compiling, and submitting to **CHED** the Final Terminal Report. Failure by a participating HEI to submit the Final Terminal Report to **CHED** within sixty (60) calendar days from the date of completion of the research project shall result in the return of one percent (1%) of the Administrative Overhead Cost for each day of delay.

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4.13 Be accountable for the payment of all taxes, duties and fees of whatever nature connected with or related to this RFA where applicable.

4.14 Remain accountable to the **CHED** for the successful implementation and completion of the project.

4.15 Contribute to the identification of annual research opportunities by providing up-to-date information on state of the art technologies that may be developed, applied, or adapted to the Philippine national development priorities;

4.16 Identify or nominate staff who can assist in managing any of the research and development activities of the project, including the monitoring of collaborative projects undertaken and/or led by the implementing institution researchers;

4.17 Open a separate Trust Account with any government authorized depository bank nearest to the project site where project funds will be deposited.

4.18 Submit final financial report to the **CHED**, certified correct by the respective **UNIVERSITY** Accountant and approved by the respective **UNIVERSITY** Head within sixty (60) calendar days after the completion of the project and provide full access to the **CHED** to all data, observations and project results/findings. The Project Leader shall be responsible in collecting, compiling, and submitting to **CHED** the Financial Reports. Failure by a participating HEI to submit the Financial Report to **CHED** within sixty (60) calendar days from the date of completion of the research project shall result in the return of one percent (1%) of the Administrative Overhead Cost for each day of delay.

4.19 Keep all physical and electronic documents and records related to all approved research projects, including the scientific and laboratory notebooks, databases, and research notes for a period of ten (10) years after termination of the project. The **CHED** shall, during this period, have access to any or all of such documents for review, reference and possible IP protection;

4.20 Keep and maintain separate accounting books and records for the project which shall be voluntarily submitted whenever required and subjected to monitoring and evaluation of the **CHED's** Authorized Representative/s. The **CHED** shall be fully furnished certified true copies of any/all required documents upon request.

4.21 Return to the **CHED** any/all unused balance of the project funds and interests earned upon pre-termination or completion of the project within sixty (60) calendar days from date of termination, except in the case of Philippine public HEIs which are authorized to utilize such interests earned pursuant to the provisions of their respective charters and applicable government accounting and auditing laws, rules and regulations;

All project income other than income from commercialization of IP/IPRs governed by Section 4.22 below, that may be generated in the course of project implementation shall be duly accounted for and returned to the **CHED** within sixty (60) calendar days from date of termination of the project except in the case of Philippine public HEIs which are authorized to utilize their internally generated project income pursuant to the provisions of their respective charters and applicable government accounting and auditing laws, rules and regulations;

Project income refers to gross income earned by the HEIs that is directly generated by a supported activity or earned as a result of the approved research project; it includes, but is not limited to, royalty payments and other intellectual property rights remuneration received from results/products, processes, and technology systems arising from a project; rental fees, and related types of remuneration received from the use of equipment/facilities funded by the LAKAS Program; sale of produce and other products, services and publications from project activities; and training fees (net of expenses) collected from training packages developed from the project activities.

Interests earned on advances of LAKAS Program funds is not project income. Except as otherwise provided in the terms and conditions of the Project funding, project income does not include the receipt of principal on loans, rebates, credits, discounts, etc., or interest earned on any of them.

4.22 Be vested with the authority to use its share of the revenues derived from commercialization of IP generated from the research project as a state university undertaking technology transfer, which enjoys fiscal autonomy under its charter, the Philippine Technology Transfer Act of 2009, Republic Act No. 10055, and the applicable government accounting and auditing laws, rules and regulations;

4.23 Receive funds from the **CHED** for the project subject to the visitorial audit of the Philippine Commission on Audit (COA);

4.24 Abide by the provisions of COA Circular No. 94-013 dated 13 December 1994. This circular directly or indirectly pertains to projects funded either fully or partly by government agencies;

4.25 Comply with the provisions of Republic Act No. 9184 and its Implementing Rules and Regulations in any procurement the **UNIVERSITIES** shall undertake.

ARTICLE 5. RESEARCH PERSONNEL

5.1 The Project Leader and staff shall be composed of the following:

For UP	
Project Leader:	**Michael Y. Roleda, Dr. rer. nat.**
Co-Project Leader:	**Irene B. Rodriguez, Dr. rer. nat.**
For MSU-Naawan	
Collaborator/Project Staff:	**Rey Y. Capangpangan, Ph.D.**
For ParSU	
Collaborator/Project Staff:	**Michael A. Clores, Ph.D.**
For the SorSU	
Collaborator/Project Staff:	**Richard V. Dumilag, Ph.D.**

For UEP
Collaborator/Project Staff: **Ronelie C. Salvador, Ph.D.**

5.2 The Project Leader and the project staff are directly accountable for the implementation and completion of the project. The **UNIVERSITIES** shall ensure that an Officer-in-Charge will be available in the absence of the Project Leader.

5.3 The designation of the Project Leader and research project staff shall be co-terminus with the project.

5.4 Subcontracting of any part of the research shall require prior written approval from the **CHED**. However, in the event that such subcontracting is allowed, the **UNIVERSITIES** shall remain liable and responsible for the output of the subcontracted party.

ARTICLE 6. FAILURE TO COMPLETE THE RESEARCH PROJECT

It is understood that the Pre-termination Clauses of this RFA shall apply should the collaborating HEIs fail to complete the project for cause or prematurely voluntarily terminate the same. Such failure to satisfactorily complete the research project within the period stated in the approved Project Work Plan and Budget or any extension granted shall mean forfeiture or discontinuance of the project. The remaining balance of the research fund shall revert automatically to the **CHED** subject to the applicable Philippine government accounting and auditing laws, rules and regulations.

ARTICLE 7. EQUIPMENT AND OWNERSHIP

7.1 All equipment purchased for the project and funded by the **CHED** shall be reported to **CHED.** The **CHED** reserves the right to ownership over the purchased equipment and facilities upon submission of the required documents for the transfer of ownership as may be determined by **CHED**.

7.2 The equipment may be shared with other ongoing research programs/projects subject to the guidelines drawn up by the **CHED** for this purpose. The Property Officer of the **CHED AFMS-GSD** shall keep an official record of all equipment purchased. The **UNIVERSITY** Project Leader shall be primarily responsible for the proper utilization and maintenance of the equipment for the project.

ARTICLE 8. INTELLECTUAL PROPERTY RIGHTS

8.1 **Law Governing IP/IPR Ownership**. The employees of the **UNIVERSITIES** are required to assign IP rights to their respective institutions, in accordance with the applicable policies of each institution. Considering that the budget for the approved collaborative research project under the LAKAS Program shall be funded by the Philippine Government, through the **CHED**, the provisions of Republic Act No. 8293, the Intellectual Property Code of the Philippines, as amended, and Republic Act No. 10055, the Philippine Technology Transfer Act of 2009, shall govern, where applicable, on the inventorship and ownership of intellectual properties (IP) and intellectual property rights (IPR) arising from and/or as a result of the implementation of the approved collaborative research project.

8.2 **Use of HEI Resources**. In the joint creation of IP and IPR as output and/or result of the collaborative research projects funded by the **CHED** under this RFA, the use of any laboratory/research facility or utilization of any faculty or staff or resources of the **UNIVERSITIES** shall not grant either party the sole ownership over the IP/IPR.

8.3 **No Harm Policy**. The **UNIVERSITIES** may utilize and exploit the IP/IPR in accordance with the purposes for which the same are intended and in such a way as not to injure their respective interest therein or prevent them from utilizing and exploiting the IP in accordance with their rights.

8.4 **Territorial Exploitation**. Unless there is a written agreement between the **UNIVERSITIES**, they shall each have the right to license the use, offer to sell, sell and/or manufacture their solely owned IP/IPR which were conceived and first reduced to practice in the course of their collaboration within their respective territories. The HEIs may also agree on specific areas or countries outside of their respective territories where each of them will license the IP/IPR.

8.5 **Recognition of IP**. If a particular collaborative research project has led to the creation or generation of IP/IPR, the rights shall be governed by this RFA.

8.6 **Licensing and Assignment of IP Rights**. For IP rights that are conceived and reduced to practice in the course of and under a RFA, each IP license between the **UNIVERSITIES** and a commercial licensee will: (i) reserve for the IP owner(s) and others in the non-profit sector, the right to make and use the rights on their own behalf for their education and research purposes; (ii) reserve a sublicensable, non-exclusive, non-commercial license for the **CHED**, for the life of the IP rights; and (iii) acknowledge the **CHED** as a funder of the research that led to the invention.

Jointly owned IP rights will be governed by the inter-institutional agreements (IIAs) between the joint owners. Such IIAs will specify, at a minimum, how licensing of the joint property is to take place, details regarding patent prosecution and management, expense and revenue sharing, communications with law firms, and the like.

8. 7 **Ownership of Background IP**. The **UNIVERSITIES** own and shall continue to own their respective Background IP which is herein collectively defined as those IP that they have created prior to the implementation of the collaborative research project/s under the LAKAS Program covered by the RFA. The ownership of the Background IP notwithstanding, the HEIs shall grant each other a non-exclusive, compensation-free license to use its Background IP to the extent necessary for the purpose of conducting the collaborative research projects for internal non-commercial research purposes only. For the avoidance of doubt, Background IP rights of the researchers who do not receive funding through a RFA are excluded from this Section and can only be included if the concerned party obtains the (future) written consent of the relevant inventor.

8.8 **Recognition of IPR of Authors and Inventors**. The **UNIVERSITIES** accept that the moral recognition of the IPR of authors and inventors are effective ways of ensuring accountability and accessibility of knowledge and technologies. The names of the authors of the works shall be indicated unless the author himself/herself expressly declines to be named. Inventorship shall be determined in accordance with Section 8.9 below. The compensation, royalty income or revenue share of individual authors and inventors who are acknowledged as such in projects covered by this RFA shall be based on the policy of the HEI to which they belong, and subject further to the provisions of Section 8.6 above.

8.9 **Inventorship, IP/IPR Disclosures and Protection.**The **UNIVERSITIES** shall promptly notify each other and the **CHED** in writing after receipt of a written invention or copyright disclosure that names the **CHED** as a funder of the research. Each HEI agrees to take appropriate steps to cause all personnel assigned to the research projects not to disclose any and all inventions until the IP/IPR are protected. Unless otherwise agreed upon in writing by the HEIs, inventorship is subject to their respective IP policies.

8.10 **Recognition of the Indigenous Knowledge and Biodiversity and Genetic Resources of the Philippines**. The **UNIVERSITIES** agree that any and all research materials from the Philippines which use, incorporate, build on, and/or comprise the development of any indigenous knowledge, resources, practices and systems, as well as biodiversity and genetic resources of the Philippines shall comply with the legal mechanisms and processes under Republic Act No. 8371 (Indigenous Peoples Rights Act) and Republic Act No. 9147 (The Wildlife Act) and such other related laws and international conventions, treaties and agreements.

8.11 **Public Good**. The **CHED** and the **UNIVERSITIES,** in accordance with their respective institutional policies, programs and processes shall agree, depending on the nature and outcome of the research, what IP/IPR shall be donated, assigned, transferred, adapted, or licensed without compensation.

8.12 **No Implied License.** Neither HEI shall grant an express or implied license to any of its IP rights to the other HEI except as expressly provided herein.

ARTICLE 9. CONFIDENTIAL INFORMATION

9.1 **Publication.** The Parties share a common interest in making the significant results of the collaborative researches funded by the **CHED** under the LAKAS Program available to the public. Subject only to the conditions stipulated herein, either HEI may publicly disclose the results of (i) said research (as well as related information about equipment, instruments, materials and methods used in acquiring such results), and (ii) any other research and/or development activities by the disclosing HEI which involves the use of materials, results, or other deliverables resulting from the collaborative research, or otherwise obtained under this RFA.

For purposes of public disclosure of information and data described above, both Parties acknowledge that the **UNIVERSITIES** which conducted and implemented the collaborative research is considered as either Disclosing or Reviewing Party under this Section.

9.1.1. **Period of Submission**. The Disclosing and Reviewing Parties share a common interest in making research conducted in connection with this RFA publicly available. Subject to their obligations of non-disclosure set forth in Section 9.2 (Confidentiality), the Parties have the right to copyright, publish, disclose, disseminate and use, in whole or in part, any data or information received or developed under this RFA. A Disclosing Party agrees to provide the Reviewing Party with an advanced courtesy copy of any publication, presentation or other dissemination for the purpose of review for patentable subject matter and Confidential Information not less than fifteen (15) calendar days prior to the Disclosing Party's planned release for publication or other dissemination.

9.1.2. **Period of Review**. The Reviewing Party shall have: (i) fifteen (15) calendar days to review, comment on, and notify the Disclosing Party of

the existence of the Reviewing Party's Confidential Information that it wishes to have deleted. The Disclosing Party agrees to remove such Confidential Information of the Reviewing Party upon request and; (ii) to identify patentable inventions for which it reasonably believes a patent application should be filed and to promptly cause written invention disclosures to be submitted to the Disclosing and Reviewing Parties' respective Technology Licensing departments. In such cases, the IPR owners agree to delay publication, presentation, or other dissemination for up to an additional sixty (60) calendar days in order to enable IIAs to be executed (where relevant) and a patent application to be prepared and filed. If no patent application will be filed, the IPR owner may waive all or a portion of the sixty (60) days extension period.

9.1.3 **Period of Post Review.** For the avoidance of doubt, publication, presentation or other dissemination may occur: (i) the day following the date on which the patent application is filed or after the ninety (90) calendar day period elapsed (whichever occurs first) or; (ii) the date within the ninety (90) calendar day period on which the IP/IPR owner(s) has/have given its/their written consent. The consent of both owners must be obtained in the case of joint ownership.

9.1.4. **Publication.** In any publication of the results of research, the Disclosing Party shall acknowledge the Reviewing Party's contribution to and/or participation in the research, unless the Reviewing Party desires otherwise and so advises the Disclosing Party in writing.

9.2 **Confidentiality**. During the course of the conduct and implementation of this collaborative research funded by the **CHED** under the LAKAS Program, a Providing Party may disclose to the Receiving Party certain information that the Providing Party regards as confidential or proprietary in nature (hereinafter collectively referred to as "Confidential Information") excluding trade secrets. Confidential Information shall be disclosed in writing or other tangible form to the **UNIVERSITY** Project Leader and shall be clearly marked "Confidential Information," or if disclosed orally shall be summarized in writing and marked "Confidential Information" by the Providing Party and submitted to the Receiving Party within thirty (30) calendar days of the oral disclosure. The obligations of confidentiality under this Section do not apply to any Information that: (i) was known to the Receiving Party prior to receipt thereof from the Providing Party; (ii) was or becomes a matter of public information or publicly available through no act or failure to act on the part of the Receiving Party; (iii) is acquired by the Receiving Party from a third party entitled to disclose the information to it and; (iv) is required to be disclosed in order to comply with an applicable law, regulation or court order (including, but not limited to any law or regulation governing the disclosure of information in or in conjunction with a patent application or patent). The Receiving Party agrees (i) to use Confidential Information only for purposes of performing research and (ii) to exercise reasonable efforts to prevent disclosure to third parties of Confidential Information in the absence of any written consent by the Providing Party to such disclosure. However, neither the Providing nor the Receiving Party shall be held liable for inadvertent disclosure of Confidential Information provided that it has exercised reasonable care and adequate security aimed at maintaining the confidentiality of the information, including notifying those of its employees who have access to Confidential Information of the confidentiality obligations imposed herein.

The Providing and Receiving Parties shall take all necessary steps to ensure that their directors, officers, agents, employees, staff, and authorized representatives shall be bound by this Section.

Upon termination, or pre-termination of this RFA as provided under Articles 20 and 23 hereof, the Receiving Party shall return to the Providing Party all Confidential Information received from the Providing Party.

Except as required by law and/or as may be stipulated by the **UNIVERSITIES** and/or evidenced by the RFA, the Receiving Party shall not disclose Confidential Information for a period of five (5) years from the receipt of such Confidential Information under this RFA. Nothing in this RFA shall prevent or limit either the Providing or Receiving Party from using or disclosing its own Confidential Information.

ARTICLE 10. RECORDS

10.1 The **UNIVERSITIES** shall retain all financial documents and records for a period of ten (10) years after termination of the project. The **CHED,** or its duly authorized representative(s) shall, within the project and for the aforesaid period, have access to such records for the purpose of conducting audits as provided under applicable Philippine government accounting and auditing laws, rules and regulations.

10.2 The **CHED** through its duly authorized representative(s) shall, for the purpose of the **CHED**'s statutory obligations to the Philippine Commission on Audit (COA), have access to any of the implementing Parties' records related to this RFA to monitor and conduct evaluation of operations which may include: a visit by personnel or designee to observe the implementing Parties' project structure; and, a review of financial and other records for the purpose of conducting audits as provided under applicable Philippine government accounting and auditing laws, rules and regulations.

ARTICLE 11. USE OF NAMES AND TRADEMARKS

Nothing contained in this RFA will be construed as conferring any right to either Party to use in advertising, publicity, or other promotional activities any name, trade name, trademark, or other designation of the other Party (including a contraction, abbreviation, derivation or simulation of any of the foregoing). Unless consented to in writing by the authorized licensing office of the **UNIVERSITIES** or the Office of the Chairperson of the **CHED**, the use of the name, trademarks, marks, icons, images and symbols of the other Party in advertising, publicity, or other promotional activities is expressly prohibited.

ARTICLE 12. DISCLAIMER OF WARRANTY

Materials, information, results, and data provided by one party to the other party under this RFA are provided without warranty of merchantability or fitness for a particular purpose or any other warranty, express or implied, including any warranty of non-infringement of any third party intellectual property right. Nothing in this RFA is or shall be construed as a warranty or representation by either party as to the validity, enforceability, or scope of any patent, patent application, or other IPRs.

ARTICLE 13. INDEMNIFICATION

The **UNIVERSITIES** hereby agrees to indemnify and hold harmless the **CHED** against any and all liability, claims, suits, losses, costs and legal fees caused by, arising out of, or resulting from any negligent act or omission of the project assigned scientist/staff in the performance and/or non-performance of the services stipulated under the RFA.

ARTICLE 14. COMPLIANCE WITH LAWS

In the performance of the research, each Party will comply with all applicable laws, codes, regulations, rules, and orders pertaining to research including, but not limited to, those pertaining to research involving the use of animals and recombinant DNA. These shall include all relevant international conventions, treaties and agreements.

The transfer of all research materials used shall be covered by Material Transfer Agreements (MTA). Further, the transfer of biological materials including breeding materials and germplasm will be subject to pertinent biosafety and bioprospecting laws, rules and regulations of the country of origin and the receiving country. Either Party may use such materials but will give full credit to and comply with other required conditions of the source of the materials.

All applications for IP protection which utilize indigenous knowledge, resources, practices and systems, as well as biodiversity and genetic resources shall comply with the legal mechanisms and processes under applicable laws.

ARTICLE 15. RELATIONSHIP OF PARTIES

The **CHED** and the **UNIVERSITIES** are independent parties in performing the obligations under this RFA. Neither the **UNIVERSITIES** nor the **CHED** shall be considered nor represent themselves as a joint venturer, partner, agent, fiduciary, or employee of the other.

ARTICLE 16. INSURANCE

All collaborating HEIs/SUCs shall maintain sufficient insurance to meet their indemnification and other liability obligations under this RFA.

ARTICLE 17. LIABILITY

Notwithstanding any other provision of this Agreement, in no event shall any Party be liable to the other Parties in connection with this agreement for any indirect, incidental, special, exemplary, punitive or consequential damages (including but not limited to loss of revenue, loss of profit, loss of business reputation or loss of opportunity) suffered by another Party whether arising out of contract, tort (including negligence), strict liability, or other legal theory and whether at law, in equity or otherwise.

ARTICLE 18. CONFLICT OF INTEREST

The **UNIVERSITIES** represent that there is no conflict of interest that might prejudice its independence and objectivity in carrying out its responsibilities under this RFA. The **UNIVERSITIES** also undertakes that upon becoming aware of any such conflict of interest during the performance of this RFA (whether the conflict existed before or arises during its performance), it shall immediately notify the **CHED** in writing pursuant to Section 22.6 hereof. This written notice shall give particulars of the nature of

the conflict of interest and the circumstances in which it exists or arises and shall furnish such further information as the **CHED** may reasonably require.

ARTICLE 19. DISPUTE SETTLEMENT

19.1 In the event of any dispute, controversy or claim arising out of or relating to this RFA, the Parties shall exert utmost efforts to amicably settle through negotiation and/or mediation within sixty (60) calendar days from receipt of a written notice by one Party from the other.

19.2 All such conflicts which cannot be resolved by negotiation and/or mediation shall be decided by arbitration in accordance with the Rules of Arbitration contained in the UNCITRAL Model Law as adopted in Republic Act No. 9285 or the Alternative Dispute Resolution (ADR) Law of 2004. For this purpose, there shall be three (3) arbitrators with each Party appointing one arbitrator each and those appointed selecting the third arbitrator who shall be the chair of the Arbitral Panel. The appointing authority mentioned in Article 6 of the UNCITRAL Model Law and Article 6 of Republic Act No. 9285 shall be the Chair of the **CHED**.

19.3 The place of arbitration shall be Metro Manila without prejudice to holding hearings in another suitable place for the convenience of witnesses who may be presented by the Parties. The language of the arbitration shall be in English or in Filipino.

19.4 The decision of the Arbitral Panel shall be final.

ARTICLE 20. PRETERMINATION

20.1 **Termination for Cause**. In the event that the **UNIVERSITIES** commit a material breach of the RFA or refuses and/or fails to perform any provision thereof, with such diligence as would ensure the research project's completion within the time specified in the approved Work Plan Schedule/s, or fails to complete the project and deliver the agreed output on time, the **CHED** may, by way of a thirty (30) calendar days First Notice in writing pursuant to Section 22.6 hereof, require the **UNIVERSITIES** to place the project back on track, failing which, this RFA shall be terminated in whole or in part, after thirty (30) calendar days Final Notice of Termination pursuant to Section 22.6 hereof counted from the date of expiration of the thirty (30) day calendar period of the First Notice. In case of such termination, the **UNIVERSITIES** will provide the **CHED** with copies of any deliverables in process and the **CHED** shall retain rights in deliverables previously delivered in accordance with the terms of this RFA.

20.2 **Voluntary Termination**. Each of the participating HEIs may terminate the contract voluntarily provided that the pre-terminating HEI return all project funds released and refund all project funds used to **CHED** as a precondition for pre-terminating this agreement. The preterminating HEI shall also pay damages to **CHED** in the amount as **CHED** may deem fit.

20.3 In both cases, it is further agreed that the **CHED** shall not only retain such project outputs already completed and delivered by the **UNIVERSITIES** but shall also enjoy the right to use the intellectual property rights over said outputs.

20.4 It is likewise understood that upon project termination in either case, the remaining balance of the research fund and project income generated shall be returned to the **CHED** subject to the provisions of the Republic Act No. 10055, the Philippine Technology Transfer Act of 2009, except in the case of Philippine Public HEIs which are

authorized to utilize internally generated project income pursuant to the provisions of their respective charters and other applicable laws, accounting and auditing laws, rules and regulations.

20.5. In both cases, Project Leaders and such other project staff as **CHED** may identity shall be suspended from receiving research grants from **CHED** as prescribed in the LAKAS Manual of Procedures.

ARTICLE 21. FORCE MAJEURE

A delay in or failure of performance of any of the Parties hereto shall not constitute default under this Agreement nor give rise to any claim for damages, if to the extent such delay or failure is caused by force majeure which shall mean an event which is unforeseen and outside the reasonable control of the Party which invokes it and which renders such Party unable to comply totally or partially with its obligations under the Agreement and including but not limited to, if the forgoing criteria are satisfied, fire, flood, earthquake, tornado, hurricane, strike, including labour trouble or other disturbance with a duration of more than seven (7) calendar days, war (declared or undeclared), acts of terrorism, embargoes, blockades, acts of any governmental authority (whether or not legally valid), riots, and insurrections. Any Party claiming force majeure under this Article shall use its commercially reasonable efforts to remove or terminate the event subject of such claim, provided that nothing herein shall obligate a Party to settle any strikes or labor disturbances. Such Party claiming force majeure shall, as soon as it is reasonably possible, notify the other Party in writing of the occurrence of the force majeure event and of the resolution of the force majeure event.

ARTICLE 22. MISCELLANEOUS

22.1 **Assignment:** Neither Party shall assign or transfer any interest in this RFA, nor assign any claims for money due or to become due under this RFA, without the prior written consent of the other Party.

22.2 **Entire Agreement:** This RFA together with attachments hereto, represent the entire understanding between the Parties as of the date of final signature below in respect to the subject matter of the RFA and supersede any previous or contemporaneous oral or written representations, statements, negotiations, agreements, or understandings between the **UNIVERSITIES** and the **CHED**.

22.3 **Severability:** The provisions of this RFA are severable, and if any provision, or any portion of a provision, is determined by an Arbitral Panel of competent jurisdiction to be invalid, illegal, or unenforceable for any reason, it will be adjusted rather than voided, if possible, to achieve the intent of the Parties to this RFA to the fullest extent possible. In any event, any remaining portion of that provision, and all other provisions of this RFA, will remain valid and enforceable to the fullest extent permitted by law in order to give effect to the Parties' intentions, unless, however, the provision is of such importance that the Parties or Party to the benefit of which such provision is made would not have entered into the RFA if they/it had known that the provision would not be valid.

22.4 **Waiver:** The waiver by a Party of any breach of or default under any of the provisions of this RFA or the failure of a Party to enforce any of the provisions of this RFA or to exercise any right hereunder shall not constitute or be construed as a waiver of any other breach or default or as a waiver of any such rights or provisions hereunder.

No waiver under this RFA will be effective unless expressly made in writing and signed by the authorized representative of the waiving Party.

22.5 **Successors and Assigns:** This RFA shall be binding upon and inure to the benefit of the Parties, their successors and permitted assigns.

22.6 **Notices:** Except for payments, any notice required or permitted to be given to either Party hereto will be in writing and deemed to have been properly given and effective; a) on the date of delivery if delivered in person during recipient's normal business hours; or b) on the date of receipt if delivered by courier, express mail service or first class mail, registered or certified, return receipt request. Such notice will be delivered or sent to the respective address given to the other Party as follows:

For the CHED:
The Commission on Higher Education
The LAKAS Program Management and Coordination Office (PMCO)
Attention: The LAKAS Program Manager
Higher Education Development Center
C.P. Garcia Avenue, U. P. Campus
Diliman, Quezon City, Philippines
Telephone: (632) 83761758
Email: lakaspmco@ched.gov.ph

For UP/Lead HEI:
Name: **Michael Y. Roleda, Dr. rer. nat.**
Designation: Professor / Dr. rer. nat
Address: Marine Science Institute, University of the Philippines Diliman, Velasquez Str., Quezon City
Contact Number: 09998486567/ (632) 8922-3962
Email Address: myroleda@up.edu.ph

For MSU-Naawan:
Name: **Rey Y. Capangpangan, Ph.D.**
Designation: Professor VI/ Director for International Affairs and Linkages
Address: Mindanao State University at Naawan, Poblacion, Naawan, Misamis Oriental, 9023, Philippines
Contact Number: +63 977 820 3227
Email Address: rey.capangpangan@msunaawan.edu.ph

For ParSU:
Name: **Michael A. Clores, Ph.D.**
Designation: Associate Professor II
Address: Partido State University, Caramoan Campus, Barangay Cadong, Caramoan, Camarines Sur, 4429, Philippines
Contact Number: +63 9175215772
Email Address: michael.clores@parsu.edu.ph

For SorSU:
Name: **Richard V. Dumilag, Ph.D.**
Designation:Associate Professor V
Address: Sorsogon State University, Magallanes Campus, Aguada Norte, Magallanes, Sorsogon, 4705, Philippines
Contact Number: +63 9361585522
Email Address: richard.dumilag@sorsu.edu.ph

For UEP:
Name: **Ronelie C. Salvador, Ph.D.**
Designation: Professor VI/Dean, College of Agriculture, Fisheries and Natural Resources
Address: University of Eastern Philippines, University Town, Catarman N. Samar, 6400
Contact Number: 09084303550
Email Address: roneliesalvador@yahoo.com

ARTICLE 23. TERM, AMENDMENT AND TERMINATION

23.1 This RFA shall take effect as approved by the proper authorities and shall remain valid for a period of two (2) years unless sooner terminated when one of the Parties notifies the other in writing of its intention to terminate the RFA, in which case, the RFA will terminate sixty (60) calendar days from date of notification. Such termination will not relieve the Parties of any obligation or liability accrued prior to termination, including provision of any payments due to the **UNIVERSITIES** by the **CHED** up to and including the date of termination and including any outstanding obligations incurred prior to the date of termination

23.2 The termination of the RFA of one of the above Party HEIs shall be without prejudice to the continuance of certain phases of the project that are ongoing, which should be allowed to continue until full completion as much as possible under such terms and conditions to be set forth in writing within the sixty (60) calendar day period. Further, the Party HEI whose RFA is prematurely terminated shall undertake to complete the expected outputs within sixty (60) calendar days, else Article 6 shall apply. The **UNIVERSITIES** will exert reasonable efforts to limit any expenses that may accrue during the termination period.

23.3 This RFA may be amended, modified, altered or supplemented subject to the mutual agreement of the Parties, which must be set forth in writing to be enforceable.

23.4 This RFA may be further renewed or extended upon mutual and formal agreement of the Parties.

IN WITNESS WHEREOF, the parties affix their signatures this JUL 11 2022 day of QUEZON CITY.

For the
Commission on Higher Education (the **"CHED"**):

J. PROSPERO E. DE VERA III, DPA
Chairman

For the
University of the Philippines
(the **"UP"**):

CARMENCITA D. PADILLA, M.D
Chancellor

For the
Mindanao State University- Naawan Campus
(the **"MSU-Naawan"**):

ELNOR C. ROA, Ph.D.
Chancellor

For the
Partido State University
(the **"ParSU"**):

RAUL G. BRADECINA, Ph.D.
SUC President III

For the
Sorsogon State University
(the **"SorSU"**):

ATTY. RYAN V. ESTEVEZ
OIC President

For the
University of Eastern Philippines
(the **"UEP"**):

CHERRY I. ULTRA, Ph.D.
University President

SIGNED IN THE PRESENCE OF:

ANTONIO S. MAGBANUA JR.
Project Technical Staff III (Admin and Finance)
LAKAS-PMCO

MICHAEL Y. ROLEDA, Dr. rer. nat.
Project Leader

REY Y. CAPANGPANGAN, Ph.D.
MSU-Naawan Collaborator

MICHAEL A. CLORES, Ph.D
ParSU Collaborator

RICHARD V. DUMILAG, Ph.D.
SorSU Collaborator

RONELIE C. SALVADOR, Ph.D
UEP Collaborator

CERTIFIED AS TO AVAILABILITY OF FUNDS:

MIRRIAM V. MIRANDA
CHED Chief Accountant

1.) UP 02-01102101-2022-06-01832
6/30/22 ₱ 24,015,809.98 cont.

2) UP 06-01101101-2022-06-01834
6/30/22 ₱ 34,170,000

3.) UNIVERSITY OF EASTERN PHILIPPINES
02-01102101-2022-06-01835
6/30/22 ₱1,491,072.00 cont.

4.) PARTIDO STATE UNIVERSITY
02-01102101-2022-06-01836
6/30/22 ₱ 1,011,072.00 cont.

5.) MSU naawan
02-01102101-2022-06-01840
6/30/22 ₱ 1,071,072.00

6.) SORSOGON STATE UNIVERSITY
₱ 1,731,072.00 cont. 6/30/22

ACKNOWLEDGMENT

REPUBLIC OF THE PHILIPPINES)
QUEZON CITY) s.s.

BEFORE ME, a Notary Public for and in Quezon City personally appeared:

Name	ID No	Date and Place of Issuance
J. PROSPERO E. DE VERA III, DPA	CO-0394	04 March 2019/ CHED-CO
CARMENCITA D. PADILLA, M.D	Passport No. P7167704B	Valid until 7/08/2031
ELNOR C. ROA, Ph.D.	S0018103A	02 December 2019, DFA Manila
ATTY. RYAN L. ESTEVEZ	CO - 0481	09 JULY 2020 / CHED - CO
RAUL G. BRADECINA, Ph.D.	P7648090B	21 September 2021 / DFA Legazpi
CHERRY I. ULTRA	UEPB-SUCPRES3-1-1998	10 December 2019 / Catarman N. Samar

known to me to be the same persons who executed the foregoing instrument and acknowledged to me that the same is their free and voluntary act and deed and that of the institutions represented.

This instrument refers to a Research Funding Agreement for the implementation of the project titled **LAKAS 2021-035 "Phytochemical Characterization of Macroalgae for Food and High Value Products (PhycoPRO)"** under the LAKAS Program consisting of _____ pages and the page on which this Acknowledgement is written including Annex "A" with ____ pages and Annex "B" with ____ pages signed by the parties and their instrumental witnesses on the signature page and initialed on the rest of the pages, and sealed with my notarial seal.

WITNESS MY HAND AND SEAL on this JUL 11 2022 at **Quezon City, Philippines**.

NOTARY PUBLIC

Doc. No. 342;
Page No. 70;
Book No. XII;
Series of 2022

ATTY. IZZITEL CHRISTINE DG. CRUZ
NOTARY PUBLIC FOR AND IN QUEZON CITY
Unit 7 Bldg. 2, Kamagong Centennial Dorm, E. Jacinto St.,
UP Diliman, QC. Tel No. (02)85463754; (02)89818500 loc 8680
ADM. MATTER NO. NP-001; Roll of Attorney No. 65418
IBP Lifetime No. 017849; OR No. 009899; 6-21-2017; Pasig City
PTR O.R. No. 2463174 D; 1-3-2022; Quezon City
MCLE Compliance No. VII- 0006084; Nov.23, 2021; Pasig City
My Commission Expires December 31, 2023

ANNEX "A"

APPROVED COLLABORATIVE RESEARCH PROJECT

ANNEX "B"

SUMMARY OF SCHEDULE OF PROJECT FUND RELEASES

	HEI/SUC	*Amount*
B1	University of the Philippines	**PhP137,687,132.54**
B2	Mindanao State University - Naawan	**PhP11,654,288.00**
B3	Partido State University	**PhP12,054,288.00**
B4	Sorsogon State University	**PhP12,674,288.00**
B5	University of Eastern Philippines	**PhP8,984,288.00**
Total		**PhP 183,054,284.54**

ANNEX "B1"

SCHEDULE OF PROJECT FUND RELEASES
[For UP]

Project Title	:	LAKAS 2021-035 Phytochemical Characterization of Macroalgae for Food and High Value Products (PhycoPRO)
Name of the LEAD UNIVERSITY	:	University of the Philippines
Name of the Project Leader:		Michael Y. Roleda, Dr. rer. nat.
Name of the Collaborating UNIVERSITIES	:	Mindanao State University – Naawan Sorsogon State University Partido State University University of Eastern Philippines
Name of Collaborators	:	Rey Y. Capangpangan, Ph.D. Michael A. Clores, Ph.D. Richard V. Dumilag, Ph.D. Ronelie C. Salvador, Ph.D.
Period Covered (in months)	:	24 months (July 01, 2022 to June 30, 2024)
Total Project Cost	:	**PhP137,687,132.54**

Release	Period Covered	Amount (PhP)	Documentary Requirements
1st	Tranche 1 Month 1 to 6	**PhP58,188,809.98**	1. Approved Joint Research Proposal, Project Work Plan & Budget, Schedules; 2. Research Funding Agreement (RFA) 3. Fund Request for first 6 months
2nd	Tranche 2 Month 7 to 18	**PhP66,591,466.27**	1. Month 3 PMCO Monitoring Report 2. Acceptance of Monitoring Report by PMCO 3. Fund Utilization Report of 75% of Tranche 1
3rd	Tranche 3 Month 19 to 24	**PhP12,906,856.30**	1. Month 16 Progress Report 2. Fund Utilization Report of 75% of Tranche 2
	60 days after Month 24 or after project completion or project termination (including approved extension)		1. Final Terminal Report for acceptance of CHED Commission En Banc 2. Consolidated Fund Utilization Report
Total Amount: PhP137,687,132.54			

Prepared and Submitted by:

IRENE RODRIGUEZ, Dr. rer. nat.
CO-Project Leader

Noted by:

ATTY. CINDERELLA FILIPINA S. BENITEZ-JARO
Executive Director IV
Office of the Executive Director
Commission on Higher Education

ANNEX "B2"

SCHEDULE OF PROJECT FUND RELEASES
[For MSU-Naawan]

Project Title	:	LAKAS 2021-035 Phytochemical Characterization of Macroalgae for Food and High Value Products (PhycoPRO)
Name of the Lead UNIVERSITY	:	University of the Philippines
Name of the Project Leader	:	Michael Y. Roleda, Dr. rer. nat.
Name of the Collaborating UNIVERSITY	:	Mindanao State University - Naawan
Name of Collaborator	:	Rey Y. Capangpangan, Ph.D.
Period Covered (in months)	:	24 months (July 01, 2022 to June 30, 2024)
Total Project Cost	:	**PhP11,654,288.00**

Release	Period Covered	Amount (PhP)	Documentary Requirements
1st	Tranche 1 Month 1 to 6	**PhP1,071,072.00**	1. Approved Joint Research Proposal, Project Work Plan & Budget, Schedules; 2. Research Funding Agreement (RFA) 3. Fund Request for first 6 months
2nd	Tranche 2 Month 7 to 18	**PhP9, 512,144.00**	1. Month 3 PMCO Monitoring Report 2. Acceptance of Monitoring Report by PMCO 3. Fund Utilization Report of 75% of Tranche 1
3rd	Tranche 3 Month 19 to 24	**PhP1,071,072.00**	1. Month 16 Progress Report 2. Fund Utilization Report of 75% of Tranche 2
	60 days after Month 24 or after project completion or project termination (including approved extension)		1. Final Terminal Report for acceptance of CHED Commission En Banc 2. Consolidated Fund Utilization Report
Total Amount: PhP11,654,288.00			

Prepared and Submitted by:

REY Y. CAPANGPANGAN, Ph.D.
Project Collaborator

Noted by:

ATTY. CINDERELLA FILIPINA S. BENITEZ-JARO
Executive Director IV
Office of the Executive Director
Commission on Higher Education

ANNEX "B3"

SCHEDULE OF PROJECT FUND RELEASES
[For ParSU]

Project Title	:	LAKAS 2021-035 Phytochemical Characterization of Macroalgae for Food and High Value Products (PhycoPRO)
Name of the Lead UNIVERSITY	:	University of the Philippines
Name of the Project Leader	:	Michael Y. Roleda, Dr. rer. nat.
Name of the Collaborating UNIVERSITY	:	Partido State University
Name of Collaborator	:	Michael A. Clores, Ph.D.
Period Covered (in months)	:	24 months (July 01, 2022 to June 30, 2024)
Total Project Cost	:	**PhP12,054,288.00**

Release	Period Covered	Amount (PhP)	Documentary Requirements
1st	Tranche 1 Month 1 to 6	**PhP1,011,072.00**	1. Approved Joint Research Proposal, Project Work Plan & Budget, Schedules; 2. Research Funding Agreement (RFA) 3. Fund Request for first 6 months
2nd	Tranche 2 Month 7 to 18	**PhP10,032,144.00**	1. Month 3 PMCO Monitoring Report 2. Acceptance of Monitoring Report by PMCO 3. Fund Utilization Report of 75% of Tranche 1
3rd	Tranche 3 onth 19 to 24	**PhP1,011,072.00**	1. Month 16 Progress Report 2. Fund Utilization Report of 75% of Tranche 2
	60 days after Month 24 or after project completion or project termination (including approved extension)		1. Final Terminal Report for acceptance of CHED Commission En Banc 2. Consolidated Fund Utilization Report
Total Amount: PhP12,054,288.00			

Prepared and Submitted by:

MICHAEL A. CLORES, Ph.D.
Project Collaborator

Noted by:

ATTY. CINDERELLA FILIPINA S. BENITEZ-JARO
Executive Director IV
Office of the Executive Director
Commission on Higher Education

ANNEX "B4"

SCHEDULE OF PROJECT FUND RELEASES
[For SorSU]

Project Title	:	LAKAS 2021-035 Phytochemical Characterization of Macroalgae for Food and High Value Products (PhycoPRO)
Name of the Lead UNIVERSITY	:	University of the Philippines
Name of the Project Leader	:	Michael Y. Roleda, Dr. rer. nat.
Name of the Collaborating UNIVERSITY	:	Sorsogon State University
Name of Collaborator	:	Richard V. Dumilag, Ph.D.
Period Covered (in months)	:	24 months (July 01, 2022 to June 30, 2024)
Total Project Cost	:	**PhP12,674,288.00**

Release	Period Covered	Amount (PhP)	Documentary Requirements
1st	Tranche 1 Month 1 to 6	**PhP1,731,072.00**	1. Approved Joint Research Proposal, Project Work Plan & Budget, Schedules; 2. Research Funding Agreement (RFA) 3. Fund Request for first 6 months
2nd	Tranche 2 Month 7 to 18	**PhP9,812,144.00**	1. Month 3 PMCO Monitoring Report 2. Acceptance of Monitoring Report by PMCO 3. Fund Utilization Report of 75% of Tranche 1
3rd	Tranche 3 Month 19 to 24	**PhP1,131,072.00**	1. Month 16 Progress Report 2. Fund Utilization Report of 75% of Tranche 2
	60 days after Month 24 or after project completion or project termination (including approved extension)		1. Final Terminal Report for acceptance of CHED Commission En Banc 2. Consolidated Fund Utilization Report
Total Amount: PhP12,674,288.00			

Prepared and Submitted by:

RICHARD V. DUMILAG, Ph.D.
Project Collaborator

Noted by:

ATTY. CINDERELLA FILIPINA S. BENITEZ-JARO
Executive Director IV
Office of the Executive Director
Commission on Higher Education

ANNEX "B5"

SCHEDULE OF PROJECT FUND RELEASES
[For UEP]

Project Title	:	LAKAS 2021-035 Phytochemical Characterization of Macroalgae for Food and High Value Products (PhycoPRO)
Name of the Lead UNIVERSITY	:	University of the Philippines
Name of the Project Leader	:	Michael Y. Roleda, Dr. rer. nat.
Name of the Collaborating UNIVERSITY	:	University of Eastern Philippines
Name of Collaborator?	:	Ronelie C. Salvador, Ph.D.
Period Covered (in months)	:	24 months (July 01, 2022 to June 30, 2024)
Total Project Cost	:	**PhP8,984,288.00**

Release	Period Covered	Amount (PhP)	Documentary Requirements
1st	Tranche 1 Month 1 to 6	**PhP1,491,072.00**	1. Approved Joint Research Proposal, Project Work Plan & Budget, Schedules; 2. Research Funding Agreement (RFA) 3. Fund Request for first 6 months
2nd	Tranche 2 Month 7 to 18	**PhP6,602,144.000**	1. Month 3 PMCO Monitoring Report 2. Acceptance of Monitoring Report by PMCO 3. Fund Utilization Report of 75% of Tranche 1
3rd	Tranche 3 Month 19 to 24	**PhP891,072.00**	1. Month 16 Progress Report 2. Fund Utilization Report of 75% of Tranche 2
	60 days after Month 24 or after project completion or project termination (including approved extension)		1. Final Terminal Report for acceptance of CHED Commission En Banc 2. Consolidated Fund Utilization Report
Total Amount: PhP8,984,288.00			

Prepared and Submitted by:

RONELIE C. SALVADOR, Ph.D.
Project Collaborator

Noted by:

ATTY. CINDERELLA FILIPINA S. BENITEZ-JARO
Executive Director IV
Office of the Executive Director
Commission on Higher Education

Project Code	LAKAS 2021-035
Project Title	Phytochemical characterization of macroalgae for food and high value products (**PhycoPRO**)
Proponents	Michael Y. Roleda, Ph.D.; Irene B. Rodriguez, Ph.D.; William H. Gerwick, Ph.D.; Lena S. Gerwick, Ph.D.
Implementing Agencies	University of the Philippines Diliman University of California San Diego
Funding Agency	CHED-PCARI
Collaborating HEIs	University of Eastern Philippines: Dr. Ronelie Salvador Mindanao State University – Naawan: Dr. Rey Capangpangan Sorsogon State University: Dr. Richard Dumilag Partido State University: Dr. Michael Clores
Proposed Budget	Total: **183,054,284.54**
Project Duration	2 years
Project Rationale/ Description	The Philippines is host to 1079 identified seaweed species and still have many unidentified species to be catalogued. Among the taxonomically identified species, only a small fraction (>5%) is substantially used for food and industrial applications. This can be attributed to poor understanding of its biochemical composition, health benefit and safety for consumption. Among the marketed and currently farmed species, the Philippines has an economic advantage in the global seaweed industry with an available farmable area of ~200,000 hectares along the coastlines. Currently, roughly 12% of this area is utilized for farming of seaweed species and strains from 4 genera namely: *Kappaphycus*, *Eucheuma*, *Gracilaria* and *Caulerpa*. Seaweed farming sustains the livelihood of more than 200,000 fishing families and generates about 250 million US dollars in export earnings. The diversity of seaweed species available in the Philippines and its available farmable area represent a massive potential for seaweed farming in the country. However, **there are several challenges that must be addressed to maximize the potential of the seaweed industry in the country. These include multi-disciplinary basic research focused on comprehensive phytochemical characterization of seaweeds and research into associated epiphytes of seaweeds to understand their role on the quality of seaweed-derived products.** This project is envisioned to enhance the status of the seaweed industry in the Philippines by **providing critical knowledge through a comprehensive phytochemical characterization of seaweeds to identify species that have higher economic potential and pinpoint locations where their growth will be maximal**. The major output of this project is a handbook of seaweeds outlining their phytochemical properties, which will be made available to the Bureau of Fisheries and Aquatic Resources (BFAR), seaweed farmers, and other stakeholders involved in the seaweed industry as a guide for mariculture activities. **The publication of a handbook of seaweeds outlining its phytochemical properties will have societal impacts in long-, medium-, and short-term scales. For long-term impact, the knowledge gained from this project will serve as the starting point of future efforts into product development for various applications including food and food additives and high-value compounds such as pigments, nutraceuticals, and pharmaceuticals. For medium-term impact, identification of seaweed species that are more profitable for a specific location will guide seaweed farmers on what species they will prioritize for farming and improvement of mariculture methods, which may result in poverty alleviation in coastal communities. For short-term impact, the identification of seaweed species with high nutritional content and safe for direct consumption will provide additional or alternative food sources for the general population, who are clamoring for nutritious dietary sources.** The project is envisioned to contribute to knowledge generation about potential uses of seaweeds that will have societal and economic impacts to coastal communities and the Philippine population in general. Millions of marginalized small-scale fisherfolk are dependent upon seaweed farming for their livelihoods. However, they are faced with problems of declining productivity and associated incomes due to a dependence on old and outdated seaweed farming methods that limit people to very shallow (<2m) coastal waters, old varieties of seaweed that have been vegetatively propagated for the last five decades, and the impacts of climate change. These factors interact to result in low growth rates and high mortality from epiphytism and ice-ice disease. As a result, seaweed farmers are increasingly returning to fishing, resulting in unsustainable levels of exploitation that diminish natural resources and trap people in poverty. This project proposal will build upon our expertise in the production of new and high-performing *Kappaphycus* cultivars generated from spores of wild local stocks through sexual reproduction and growing them up in local seaweed nurseries for distribution to seaweed farmers. Promising new strains will be tested for carrageenan content to preselect those with the highest quality and yield. Good quality biomass will command higher buying price that will effectively increase household incomes of small-scale fisherfolk/ seaweed farmers to at least P15-20,000 per month.

Project Objectives	The specific objectives of the proposed project are: 1. To perform systematic phytochemical characterization (including lipid, fatty acid, amino acid, and total phenolic content; proximal analysis; β-carotene, fucoxanthin, fucosterol, and iodine content of at least 50 seaweed species from the Philippines; 2. To assess the potential anti-cancer activity of extracts and purified compounds from seaweeds and associated epiphytes; and 3. To conduct training workshops and seminar on seaweed farming techniques, seaweed collection and preservation methods, and seaweed propagation strategies for faculty and researchers of partner HEIs, farming communities, and other stakeholders. The partner HEIs will actively participate in the conduct of research activities from site selection, sampling, sample processing and analysis, data collection and analysis, manuscript and report writing, and in training/workshops. The specific roles and participation of partner HEIs are outlined in the document "Summary of Role and Participation of HEIs" and also inserted in this document.
OUTCOME/ Socio-Economic Impacts	The project is expected to contribute towards poverty alleviation initiatives by providing additional or alternative sources of income for fishing families in coastal communities. Diversified income sources from introduction of seaweed farming and trading will reduce the pressure on fisheries, which may lead to replenishment of standing stocks of critical fish species. This will have positive effects on fish catch volumes of small-scale fishermen in the long run. The potential improvement in income sources will trigger positive effects like dominos starting with better livelihood, improved health, education of children, women empowerment, and many other socio-cultural aspects that are linked to the financial burden of coastal families. To guarantee that this initiative is continued beyond the duration of the project, the partner HEIs will be guided on how to conduct the training workshops in seaweed farming to ensure that information about new strains or up-to-date farming techniques are disseminated to seaweed farmers and community organizations in their respective provinces. The continuity of training that will be carried out by partner HEIs will guarantee that issues in mariculture activities will be addressed in a prompt and adequate manner due to proximity of HEIs to the communities requiring them.
Benefits to Partner HEIs	The PhycoPRO project is envisioned to be the foundation of long-term research collaborations between the UP-MSI and all partner HEIs. The research on seaweeds is planned to branch out and expand to other related research and development initiatives including seaweed biodiversity and mariculture studies at the University of Eastern Philippines (through Dr. Ronelie Salvador), Sorsogon State University (through Dr. Richard Dumilag), and Partido State University (through Dr. Michael Clores), and seaweed polymer chemistry research in Mindanao State University – Naawan (through Dr. Rey Capangpangan). These partnerships are aimed towards increasing the research productivity of the HEIs and promote global competitiveness of the faculty, researchers, and degree programs in these institutions. The outcome that the PhycoPRO proponents would like to achieve from this partnership is for the partner HEIs to be the focal SUCs in their respective provinces for seaweed research and in training of farmers and coastal community organizations. Faculty members and researchers in all HEIs identified will be trained on current seaweed mariculture techniques, seaweed culture maintenance, and basic physicochemical monitoring protocols. For those with existing marine biology programs and seaweed/seagrass resident scientists, e.g. Sorsogon State University and Partido State University, efforts towards more complex research initiatives will be sought. This initiative will be supported by having a separate budget for each partner HEI to fund salaries of research staff and for the procurement of equipment and materials they will require to conduct research activities in collaboration with the PhycoPRO proponents and on their own as lead researchers. While the construction and facilities required for a fully-functional seaweed nursery is dependent on the research thrust of the SUC and inhouse expertise, the SUC president and CHED Commissioner-in-Charge will consider the feasibility of establishing a seaweed nursery, and apply for separate funding as a long-term impact beyond the duration of the PhycoPRO project. Moreover, it is hoped that the financial support for equipment provided to kickstart this collaborative research project will lead to the eventual independent research initiatives of the collaborating institutions. The equipment housed in the partner institutions secured through this funding will enable researchers in the partner HEIs to continue research on seaweeds and other research directions in the future. Moreover, research initiatives of faculty/researchers at the partner HEIs that require the use of the sophisticated equipment (e.g., ICPMS, GCMS, and HPLCMS) will be supported by providing access to the instruments through research collaboration or through an agreement (e.g., MOA) that may be drafted between the UP-MSI and the partner HEIs.

Outputs/ Deliverables and Contribution to CHED initiatives	The outputs of the PhycoPRO project will contribute to knowledge generation and economic improvement in various aspects as outlined below: 1. The **handbook of seaweeds describing their phytochemical properties and potential commercial applications** may be used in the following: a. As a reference material for seaweed biodiversity studies by HEIs offering programs in Biology, Marine Biology, Ecology, Fisheries, and related programs; b. As a reference material for studies on marine natural products by HEIs offering programs in Chemistry, Biochemistry, Biotechnology, and related programs; c. As a reference material for Elementary and High School science classes to introduce students to seaweeds and its ecological or economic importance thus ensuring that the next generation of Filipinos will be educated in the biodiversity and richness of seaweed resources in the country; d. As a reference material for education and information of people's organizations especially within coastal communities to engage them in the conservation and, when techniques are available, in farming of seaweed species native to their coastal environments; e. As a reference material for various stakeholders including the Bureau of Fisheries and Aquatic Resources (BFAR), Seaweed Industry Association of the Philippines (SIAP), seaweed farmers, local government units, and others, who are involved in drafting policies on seaweed mariculture, development, and its utilization; and f. In cases where sufficient unique and/or endemic seaweed species with enough biomass for commercial use are encountered in coastal areas close to the location of partner institutions, a special edition handbook will be prepared featuring these seaweeds highlighting their potential for food and other applications and their role in the biodiversity of the region. 2. The **standard methods that will be optimized and validated for the determination of various phytochemical properties of seaweeds** may be used in the following: a. As reference materials for studies in natural products and analytical chemistry by HEIs offering programs in Chemistry, Biochemistry, Biotechnology, and related programs; and b. As reference materials to be used by laboratories to assess quality of raw materials and processed food products derived from seaweeds and other related plant products. 3. The project will focus on phytochemical characterization of at least 50 seaweed species from the country and this may lead to identification of bioactive compounds with potential medical application. When this happens in the conduct of the PhycoPRO project, a patent application will be filed and subsequent research and development work will be performed. Further, when other **seaweed species that are currently not farmed are characterized to be of potential high economic value**, the **suitable cultivation techniques for this species** will be developed so that it may be utilized in mariculture activities and further processing (e.g. harvesting, biomass post-harvest processing and pretreatment, among others) of the seaweed will be studied. The cultivation techniques, including current practices and new techniques, **will be taught to new seaweed farmers to introduce them to seaweed farming to diversify their livelihood sources**. 4. The PhycoPRO project will support the establishment of two core facilities at the UP-Marine Science Institute: the ICPMS facility and Seaweed Chemistry laboratory, both of which will support not just the PhycoPRO project but also other related projects at the UP-MSI and collaborating institutions. In addition, the core facilities will support the following: a. the training of graduate students who will carry out their research and will require use of the facilities. The PhycoPRO project will involve and support thesis or dissertation research of MS/PhD students to increase the number of experts in the field; and b. thesis/dissertation research of interested faculty members from the partner HEIs (University of Eastern Philippines in Northern Samar, Caraga State University in Agusan Del Norte, and Mindanao State University Tawi-Tawi) will be supported by the project through use of laboratories including the core facilities. This may lead to co-authored publications in high-impact factor scientific publications and participation in national and international conferences by researchers from partner HEIs. c. research initiatives of faculty/researchers at the partner HEIs that require the use of the sophisticated equipment (e.g., ICPMS, GCMS, and HPLCMS). The access to the instruments may be through research collaboration or through an agreement (e.g., MOA) that may be drafted between the UP-MSI and the partner HEIs.

	5. The partner HEIs will have separate budget to support procurement of equipment and materials that will kickstart research that will be led by faculty members of the respective institutions in collaboration with the proponents of the PhycoPRO project initially, but will eventually lead to research projects that will be implemented by researchers/faculty members in the collaborating HEIs. This is envisioned to contribute, if not start, to the research culture in the partner HEIs that may lead to research productivity of faculty and researchers in the partner HEIs.

Summary of the involvement of the collaborating SUC in the implementation of the activities under the PhycoPRO

MAJOR DELIVERABLES	ACTIVITIES	Lead HEI	Partner/Collaborating HEIs			
		University of the Philippines Diliman	University of Eastern Philippines	Mindanao State University – Naawan	Sorsogon State University	Partido State University
Seaweed collection and identification for herbarium collection and method optimization	Seaweed collection for optimization of methods					
	Deposit samples in herbarium					
	Pre-treatment of samples for method optimization					
Optimization and validation of methods	Optimization and validation of methods for: proximal analysis, total lipid, fucoxanthin, and fucosterol analyses, fatty acid extraction and analysis, amino acid analysis and specific component MAAs, β-carotene extraction and analysis, iodine analysis and estimation of iodine intake from seaweeds, analysis of phenolic content, determination of antioxidant activity, total metal analysis, speciation analysis of specific trace metals of interest, and in-vitro dialyzability procedure to determine bioavailability and bioavailability of trace metals.					
Screening of seaweed and associated epiphytes for bioactivities	Training of staff for seaweed handling and preservation for drug discovery related studies					
	Extraction for drug discovery related work, sending of materials to CA					
	Analysis and evaluation of extracts for activity and other parameters					

	Identification of isolated compounds with activity					
	Training of staff for isolation, purification, and identification of potential compounds					
Seaweed collection and evaluation of phytochemical parameters	Sampling for identification and preliminary testing					
	Deposit samples in herbarium					
	Sampling for analysis of various parameters					
	Analyses of seaweed and seaweed extracts using optimized methods for trace metals and lipid content					
	Basic phytochemical profiling of seaweeds					
	Data processing and analyses					
Coordination with LGUs, NGOs, and people's organizations	Organize local stakeholders' meetings with seaweed farmers, traders, and other relevant actors in the seaweed value chain to: • address risks associated with seaweed farming and corresponding risk management strategies. • mainstream gender equality • introduce idea of forming a cooperative for seaweed farmers • introduce added value chain for produce (e.g., seaweed for food production and fertilizer)					
	Conduct interviews with local stakeholders					
	Data analysis					
Training/Workshop on seaweed farming	Preparation of seaweed farming information materials					
	Conduct of training/workshop for local seaweed farmers, faculty members, and research staff on: • Philippine National Standard (PNS) on good aquaculture practices and PNS on raw dried seaweed					

	• Production-environmental sustainability model that can provide premium on the raw dried seaweed					
Publication	Write research papers					

PROJECT PROPOSAL

FOR CONSIDERATION UNDER THE

Phytochemical characterization of macroalgae for food and high value products (PhycoPRO)

SUBMITTED BY

Michael Y. Roleda
Marine Science Institute
University of the Philippines Diliman

Irene B. Rodriguez
Marine Science Institute
University of the Philippines Diliman

William H. Gerwick
Scripps Institution of Oceanography
Skaggs School of Pharmaceutical Sciences
University of California San Diego

Lena S. Gerwick
Scripps Institution of Oceanography
University of California San Diego

June 7, 2022

DATA SHEET

1. **Project Title: Phytochemical characterization of macroalgae for food and high value products (PhycoPro)**

2. **Proponents:**

PH Project Leader (PL)

Name: **Michael Y. Roleda**

Designation: Professor

Institution: Marine Science Institute, University of the Philippines Diliman

Address: Quezon City, Philippines

Telephone No: +63-2-89223962
E-mail: myroleda@up.edu.ph

3. **Co-Project Leader/Co-Principal Investigator/Collaborator:**

PH Co-Project Leader
Name: **Irene B. Rodriguez**
Designation: Associate Professor

Institution: Marine Science Institute, University of the Philippines Diliman
Address: Quezon City, Philippines
Telephone No: +63-2-89223962
E-mail: ibrodriguez@msi.upd.edu.ph

CA Collaborator
Name: **William H. Gerwick**
Designation: Professor
Institution: Scripps Institution of Oceanography and
Skaggs School of Pharmaceutical Sciences, University of California San Diego
Address: 9500 Gilman Drive, MC 0212, La Jolla, California, USA
Telephone No: +1-858-534-0578
E-mail: wgerwick@ucsd.edu

CA Co-Collaborator
Name: **Lena S. Gerwick**
Designation: Professor
Institution: Scripps Institution of Oceanography, University of California San Diego

Address: 9500 Gilman Drive, MC 0212, La Jolla, California, USA
Telephone No: +1-858-534-0578
E-mail: lgerwick@ucsd.edu

4. **Names and Addresses of Other Collaborating Institutions:**

University of Eastern Philippines: Dr. Ronelie Salvador
Mindanao State University – Naawan: Dr. Rey Capangpangan
Sorsogon State University: Dr. Richard Dumilag
Partido State University: Dr. Michael Clores

5. **Duration of the Project: Two years**

6. **Total Budget Requirement: PhP 183,054,284.54**

7. **Deliverables (In bullet form):**

- Handbook of seaweeds describing their phytochemical properties and potential commercial applications.
- Standard procedures for evaluating phytochemical properties (including lipid, fatty acid, amino acid, and total phenolic content; proximal analysis; β-carotene, fucoxanthin, fucosterol, and iodine content; and content of metals such as arsenic, cadmium, lead, iron, zinc, manganese, cobalt, copper, nickel, and selenium) of seaweeds.
- Standard procedures for seaweed extraction and preservation for subsequent characterization and bioassays
- Data on potential anti-cancer drug leads
- Scientific outputs including publications
- Professional development of research staff

8. **Expected Outcome(s) in Physical Terms (indicate all that apply):**

i) *New/ Upscaled Process*: Standard procedures for evaluating phytochemical properties of seaweeds; and standard procedures for seaweed extraction and preservation for subsequent characterization and bioassay of extracts for drug development.

ii) *Services*: Analytical services of the ICPMS facility and Seaweed Chemistry Laboratory employing optimized standard procedures for evaluation of raw materials and seaweed products.

iii) Handbook of seaweeds describing their phytochemical properties and potential commercial applications.

iv) Herbarium samples of seaweeds species from different areas or sampling locations in the Philippines.

v) Scientific outputs including publications.

9. **Human Subjects:**

Does your project involve human subjects, tissue or cells from living human donors? What type of involvement will it be? List all that apply.

This project will not involve human subjects.

10. **Animal Use:**

Does your project involve the use of animals? What type of involvement will it be? List all that apply.

This project will not involve the use of animals during the period of the grant, which covers activities to determine potential of seaweed and seaweed extracts for nutraceutical and pharmaceutical applications.

11. **Intellectual Property (IP) Rights/Invention Patents:**

Are there any past/existing/future IP issues in this project? Are there any past/current/future patents (to be) filed for any portion of this project? Identify who or which institutions are affected by these issues.

Patents arising from this work will be submitted to relevant institutions following the Intellectual Property Rights (IPR) Guidelines of the UP System and UCSD.

Project Title:

Phytochemical characterization of macroalgae for food and high value products (PhycoPRO)

PH Project Leader (PL):

Name: **Michael Y. Roleda**
Designation: Professor
Institution: Marine Science Institute, University of the Philippines Diliman
Address: Quezon City, Philippines
Telephone No: +63-2-89223962
E-mail: myroleda@up.edu.ph

Co-Project Leader:

Name: **Irene B. Rodriguez**
Designation: Associate Professor
Institution: Marine Science Institute, University of the Philippines Diliman
Address: Quezon City, Philippines
Telephone No: +63-2-89223957
E-mail: ibrodriguez@msi.upd.edu.ph

CA Collaborator:

Name: **William H. Gerwick**
Designation: Professor
Institution: Scripps Institution of Oceanography and Skaggs School of Pharmaceutical Sciences, University of California San Diego
Address: 9500 Gilman Drive, MC 0212, La Jolla, California, USA
Telephone No: +1-858-534-0578
E-mail: wgerwick@ucsd.edu

CA Co-Collaborator:

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I. Project Approach

A. Abstract

Seaweeds comprise one of the most important biotic components of different marine ecosystems, including estuaries, mangrove swamps, salt marshes, and coral reefs. They also serve as the base of the

marine trophic food chain and are often used as an indicator of ecosystem health. The geographical distribution of seaweeds is largely influenced by physical, chemical, and biological factors. Among biological factors, the presence of epiphytes constitutes a major role in the growth and development of seaweeds because epiphytes secrete metabolites that are critical for the survival of the host but may be harmful in different circumstances. Centuries of human consumption have resulted in diverse uses of seaweeds from different algal groups including red (Rhodophyta), brown (Phaeophyceae), and green (Chlorophyta) algae, which represents a rich resource for a multitude of applications including products for food, nutraceuticals, pharmaceuticals, and other bioactive compounds. The worldwide seaweed industry relies on the seaweed produced by Chile (39%), China (23%), Norway (13%), Japan (9%), France (4%), Ireland (2.73%), Iceland (2%), Canada (1%), and South Africa (1%) (West et al. 2016).

Seaweeds (=macroalgae), as processed and unprocessed food, have an annual commercial value of several billion dollars. Seaweed extracts are commonly used in preparing foods and the direct consumption of seaweed has existed for centuries in the diets of East Asian and Pacific Island societies (Arasaki & Arasaki 1983). The red seaweed nori, or laver (i.e. ***Porphyra*** spp. and ***Pyropia*** spp.), is one of the most important commercial food seaweeds. In Japan, approximately 100,000 hectares of shallow bays and seas are farmed for nori production. Several species of ***Pyropia*** are found in Northern Philippines (Ilocos Norte, Cagayan, Isabela, and Batanes) where wild stocks are harvested for local consumption (Dumilag & Montilla 2018; Dumilag et al. 2017).

Species of ***Sargassum*** (a type of brown seaweed) are also harvested from wild beds along rocky shores, particularly in Ilocos Norte and Davao, where they may be eaten with fish and in soups. The green seaweed ***Ulva*** look somewhat like leaves of lettuce (sea lettuce) and are eaten as salads or in soups, relishes, and meat or fish dishes. Other seaweed species harvested and prepared as salads are the brown seaweed ***Hydroclathrus***, the green seaweed ***Caulerpa*** (sea grapes, common name "*lato*"), and several red seaweeds such as ***Gracilaria***, ***Kappaphycus***, and ***Eucheuma*** (Montaño et al. 2015).

The cell walls of many seaweeds contain phycocolloids that can be extracted by hot water. The three major phycocolloids are alginates, agars, and carrageenans (Stiger-Pouvreau et al. 2016). Alginates are extracted primarily from brown seaweeds, and agar and carrageenan are extracted from red seaweeds. These phycocolloids are polymers of chemically modified sugar molecules, such as galactose in agars and carrageenans, or organic acids, such as mannuronic acid and glucuronic acid in alginates. Most phycocolloids can be safely consumed by humans and other animals, and many are used in a wide variety of prepared foods, such as ready-mix cakes, instant puddings and pie fillings, and artificial dairy toppings (Stiger-Pouvreau et al. 2016). Alginates, or alginic acids, commercially extracted from brown seaweeds are used in ice creams to limit ice crystal formation (producing a smooth texture), in syrups as emulsifiers and thickeners, and in candy bars and salad dressings as fillers (Stiger-Pouvreau et al. 2016). Alginates can be extracted from Philippine ***Sargassum***, ***Hormophysa*** and ***Turbinaria***.

Agars extracted primarily from species of red seaweeds, such as Philippine ***Gelidium***, ***Gracilaria***, ***Gelidiella***, and ***Pterocladia***, are used in instant pie fillings, canned meats or fish, and bakery icings and for clarifying beer and wine. Agar is also used extensively in laboratory research as a substrate for growing bacteria, fungi, and algae in pure cultures and as a substrate for eukaryotic cell culture and tissue culture (Stiger-Pouvreau et al. 2016). Carrageenans are extracted from various red algae, including ***Eucheuma*** and ***Kappaphycus*** in the Philippines. Carrageenans are used for thickening and stabilizing dairy products, imitation creams, puddings, syrups, and canned pet foods and are used in the manufacture of shampoos, cosmetics, and medicines (Stiger-Pouvreau et al. 2016).

For many centuries, seaweeds around the world have been widely used as agricultural fertilizers. Coastal farmers collect seaweeds by cutting them from seaweed beds growing in the ocean or by gathering them from masses washed up on shores after storms. The seaweeds are then spread over the soil. Dried seaweed, although almost 50 percent mineral matter, contains a large amount of nitrogenous organic matter. Commercial extracts of seaweed sold as plant fertilizers contain a mixture of macronutrients, micronutrients, and trace elements that promote robust plant growth (Delaney et al. 2016). Seaweeds synthesize primary and secondary metabolites. These bioactive compounds can play a major therapeutic role in disease prevention in humans (Déléris et al. 2016). These compounds include polysaccharides, pigments, fatty acids, polyphenols and peptides that have been proven to possess various beneficial biological properties that could potentially contribute to functional food and nutraceutical development. The functional traits of these bioactive compounds include antioxidant, antibacterial, anticancer, antidiabetic, antitumor, antiviral, anti-inflammatory and anticoagulant properties (Vonthron-Sénécheau 2016).

The cosmetics industry is a key industrial sector, which is worth more than €425 billion worldwide. This industry is always searching for new ingredients and had been looking from the richness and diversity of

the marine world. Seaweeds are rich in bioactive compounds and are exploited as functional ingredients for cosmetic applications (Couteau & Coiffard 2016). Marine cosmetics, also known as phycocosmetics, are an economic reality and specific brands such as Algotherm, Phytomer, Daniel Jouvance, Science & Mer, or Thalgo focus on marine resources.

Seaweeds are of interest to the cosmetics industry not only as a source of minerals, polysaccharides, proteins, and lipids, but also because of the secondary metabolites such as phenolic compounds, terpenoids, halogenated compounds, sulfur derivatives, and nitrogen derivatives that they can produce. The cosmetic potential of different bioactive compounds found in seaweeds includes seaweed extracts as active ingredients for (1) slimming products, (2) moisturizing and antiaging products, (3) topical UV photoprotection, (4) skin whitening, and source of (5) excipients, (6) preservatives, (7) dye and pigment, and (8) aromas and fragrances (Couteau & Coiffard 2016).

The Philippines has an economic advantage in the global seaweed industry with an available farmable area of ~200,000 hectares along the coastlines. Currently, roughly 12% of this area is utilized for farming of seaweed species and strains from 4 genera namely: *Kappaphycus*, *Eucheuma*, *Gracilaria* and *Caulerpa.* The Philippines is host to 1079 identified seaweed species and may still have many unidentified species to be catalogued. Seaweed farming sustains the livelihood of more than 200,000 fishing families and generates about 250 million US dollars in export earnings (SIAP, 2017). The export destination includes Australia, China, South Korea, the USA, and others, and most of the export is of carrageenan. Among the destination countries, the US accounts for about a quarter of export earnings. However, the US National Organic Standard Board (NOSB) recommended the removal of carrageenan from the National List of Additives for Organic Products. This recent development in parallel with volatility of the seaweed local price, financing constraints, and dwindling numbers of seaweed farmers, will influence the potential earnings from seaweed production (SIAP, 2017). The reliance of the seaweed industry on the importance and demand of carrageenan poses an economic problem with the current weak international demand. Given the potential economic value and increase in demand for other seaweed-derived compounds from the international market, further research into seaweeds and seaweed culturing techniques and development of processes producing high-value products are necessitated.

Aside from carrageenan, there are other important bioactive compounds derived from seaweeds including carotenoids, fatty acids, fucoidan, and phycobiliproteins, among many others. The value of the world wide carotenoid market was nearly $1.2 billion in 2010 and is projected to reach over $1.4 billion by 2018 (BBC, 2011). β-Carotene, a major carotenoid of brown algae, has the largest share of the market as a dietary supplement and in the food market. Fucoxanthin is another orange carotenoid pigment that is a product of brown algae (Muradian et al., 2015). There is a robust literature on the health benefits of diets enriched in fucoxanthin, either taken as a supplement or through consumption of the source macroalgae. Mono-unsaturated fatty acids (MUFAs) and polyunsaturated fatty acids (PUFAs) have been shown to possess a variety of health benefits and are thus identified as important components of human nutrition. The major PUFAs of algae that exhibit health benefits are docosahexaenoic acid (DHA) and eicosapentaenoic acid (EPA). Fucoidan is the name given to a heterogeneous group of sulfated polysaccharides that can be obtained from various brown algae (Atashrazm et al., 2015). The variable linkages between these sugar units, location of sulfate groups and the trace components of other sugars give these biopolymers very different biological and pharmacological properties, depending on the source organism, time of collection, and processing parameters. The reported useful biological properties of fucoidan are largely in anticancer and antiproliferation (Atashrazm et al., 2015). The distinctive pigments of red algae, known as phycobiliproteins, have a growing number of biomedical applications. These pigments, including phycoerythrin, phycocyanin, and allophycocyanin, are exceptionally bright and have been used in such applications as cell marking for use in fluorescent activated cell sorting (FACS) (Kronick, 1986). Because of their light absorbing characteristics, they have also been used in photodynamic therapies (Huang et al., 2002), as well as for their antioxidant and anti-inflammatory properties (Romay et al., 2003). Finally, there is a growing use of these pigments in the cosmetic and food industries.

The **numerous important compounds derived from seaweeds** highlight the potential of developing it as a **renewable resource for a biobased economy**. Unlike Japan, Korea, and Chile, where seaweed is commercially available as dried products for food, in the Philippines its use is mostly limited to consumption of freshly harvested seaweed, which has a very short shelf life. Moreover, there has been no systematic analysis of the food quality of Philippine seaweed species [content such as amino acids (AA), fatty acids (FA), carbohydrates, vitamins, minerals, and heavy metals]. With these diverse seaweed species and massive untapped potential in the Philippines, **this project will focus on the phytochemical characterization of at least 50 seaweed species in the country. The phytochemical characterization of seaweed species will serve as the foundation for development of processes for seaweed utilization for food or as a source of material for nutraceuticals, which will be pursued by the proponents in future projects. The proposed widespread sampling is a great opportunity for bioprospecting to discover**

biomedicinal products from Philippine seaweeds through a comprehensive screening program for high value materials with a specific focus on anti-cancer agents.

The natural products of marine organisms are of high structural diversity and are produced in biological systems for inherent biological purposes; thus, they are best screened in mechanism-agnostic screens such as the bioassays in the Gerwick laboratory at UCSD. These natural products have a high potential to effect cells in new and diverse ways of relevance to cancer chemotherapy. Certainly, diversity in discovery programs should be a high international priority given the dismal therapeutic outcomes against the major human solid tumors over the past two decades of research that have featured approaches other than those proposed here. The future of cancer treatment is envisioned to revolve around the identification of specific DNA mutations in a given cancer patient's tumor cells followed by the development of a specific cocktail of anticancer drugs "targeted' to those modifications. Unfortunately, there are few such drugs available to mix into these futuristic cocktails, and there are literally thousands of potential targets. Through projects such as the one proposed here that throw a wide net around unique natural products for their anticancer activity and then define the mechanisms of the therapeutically active ones, we have the highest probability of finding agents that work by new and specific mechanisms. These advances are bringing the future of cancer therapy into the present. It is envisioned that natural products screening capability in UP-MSI will be further developed through the training and assistance that will be performed with the involvement of the Gerwick group at UCSD.

Seaweed research in the Philippines was initiated from efforts of scientists from the UP-MSI, especially from the contributions of Drs. Gavino Trono Jr. (GCT), Edna Ganzon-Fortes (EGF), and Nemesio E. Moñtano Jr. (NEM). Dr. Trono primarily worked on marine benthic algal taxonomy, e.g., *Sargassum*, *Gracilaria*, and *Acetabularia*, among others, and on Philippine seaweed biodiversity and their utilization. Moreover, GCT worked on the biomass production and clonal propagation of different strains of commercially important eucheumatoids, the recruitment and seasonality of the standing stock of *Sargassum* species, the effects of different environmental factors on the growth and phycocolloid quality of different agarophytes, and recently on the cultivation of the red seaweed *Halymenia durvillei*. Dr. Ganzon-Fortes also worked on marine benthic algal taxonomy, e.g., *Acetabularia*, *Laurencia*, and *Hydroclathrus*, and on Philippine seaweed biodiversity. Moreover, EGF worked on different aspects, i.e., photophysiology, ecology, reproductive biology, agar yield, and quality of *Gelidiella acerosa*. EGF also worked on the physiology of diseased and healthy *Kappaphycus alvarezii*, carrageenan yield and quality of different euchuematoids, their cultivation and development of new strains from wild populations. Recently, she studied the nutrient physiology of *Padina* in Bolinao, Pangasinan. Dr. Montaño primarily worked on the chemical characterization of different phycocolloids, e.g., agar and carrageenan, and alginate extractable from different species of red and brown seaweeds, respectively. NEM also worked on different optimization methods, e.g., pre- and post-treatments to improve phycocolloid yield and quality. Recently, he is involved in screening of fucoidan content of Philippine brown seaweeds and worked on the cytotoxic activity of crude extract of *Sargassum*. **The research efforts of the pioneering scientists on seaweeds provided a sizable understanding of the current seaweed cultivars in the country. However, there is lack of knowledge about the phytochemical properties of the multitude of naturally occurring species, many of which may produce commercially viable compounds of interest. This proposed project is intended to address this knowledge gap and is specifically aimed towards identifying seaweed species that produce the highest yields of phycobiliproteins, which are of interest because of their applications in food, medical, and cosmetic applications, and wide use in research and development. Currently, the market price for C-phycocyanin is about PhP 12,000.00/10 mg and B-phycoerythrin is about PhP 15,000.00/10 mg. The high cost of these pure phycobiliproteins highlights the economic potential of seaweeds in the Philippines.**

This project is envisioned to enhance the status of the seaweed industry in the Philippines by **providing critical knowledge through a comprehensive phytochemical characterization of seaweeds to identify species that have higher economic potential and pinpoint locations where their growth will be maximal**. The major output of this project is a handbook of seaweeds outlining their phytochemical properties, which will be made available to the Bureau of Fisheries and Aquatic Resources (BFAR), seaweed farmers, and other stakeholders involved in the seaweed industry as a guide for mariculture activities. **The publication of a handbook of seaweeds outlining its phytochemical properties will have societal impacts in long-, medium-, and short-term scales. For long-term impact, the knowledge gained from this project will serve as the starting point of future efforts into product development for various applications including food and food additives and high-value compounds such as pigments, nutraceuticals, and pharmaceuticals. For medium-term impact, identification of seaweed species that are more profitable for a specific location will guide seaweed farmers on what species they will prioritize for farming and improvement of mariculture methods, which may result in poverty alleviation in coastal communities. For short-term impact, the identification of seaweed species with high nutritional content and safe for direct consumption will provide additional or alternative food sources for the general population, who are clamoring for nutritious dietary sources.** The project is

envisioned to contribute to knowledge generation about potential uses of seaweeds that will have societal and economic impacts to coastal communities and the Philippine population in general.

Millions of marginalized small-scale fisherfolk are dependent upon seaweed farming for their livelihoods. However, they are faced with problems of declining productivity and associated incomes due to a dependence on old and outdated seaweed farming methods that limit people to very shallow (<2m) coastal waters, old varieties of seaweed that have been vegetatively propagated for the last five decades, and the impacts of climate change. These factors interact to result in low growth rates and high mortality from epiphytism and ice-ice disease. As a result, seaweed farmers are increasingly returning to fishing, resulting in unsustainable levels of exploitation that diminish natural resources and trap people in poverty. This project proposal will build upon our expertise in the production of new and high-performing *Kappaphycus* cultivars generated from spores of wild local stocks through sexual reproduction and growing them up in local seaweed nurseries for distribution to seaweed farmers. Promising new strains will be tested for carrageenan content to preselect those with the highest quality and yield. Good quality biomass will command higher buying price that will effectively increase household incomes of small-scale fisherfolk/ seaweed farmers to at least P15-20,000 per month.

This project represents the fusion of high scientific rigor and experimental innovation focused on the phytochemical characterization of various seaweeds species that will eventually lead to development of novel food and biotechnology products from Philippine marine algae and their epiphytes. As such, we expect a number of high impact international publications on the unique marine resources present in these marine algae. These publications will provide the scientific underpinnings for new utilization of some species of algae as a source of healthy foods, as well as validation of the nutritive value of several other seaweeds that are currently used as food. Potentially harmful properties of the seaweeds to human health will also be characterized, providing a holistic view to their utilization for these purposes. Complementing this, we expect to also define a number of promising lead compounds for preclinical development in biological assays focused on anticancer activities.

The specific objectives of the proposed project are:

1. To perform systematic phytochemical characterization (including lipid, fatty acid, amino acid, and total phenolic content; proximal analysis; β-carotene, fucoxanthin, fucosterol, and iodine content of at least 50 seaweed species from the Philippines;
2. To assess the potential anti-cancer activity of extracts and purified compounds from seaweeds and associated epiphytes; and
3. To conduct training workshops and seminar on seaweed farming techniques, seaweed collection and preservation methods, and seaweed propagation strategies for faculty and researchers of partner HEIs, farming communities, and other stakeholders.

The partner HEIs will actively participate in the conduct of research activities from site selection, sampling, sample processing and analysis, data collection and analysis, manuscript and report writing, and in training/workshops. The specific roles and participation of partner HEIs are outlined in the document "IHITM_2018 035_Summary of Role and Participation of HEIs_PhycoPRO 26 May 2022."

OUTCOME/SOCIO-ECONOMIC IMPACTS

The project is expected to contribute towards poverty alleviation initiatives by providing additional or alternative sources of income for fishing families in coastal communities. Diversified income sources from introduction of seaweed farming and trading will reduce the pressure on fisheries, which may lead to replenishment of standing stocks of critical fish species. This will have positive effects on fish catch volumes of small-scale fishermen in the long run. The potential improvement in income sources will trigger positive effects like dominos starting with better livelihood, improved health, education of children, women empowerment, and many other socio-cultural aspects that are linked to the financial burden of coastal families.

To ensure that this initiative is continued beyond the duration of the project, the partner HEIs will be guided on how to conduct the training workshops in seaweed farming to ensure that information about new strains or up-to-date farming techniques are disseminated to seaweed farmers and community organizations in their respective provinces. The continuity of training that will be carried out by partner HEIs will guarantee that issues in mariculture activities will be addressed in a prompt and adequate manner due to proximity of HEIs to the communities requiring them.

Approach

Background

Notable examples of isolated natural products from seaweeds. Human consumption of seaweeds has been a tradition for several centuries in China, Japan and Korea (McHugh, 2003). In the tropics, fresh

seaweed prepared as salads is a delicacy among coastal dwellers from Indonesia, Malaysia and the Philippines. Today, the use of seaweed, a macroalgae, as food and functional ingredient is known worldwide. Macroalgae have traditionally been an extremely rich source of bioactive small molecules, some of which have potential biomedical utility. For example, others as well as ourselves have characterized small halogenated monoterpenes as possessing anticancer type properties, such as those from *Plocamium* spp. and *Portieria hornemannii*, and these have been a source of great interest by the National Cancer Institute in the US (Andrianasolo et al., 2005). Other red algal natural products have been examined for their high potential as antineoplastic agents; prominent among these is the highly active triterpene polyether thyrsiferol and related compounds from *Laurencia viridis* (Blunt et al., 1978).

The brown algae or Phaeophyceae, are similarly rich in bioactive natural products, including terpenes and meroterpenes, polyphenolics, and oxylipins. For example, we previously discovered stypoldione and related compounds, mero-diterpenoids with unique orthoquinone functionalities, to have powerful cancer cell toxicity as well as neurotoxic properties (Sabry et al., 2005). Eckol is a brown algal polyphenolic metabolite with a variety of biological activities, including inhibition of adipogenesis (Kim et al., 2017), anti-inflammation (Eom et al., 2017), and antibacterial properties (Eom et al., 2013). Finally, brown algae are rich in unique prostaglandin-like substances, known as oxylipins, and these have a variety of biological properties in mammalian systems. The egregiachlorides are one such class of unique oxylipin (Todd et al., 1993); prostanoids of this sort have some of the most potent and profound hormone-like properties.

Finally, the green algae or Chlorophyta are also a chemically rich group of marine algae that are important in tropical reef systems in a variety of ways. For example, several calcium carbonate aggregating species of the genus *Halimeda*, which are important to the construction of coral reefs, are chemically defended against predation by producing the extremely potent metabolite halimedatrial, a compound with additional biological activities such as cytotoxicity to cancer cells (Paul et al., 1988). A notable green algal metabolite of relevance to the Philippines is caulerpin which was shown to be a plant growth regulator in a root growth test of lettuce seeds and to exhibit in vitro antitumor activity (Raub et al., 1987). Caulerpin has a unique bis-indole structure which is related to the anti-inflammatory natural products topsentin (Alvarado et al., 2013) and scytonemin (Stevenson et al., 2002).

Potential of seaweed-derived high value products. Seaweeds thrive in habitats where conditions, e.g. high irradiance and ultraviolet radiation, temperature fluctuation, and desiccation among others, favor the formation of free radicals and other oxidizing agents. Their ability to minimize oxidative damage and maintain cellular integrity against these adverse conditions is related to the presence of different antioxidant systems. In living organisms, failure to activate the defense mechanism can lead to the accumulation of reactive oxygen species (ROS) and other free radicals causing irreversible damage to proteins, amino acids, lipids and DNA. In food products, ROS can promote the degradation of oil and fats contributing to the decrease in nutritional quality and security. Polyphenols and sulfated polysaccharides are among the powerful antioxidant compounds found in different seaweed species (e.g. Valentão et al., 2010; Balboa et al., 2013; Di et al., 2017). These naturally occurring compounds in seaweeds promote its use as food or as a functional ingredient that may prevent chronic diseases and improve food shelf life and quality with health-related beneficial properties (Roohinejad et al., 2017). The same compounds in seaweeds, such as cell wall polysaccharides and proteins with biomedical potentials and health benefits, also have several metal binding sites, which may contribute to the complexity of the ROS homeostasis in the cells (Kuyucak and Volesky, 1989; Reddy and Prasad, 1990; Güven et al. 1995).

Current limitations of seaweed products development. Heavy metals are naturally occurring elements of the earth's crust. Human activities including mining and smelting operations, industrial production and use, and domestic and agricultural use of metals and metal-containing compounds have drastically altered their geochemical cycles and biochemical balance (Singh et al., 2011). Heavy metal toxicity depends on several factors such as chemical species, dose, and exposure route, as well as the age, gender, genetics, and nutritional status of exposed individuals. Arsenic (As), cadmium (Cd), lead (Pb), and mercury (Hg) are highly toxic and rank among the priority metals that are of public health significance (Tchounwou et al., 2012). Different algal polysaccharides bind with heavy metals and their binding affinity is ranked as: sodium alginate > carrageenans > agar (Güven et al. 1995). Moreover, peptides of different species of microalgae and a brown seaweed *Sargassum muticum* also bind with heavy metal through chelation (Gekeler et al., 1988). Only a few studies have conducted comprehensive investigation on different heavy metals in edible seaweeds commercialized for human consumption (e.g., Almela et al., 2006; Besada et al., 2009; Rubio et al., 2017). Considering that there is an increasing use of seaweed for food, food additives, dietary supplements, and for their medicinal properties (Liu et al., 2015; Déléris et al., 2016; Fleurence, 2016; Kang et al., 2016; Vonthron-Sénécheau, 2016), it is essential that contaminants present in different seaweed species are thoroughly investigated before they enter our daily diet.

Cyanobacterial epiphytes are especially notorious as they often go undetected on macroalgae, yet have major consequences on the chemical composition of the food material. Examples include the highly toxic polycarvernosides, products of cyanobacteria growing on red algae, and the aplysiatoxins, also products of cyanobacteria growing on red algae. In the first case, toxic outbreaks have occurred in Guam and the Philippines; in the latter location, three outbreaks of seaweed poisoning occurred from ingestion of two edible species of red alga, *Acanthophora specifera* and *Gracilaria edulis* (Yotsu-Yamashita et al., 2004). Three of 12 victims died in the first event, three of 15 died in the second event, and two of nine died in the third event, for an overall mortality of 8 deaths from 36 reported exposures (22% mortality). The causative agent was shown to be polycarvernoside A and its congeners, both in Guam and in the Philippines; however, the ultimate source of the toxins was somewhat uncertain from these studies. Subsequently, we made field collections of the free-living marine cyanobacterium *Okeania* sp. in the Caribbean, and used a cancer cell toxicity bioassay to guide the isolation of biologically active components. This resulted in the isolation of a new polycarvernoside, "D", which was closely related in structure to the previously reported polycarvernosides, and clearly showed that these metabolites are the products of marine cyanobacterial metabolism (Navarro et al., 2015).

In the Hawaiian 1994 event, consumption of the red alga *Gracilaria coronopifolia*, a seaweed which had long been safely eaten by local residents, resulted in a widespread toxic event with nausea and vomiting being the major symptom (no deaths reported). Investigation of the toxic material resulted in the isolation of a few related metabolites, clearly in the aplysiatoxin family of cyanobacterial secondary metabolites, and given the names manauealides A-C (Nagai et al., 1997). Aplysiatoxin-type metabolites are well recognized as tumor promoters due to their PKC activating properties; in the current acute poisoning situation, the toxicity appeared to be due to intestinal bleeding (Ito & Nagai, 2000). These examples demonstrate the potential for human poisonings by consumption of seaweeds that are generally safe to eat but become epiphytized by other algae and cyanobacteria, which render them toxic. **This is a principal of which the PIs of the present proposal, which is focused on exploring the nutritive value of seaweeds not yet consumed in the Philippines, are aware of. We will carefully and fully document the likelihood of seaweed species to become contaminated by epiphytes such as marine cyanobacteria, and characterize the taxonomy and chemical constituents of these contaminating species completely so as to ensure a safe food supply.**

Rationale

The diversity of seaweed species available in the Philippines and its available farmable area represent a massive potential for seaweed farming in the country. However, there are several challenges that must be addressed to maximize the potential of the seaweed industry in the country. These include:

1. Research focusing on systematic and comprehensive phytochemical characterization of seaweeds present in Philippine waters must be performed.
2. Research into potential uses and applications of seaweeds must be carried out in order to identify species that have higher economic potential and locations where their growth will be maximal.
3. Research into processes producing seaweed products must be carried out to improve their quality and stability.
4. Research focused on identification and isolation of bioactive compounds from macroalgae must be carried out to identify other potential commercial applications.
5. Research directed towards elucidating the role of associated microorganisms on the quality of seaweed-derived products and how the resident microbiome influence production of bioactive compounds and their efficacy.

This project is envisioned to address the most critical among the knowledge gaps mentioned, which is the phytochemical characterization, to enhance the status of the seaweed industry in the Philippines. This project is envisioned to enhance the status of the seaweed industry in the Philippines by providing critical knowledge through a comprehensive phytochemical characterization of seaweeds to identify species that have higher economic potential and pinpoint locations where their growth will be maximal. The major output of this project is a handbook of seaweeds outlining their phytochemical properties, which will be made available to the Bureau of Fisheries and Aquatic Resources (BFAR), seaweed farmers, and other stakeholders involved in the seaweed industry as a guide for mariculture activities. The publication of a handbook of seaweeds outlining its phytochemical properties will have societal impacts in long-, medium-, and short-term scales. For long-term impact, the knowledge gained from this project will serve as the starting point of future efforts into product development for various applications including food and food additives and high-value compounds such as pigments, nutraceuticals, and pharmaceuticals. For medium-term impact, identification of seaweed species that are more profitable for a specific location will guide seaweed farmers on what species they will prioritize for farming and improvement of mariculture methods, which may result in poverty alleviation in coastal communities. For short-term impact, the identification of seaweed species with high nutritional content and safe for direct consumption will provide additional or alternative food sources for the general population, who are clamoring for nutritious dietary sources. The project is envisioned to contribute in knowledge

generation about potential uses of seaweeds that will have societal and economic impacts to coastal communities and the Philippine population in general.

The sampling areas that will be included in this project are the following: Catarman, Northern Samar; Dinagat, Surigao Del Norte; Tandubas, Tawi Tawi; Batan, Batanes; Bolinao, Pangasinan; Currimao, Ilocos Norte; Guiuan, Eastern Samar; Biri, Northern Samar; Iba, Zambales; Busuanga, Palawan; Sorsogon, Sorsogon; Polilio Island, Quezon; Bien Unido, Bohol; and Lian, Batangas.

The successful implementation of this project is anchored on the merging of different scientific expertise and utilization of complementary approaches in addressing the various research targets and objectives. The specific approaches and methodologies are briefly described in the following section.

Brief description of methods that will be employed in this project

1. Identification of seaweed species
   Species identification will be based on the gross morphology and identification keys provided in taxonomic handbooks for Philippine and Regional Marine Flora, e.g., Field Guide and Atlas of the Seaweed Resources of the Philippines (Trono 1997), among others. Representative samples of each species will be pressed and mounted onto an herbarium paper. They will be deposited at G.T Velasquez Phycological Herbarium, the Marine Science Institute, University of the Philippines, Diliman, Quezon City. When necessary, molecular analysis, i.e. DNA Barcoding, will be employed.

2. Optimization of standard methods for analysis of target analytes/component in seaweeds or seaweed extracts
   a. *Proximal analysis*. The water, ash, and fiber content will be determined following the standard AOAC methods (water, AOAC 934.01; ash, AOAC 930.05; fiber, AOAC 985.29).
   b. *Total lipid analysis*. The lipid content of seaweed samples will be analyzed following an optimized protocol based on the method by Folch et al. (1957) and Gosch et al. (2012). A suitable amount of dried seaweed powder will be extracted using a chloroform:methanol mixture, heated at 60°C, filtered onto pre-weighed Whatman GF/A filter, washed with NaCl solution before allowing to stand. The lower phase containing the lipids will be evaporated using a gentle stream of nitrogen prior to weighing of total lipid content.
   c. *Fucoxanthin and fucosterol analysis*. Part of the lipid extract will be used to determine fucoxanthin and fucosterol by high performance liquid chromatography following the methods proposed by Maeda et al. (2005) and the liquid extraction optimized by Fung et al. (2013).
   d. *Fatty acid extraction and analysis*. The fatty acid content of seaweed samples will be determined following a method adapted from Gosch et al (2012). Briefly, fatty acids will be methylated using a mixture of methanol and acetylchloride, heated for 1 h at 100°C, allowed to cool before addition of hexane. Complete partitioning of fatty acid methyl esters (FAMEs) will be done prior to analysis using gas chromatography mass spectrometry.
   e. *Amino acid analysis*. The amino acid content will be determined by HPLC following the method proposed by Alaiz et al (1992) and modified by Ortiz et al (2006). Briefly, hydrochloric acid will be added to a suitable amount of seaweed sample, the solution will be gassed with nitrogen, sealed, and will be incubated at 110°C for 24 h. The amino acid hydrolizate will be dried and subsequently dissolved in borate buffer prior to derivatization using diethyl ethoxymethylene malonate and determination of the amino acid derivatives using HPLC with a UV-Vis detector.
   f. *Extraction and analysis of mycosporine-like amino acids (MAAs)*. The isolation of UV-absorbing MAAs will be carried out following the protocol by Rosic et al. (2015) and subsequently analyzed using HPLC.
   g. *β-carotene extraction and analysis*. The β-carotene content of seaweed samples will be extracted following the method by Norziah et al (2000) where a mixture of tetrahydrofuran:methanol followed by petroleum ether and 0.1% butylated hydroxytoluene as extraction solvents. Saponification will be carried out using a mixture of potassium hydroxide:methanol prior to drying the extract and subsequent analysis of β-carotene using reverse phase HPLC.
   h. *Analysis of iodine content*. The iodine in seaweed will be derivatized with 3-pentanone and analyzed using GC-MS following the method used by Yeh et al. (2013).
   i. *Determination of total phenolic content*. The total phenolic content of seaweed extracts will be determined according to the method by Wang et al. (2009) using the Folin-Ciocalteu reagent. The measurements will be done using phloroglucinol solution as calibration standard and using UV-Vis spectrophotometry.
   j. *Total antioxidant activity of extracts*. The antioxidant activities will be determined based on the effect of scavenging of 1,1-diphenyl-2-picrylhydrazyl (DPPH) radicals according to the method by Souza et al (2012).
   k. *Trace metal analysis*. The trace metal (essential and toxic) content of seaweed samples will be analyzed by inductively coupled plasma mass spectrometry (ICPMS). Total elemental

concentration will be determined using mineralized samples while speciation analysis will be carried out on extracts using HPLC coupled with ICPMS following optimized methods modified from Rodriguez et al. (2008 and 2009).

l. *In-vitro testing to assess bioaccessibility and bioavailability of trace metals from seaweed and seaweed extracts.* An in-vitro dialyzability procedure following protocols by Moreda-Piniero et al. (2012) and Garcia-Sartal et al. (2013) will be validated for use in seaweed and seaweed extracts. These methods simulate the chemical conditions during gastric digestion.

Strategy for Meeting Milestones

To ensure that milestones will be met, the project staff that will be hired must have requisite academic training (chemistry, biology, biochemistry, molecular biology) to deliver the outputs. In addition, the equipment that will be purchased will have built-in autosamplers to maximize the number of samples that will be analyzed. Further, proper scheduling will be implemented, e.g., seaweed sampling for preservation and method optimization will be performed while procurement of equipment is being done. The details of the project activities may be found in the GANNT chart.

3. Methods for extraction, characterization, and bioactivity assays of seaweed extracts
   a. *Extraction*. The extraction of algal tissue or cyanobacterial mats or tufts will utilize our longstanding modified Bligh and Dyer method (Crews et al., 1991).
   b. *Fractionation*. Over the past 20 years, we have found it highly effective to fractionate the extract as produced above into a series of nine "pre-fractions" (e.g. fractionated before any biological assays occur). In our experience, these pre-fractions function much more effectively in biological assays because of several factors: 1) minor constituents are concentrated in the fractions such that their activity is more easily observed, 2) nuisance compounds are removed from the extract or segregated within specific fractions so they can straightforwardly be discounted, and 3) these reduced complexity materials can be directly further purified by HPLC to yield pure compounds in a single next step, thereby accelerating the bioassay-guided process.
   c. *Dereplication*. A new innovation that we have added to our natural products discovery workflow is the use of Liquid Chromatography-Mass Spectrometry with fragment ion (LC-MS2) recording on our extracts and "pre-fractions". These data are added to the massIVE database within the Global Natural Products Social (GNPS) (https://gnps.ucsd.edu/ProteoSAFe/static/gnps-splash.jsp) molecular network and then portrayed in Cytoscape. This allows known compounds to be identified early in the process, and the relationship of new compounds to known ones to be quickly realized.
   d. *Biological Assays*. In the current project, we will focus on a couple of biological assays that will reveal the presence of small molecules that i) have potential utility in the treatment of cancer, or ii) may have mutagenic or carcinogenic properties in humans if ingested as part of the diet. The first set of assays involves antiproliferative and cytotoxicity measurements using an indicator cancer cell line, the H-460 human lung carcinoma cell line from the National Cancer Institute. *Purification*. Biologically active fractions from the above evaluations will be further fractionated by HPLC to relatively pure materials, and then re-assayed in the relevant biological system.
   e. *Mechanism of Action Studies*. Ultimately, the process outlined above will lead to the biological and chemical characterization of a novel compound from various Philippine marine algae. The relationship of the new compound to known ones will be investigated through use of comprehensive databases, such as AntiMarin and SciFinder. The pharmacological mechanism of action of the new compound can sometimes be deduced through this comparison to the literature, and for other times needs focused investigation by one of a variety of techniques with which our laboratory is experienced, for example transcriptomics, enzymatic assays and development of resistance in yeast to the isolated compound followed by sequencing.
   f. *Identification of Producing Organism*. A common question that arises from the isolation of new molecules from field collected natural assemblages is who in the assembly is actually producing the desired natural product. **In the case of macrophytic marine algae, it becomes a question whether the macrophyte is the biochemical source, or is it an epiphyte, a component of the microbiome, or an endophytic symbiont such as a fungus?** To discern this is important to both public health questions as it relates to toxins in the food supply as well as to medicinal agent discovery and subsequent commercial production. In some cases this can be deduced by culture of symbiotic or associated microorganisms, as we did in the case of unusual nucleosides from the sponge *Cryptotethia crypta* (Bertin et al, 2015). Alternatively, imaging MALDI mass spectrometry is an extremely powerful method by which to demonstrate the association of a particular substance with a specific organism present in a consortium (Simmons et al, 2008). Use of genetic signatures associated with the biosynthesis of a particular molecule can also be used, as we showed in the case of the cyanobacterium *Oscillatoria spongellae* associated with the sponge *Dysidea herbaceae* (Flatt et al, 2005). Whole genome sequencing can enhance this latter approach; this is an especially effective approach when a microorganism, such as a cyanobacterial epiphyte, is suspected to be the biosynthetic source of a secondary metabolite.

Throughput of Gerwick Laboratory

Our laboratory has considerable throughput in terms of the number of samples that can be analyzed for known as well as new secondary metabolites, whether they be toxins or potential pharmaceutical yields. This comes in large part by our exceptional infrastructure with plentiful laboratory space (8 separate laboratories with space for 15-20 researchers), great instrumentation (2 LC-MS instruments with auto samplers, one GC-MS with auto sampler, and various UV, IR and polarimeters in our labs; two 500 MHz NMRs, one with an auto sampler, in our building), and well trained and experienced staff and students. With the additional resources requested under this research grant activity, we are certain that we will be able to process and evaluate the volume of materials that will derive from this collaborative project with the Philippines.

Timeline for Meeting Milestones

On the timeline for providing the information needed from our part of the project, we expect that we will at least have quarterly joint research meetings with the rest of the members of the team, and thus we will stay up to date with the processing of samples on at least a quarterly basis. Initial evaluations of materials by chemical and biological assays occur continuously, and as noted above, we have excellent facilities and personnel to work on the project. Specific deliverables from this project include technology transfer on algal extraction and fractionation, bioassay results, structures of active and inactive compounds isolated from marine algae, mechanisms of action and biosynthetic pathway deduction for a select set of bioactive compounds, and scientific outputs including publications and patents. Our research plan involves the continuous processing of materials (extraction, fractionation) to keep pace with the number of algal species examined as sources of food or other biotechnological products. Bioassays take place on a weekly basis, and instrumental analysis of samples occurs continuously as our LC-MS and NMR instruments have auto sample changers which allow samples to be run 24 hours a day, 7 days a week. With this capacity and the appropriate level of personnel, we are confident of our ability to provide deliverables as set forth in the proposal.

B. Project Team Description & Roles

The research team members comprising the PH side will be composed of two project leaders, one project staff, and four research fellows, all of whom will be involved in the planning and implementation of the project. They will be assisted by one project assistant, whose job will focus more on the administrative aspect of the work and will be devoting 100% of his time for the success of this project.

The two project leaders, Dr. Michael Roleda and Dr. Irene Rodriguez, will divide the tasks in directing the research. Dr. Roleda will spearhead all research involving seaweed collection and identification, seaweed culturing, and other related activities. Dr. Rodriguez will lead all research involving optimization and validation of standard methods for the evaluation of physical and chemical properties of seaweeds and seaweed extracts. The project staff, Dr. Lilibeth Salvador-Reyes, is also a member of the MSI faculty. Dr. Salvador-Reyes is an expert on medicinal and natural products chemistry, biochemistry, and pharmacology. Both project leaders will be devoting 40% of their time and the project staff will be devoting 10 % of her time for the success of this project.

The science research specialists and university research associates that will be hired for this project will be identified from a pool of applicants and must have the following qualifications: a bachelor's degree in science from a reputable college or university, civil service eligibility or a license from the Professional Regulation Commission (PRC), and of good moral standing. In addition to these minimum requirements, the team members that will be chosen must demonstrate willingness to work with a team, learn new techniques and skills, work in a competitive and demanding environment, and must be interested in the pursuit of scientific excellence. All research fellows are expected to dedicate 100% of their time for the success of this project.

The Gerwick group at SIO/UCSD will be involved in teaching the Philippines group the process of organic chemical extraction and pre-fractionation of marine algae for toxin as well as medicinal agent discovery. In the Gerwick group, biological assays will be performed on these pre-fractionated materials, and then biological assay-driven isolation of active compounds will be achieved using column, flash and high-performance liquid chromatography (HPLC). Structures of bioactive compounds will be determined using spectroscopic methods including NMR and MS. Mechanisms of action of select active compounds will be explored using transcriptomics and biochemical assays. Finally, genomic data for the organisms producing the bioactive compounds will be analyzed for the presence of biosynthetic gene clusters. Deliverables from this project include technology transfer on algal extraction and fractionation, bioassay results, structures of active and inactive compounds isolated from marine algae, mechanisms of action and biosynthetic pathway deduction for a select set of bioactive compounds.

C. Annual Project Deliverables & Intermediate Key Milestones with Dates or Timeline (1-2 pages)

The specific objectives and goals of this project will be sought by the project team members simultaneously and in close coordination with each other. For the successful implementation of the project, the research team will aim to meet the following deliverables and intermediate milestones:

Deliverables	TARGET DATE
Seaweed collection for preliminary studies including optimization methods for seaweed preservation, analytical methods, bioassays, extraction methods, etc.	July 2023
Set-up of Seaweed Chem Lab	October 2023
Set of optimized methods for analysis of specific analytes (including lipid, fatty acid, amino acid, and total phenolic content; proximal analysis; β-carotene, fucoxanthin, fucosterol, and iodine content) in seaweeds and seaweed extracts	December 2023
Seaweed collection for phytochemical analysis	December 2023
Set-up of ICPMS Facility	December 2023
Set of optimized methods for analysis of specific analytes (including content of metals such as arsenic, cadmium, lead, iron, zinc, manganese, cobalt, copper, nickel, and selenium) in seaweeds and seaweed extracts	April 2023
Data and background information on the phytochemical properties of seaweed species from different sampling sites	March 2024
Seaweed collection for phytochemical analysis	April 2024
Data on compounds isolated from seaweeds with novel bioactivities	June 2024
Data on the toxicological evaluation of seaweed products	September 2024
Training/Workshop on seaweed farming	August 2024
Scientific publications, handbook	December 2024

The project research team members are also expected to present the project results in relevant international and national scientific meetings to gain feedback and insights about the work.

OUTPUTS/DELIVERABLES and CONTRIBUTION TO CHED INITIATIVES

The outputs of the PhycoPRO project will contribute to knowledge generation and economic improvement in various aspects as outlined below:

1. The **handbook of seaweeds describing their phytochemical properties and potential commercial applications** may be used in the following:
   a. As a reference material for seaweed biodiversity studies by HEIs offering programs in Biology, Marine Biology, Ecology, Fisheries, and related programs;
   b. As a reference material for studies on marine natural products by HEIs offering programs in Chemistry, Biochemistry, Biotechnology, and related programs;
   c. As a reference material for Elementary and High School science classes to introduce students to seaweeds and its ecological or economic importance thus ensuring that the next generation of Filipinos will be educated in the biodiversity and richness of seaweed resources in the country;
   d. As a reference material for education and information of people's organizations especially within coastal communities to engage them in the conservation and, when techniques are available, in farming of seaweed species native to their coastal environments; and
   e. As a reference material for various stakeholders including the Bureau of Fisheries and Aquatic Resources (BFAR), Seaweed Industry Association of the Philippines (SIAP), seaweed farmers, local government units, and others, who are involved in drafting policies on seaweed mariculture, development, and its utilization; and
   f. In cases where sufficient unique and/or endemic seaweed species with enough biomass for commercial use are encountered in coastal areas close to the location of partner institutions, a special edition handbook will be prepared featuring these seaweeds highlighting their potential for food and other applications and their role in the biodiversity of the region.

2. The **standard methods that will be optimized and validated for the determination of various phytochemical properties of seaweeds** may be used in the following:
   a. As reference materials for studies in natural products and analytical chemistry by HEIs offering programs in Chemistry, Biochemistry, Biotechnology, and related programs; and

b. As reference materials to be used by laboratories to assess quality of raw materials and processed food products derived from seaweeds and other related plant products.

3. The project will focus on phytochemical characterization of at least 50 seaweed species from the country and this may lead to identification of bioactive compounds with potential medical application. When this happens in the conduct of the PhycoPRO project, a patent application will be filed and subsequent research and development work will be performed.
   Further, when other **seaweed species that are currently not farmed are characterized to be of potential high economic value**, the **suitable cultivation techniques for this species** will be developed so that it may be utilized in mariculture activities and further processing (e.g. harvesting, biomass post-harvest processing and pretreatment, among others) of the seaweed will be studied. The cultivation techniques, including current practices and new techniques, **will be taught to new seaweed farmers to introduce them to seaweed farming to diversify their livelihood sources.**

4. The PhycoPRO project will support the establishment of two core facilities at the UP-Marine Science Institute: the ICPMS facility and Seaweed Chemistry laboratory, both of which will support not just the PhycoPRO project but also other related projects at the UP-MSI and collaborating institutions. In addition, the core facilities will support the following:
   a. the training of graduate students who will carry out their research and will require use of the facilities. The PhycoPRO project will involve and support thesis or dissertation research of MS/PhD students to increase the number of experts in the field; and
   b. thesis/dissertation research of interested faculty members from the partner HEIs (University of Eastern Philippines in Northern Samar, Mindanao State University Naawan, Sorsogon State University, and Partido State University) will be supported by the project through use of laboratories including the core facilities. This may lead to co-authored publications in high-impact factor scientific publications and participation in national and international conferences by researchers from partner HEIs.
   c. research initiatives of faculty/researchers at the partner HEIs that require the use of the sophisticated equipment (e.g., ICPMS, GCMS, and HPLCMS). The access to the instruments may be through research collaboration or through an agreement (e.g., MOA) that may be drafted between the UP-MSI and the partner HEIs.

5. The partner HEIs will have separate budget to support procurement of equipment and materials that will kickstart research that will be led by faculty members of the respective institutions in collaboration with the proponents of the PhycoPRO project initially, but will eventually lead to research projects that will be implemented by researchers/faculty members in the collaborating HEIs. This is envisioned to contribute, if not start, to the research culture in the partner HEIs that may lead to research productivity of faculty and researchers in the partner HEIs.

The project deliverables and milestones are also detailed further in the itemized Workplan.

D. Collaboration Plan

For a successful implementation of the grant and to attain the specific objectives towards harnessing the potential of seaweeds and its associated epiphytes, **our approach is to bring together scientists from different disciplines and utilize the expertise of each project leader/principal investigator.** Dr. Roleda is a phycologist with expertise in seaweed biodiversity, seaweed natural products and algal biofuel production, and seaweed culturing and farming. Dr. Gerwick is a distinguished biochemist with expertise in natural products chemistry and biosynthesis, and biomedicine. Dr. Rodriguez is a chemist with expertise in marine biogeochemistry, algal culture, and environmental analytical chemistry. The research fellows that will be hired for this project will be graduate students with academic background in biology, chemistry, molecular biology, and microbiology. In the US, postdoctoral fellows or research staff will conduct the proposed work, unless a predoctoral student with the apppropriate background and interests can be located to work on the project.

The proposed project is multifaceted research with each component being directed by specific PLs/PIs. The seaweed collection, identification, culture, farming, and process development for food applications will be led by Dr. Roleda. Characterization and further identification of the associated epiphytes will also be carried out with Dr. Roleda's direction. The collection of seaweed and associated epiphytes for screening of potential bioactive compounds or pharmaceuticals, and toxicological

evaluations, will be under the supervision of Dr. Gerwick. In addition, Dr. Gerwick and his postdoctoral associates will also supervise the training of PH-based research fellows in the collection and extraction of seaweeds for subsequent characterization and relevant assays. The optimization and validation of various methods and procedures for determination of physicochemical parameters (lipids, fatty acids, trace metal contaminant content, etc.) will be the responsibility of Dr. Rodriguez. Necessarily, part of the research work will be carried out at the MSI and part of the work will be done at Scripps.

A planning meeting will be held on the first quarter of the implementation of the project to layout sampling plans, discuss and confirm specific tasks of each research staff, carry out training of key research team members, and identify research staff who will be responsible for meeting specific research targets. Annual meetings will also be conducted to assess the progress of the project and prepare relevant reports.

A vital part of this research is to carry out systematic sampling of seaweeds from different areas in the Philippines to facilitate identification of species with the best potential for R&D. In this regard, regional higher education institutions (HEIs) including the **University of Eastern Philippines, Mindanao State University in Naawan, Sorsogon State University, and Partido State University** will be partner HEIs. As partner HEIs, representative faculty members of the institutions will contribute by helping identify seaweed sampling locations in their respective localities, participate in the sampling and preliminary preservation or drying procedures, and facilitate the conduct of training and workshops in their respective regions. The representative faculty members will also participate in the training or workshop that will be hosted in UP-MSI on how to conduct bioassays and other relevant procedures. The partner HEIs will also contribute towards the goals of the project by establishing contact with local seaweed farmers, gathering information on how local communities utilize seaweeds, and facilitating in knowledge transfer during workshops and training especially when the use of mother language is deemed more effective. In addition, faculty members, staff, and students from these institutions may also participate in the training or workshop that will be conducted in their region. As a partner HEI, faculty and students of the institutions may further benefit from the cooperation by becoming a visiting researcher to conduct academic research at the MSI laboratory funded from this project proposal.

BENEFITS TO PARTNER HEIs

The PhycoPRO project is envisioned to be the foundation of long-term research collaborations between the UP-MSI and all partner HEIs. The research on seaweeds is planned to branch out and expand to other related research and development initiatives including seaweed biodiversity and mariculture studies at the University of Eastern Philippines, Sorsogon State University, and Partido State University, and seaweed polymer chemistry research in Mindanao State University - Naawan. These partnerships are aimed towards increasing the research productivity of the HEIs and promote global competitiveness of the faculty, researchers, and degree programs in these institutions.

The outcome that the PhycoPRO proponents would like to achieve from this partnership is for the partner HEIs to be the focal SUCs in their respective provinces for seaweed research and in training of farmers and coastal community organizations. Faculty members and researchers in partner HEIs identified will be trained on current seaweed mariculture techniques, seaweed culture maintenance, and basic physicochemical monitoring protocols. For those with existing marine biology programs and seaweed/seagrass resident scientists, e.g. Sorsogon State University and Partido State University, efforts towards more complex research initiatives will be sought. This initiative will be supported by having a separate budget for each partner HEI to fund salaries of research staff and for the procurement of equipment and materials they will require to conduct research activities in collaboration with the PhycoPRO proponents and on their own as lead researchers. While the construction and facilities required for a fully-functional seaweed nursery is dependent on the research thrust of the SUC and inhouse expertise, the SUC president and CHED Commissioner-in-Charge will consider the feasibility of establishing a seaweed nursery, and apply for separate funding as a long-term impact beyond the duration of the PhycoPRO project.

Moreover, it is hoped that the financial support for equipment provided to kickstart this collaborative research project will lead to the eventual independent research initiatives of the collaborating institutions. The equipment housed in the partner institutions secured through this funding will enable researchers in the partner HEIs to continue research on seaweeds and other research directions in the future. Moreover, research initiatives of faculty/researchers at the partner HEIs that require the use of the sophisticated equipment (e.g., ICPMS, GCMS, and HPLCMS) will be supported by providing access to the instruments through research collaboration or through an agreement (e.g., MOA) that may be drafted between the UP-MSI and the partner HEIs.

All scientific outputs from the work will be attributed to collaborating parties according to the following stipulations:

(a) Authorship of scientific outputs such as publications, technical papers, and scientific presentations will be decided according to percent contributions toward the completion of the work. Mechanics on who will be listed as authors will be decided upon during the planning meeting.
(b) The Intellectual Property Rights (IPR) Policy of the UP and UC Systems will be strictly adhered to for patents or similar forms for products and processes arising from the work.

To facilitate performance of the research tasks, biological materials and extracts from biological samples need to be sent to the UCSD counterpart and this procedure will be in accordance with the Wildlife Act (Philippine Republic Act No. 9147), which provides rules for the conservation and protection of wildlife resources and their habitats. PH-based Research Staff will be responsible for the acquisition of relevant permits and licenses from pertinent government regulatory agencies.

This project is expected to lay the foundation for a long-term collaboration between the Gerwick Lab at Scripps, the Seaweed Chemistry Lab at MSI, and the partner HEIs to ensure a strong network of national and international collaborations. It is anticipated that this project will pave the way for a continual exchange of scholars between the Institutions and result in collective efforts to tackle research questions of mutual interest not just in seaweeds and their associated epiphytes but also other marine organisms.

E. Training Plan

This project is expected to contribute to the development of a critical mass of scientists, researchers, and scientifically-trained seaweed farmers.

The PH-based research fellows that will be hired for this project (total of 4 fellows) are envisioned to finish their MS degrees (at least 2 or 3) or be mid-way in their PhD dissertations (at least 1 or 2) when the project ends after two years. These fellows are expected to learn and acquire the requisite knowledge and training in seaweed ecology, algal culturing, biochemistry, biotechnological methodologies, environmental analytical chemistry, and biochemical studies. Their experience during the duration of the project is anticipated to improve their critical thinking and knowledge in carrying out rigorous experiments to tackle important scientific questions.

To ensure that the fellows will receive adequate training, the following requirements or activities will be imposed or carried out:

(a) the fellows will be required to enroll in a graduate program offered by UPMSI or any institution offering relevant degrees;
(b) *weekly laboratory meetings* will be conducted to ensure that the progress of fellows in terms of research tasks are monitored, that the senior project staff members may provide scientific guidance whenever needed, and that hurdles or problems are addressed as soon as recognized;
(c) *attendance to relevant meetings* will be encouraged and supported so that the fellows will have the opportunity to present their work and receive feedback from other researchers working in similar fields; and
(d) *publication of results in reputable journals* authored by the fellows will be strongly encouraged so that they will experience and learn from the process as well as develop their scientific credentials.

Workshops and training for seaweed farmers will be conducted in close collaboration with partner HEIs including the University of Eastern Philippines, Partido State University, and Sorsogon State University. A total of three (3) workshop and training sessions that will be conducted together with each partner HEI will ensure that seaweed farmers will be educated on proper harvesting and preservation procedures to maximize their financial earning capacity. For each session, it is projected that at least 50 participants from the local community and at least 20 representatives from the HEI faculty, staff, and student populations will be trained. When needed, the help of community development professionals will be sought to facilitate dialogue between scientists and seaweed farmers.

At Scripps/UCSD where it is anticipated that the workforce will mainly consist of postdoctoral fellows, meeting research objectives will be partnered with educational and training objectives. A postdoctoral fellow will be expected to participate in the various relevant seminar series at Scripps/UCSD, such at the Marine Chemical Biology Forum and the once monthly Natural Products Affinity Group. The Gerwick group has two meetings per week; one is a general group meeting communicating important information for the week, reviewing laboratory procedures and policies including those having to do with safety, and a more extensive research update presentation by one of the group members. The second meeting is a quick research update wherein each lab member updates the weeks progress, identifies problems

encountered, and discussion ensues to help provide solutions. Additionally, postdoctoral fellows are required to fill out NIH-style Career Development Plans; these are discussed with the postdoctoral mentor on a twice-yearly basis to ensure that development is occurring in the desired directions.

F. Institutional Research Environment

The UP Marine Science Institute (UPMSI) is one of seven academic institutes of the College of Science, University of the Philippines Diliman. It is tasked to pursue research, teaching, and extension work in marine biology, marine chemistry, physical oceanography, marine geology, and related disciplines. The UPMSI seeks to achieve the following objectives: (a) to generate basic information necessary for optimal and sustained utilization, management, and conservation of the marine environment and its resources; (b) to provide graduate-level training and extension services to develop human resource requirements in the marine sciences; and (c) to develop appropriate and environmentally-sound marine-based technologies for industrial and economic development.

The specific objectives of this project are aligned with the direction and mandate of UPMSI. Thus, logistics and infrastructure support, which includes space to house the laboratory and all equipment, necessary for the successful execution of this project will be provided by UPMSI. The equipment procured will become part of the ICPMS facility or the Seaweed Chemistry Laboratory. The ICPMS facility will house all the sophisticated equipment including ICPMS, HPLC-MS, and GC-MS, while the Seaweed Chemistry Lab will house most of the common use equipment. The core facility may be utilized for monitoring the quality of raw and processed seaweeds as a result of the proposed project, which ensures that analytical services will be available to seaweed producers for quality control and assurance of seaweed products. The institution of an ICPMS facility will pave the way for new research directions focused on trace metal analysis and speciation of metal-containing proteins and their role in aquatic ecosystems and algal biotechnology, among many scientific questions. The Seaweed Chemistry Lab will be critical for continued research into other uses and application of seaweeds and fundamental biological and ecological questions pertaining to seaweeds and associated epiphytes. Both of these laboratories will bring complementary expertise to other existing research laboratories in UP-MSI. Moreover, it is envisioned that the core facility for sophisticated and common use equipment will service the instrumentation requirements of other projects within UPMSI and other research laboratories within and outside the UP System and other researchers in the country through a collaborative agreement either as a research agreement or a MOA between institutions.

II. Project Results

The project will be implemented following a three-pronged approach to generate knowledge necessary for future biotechnological innovations that will address some of the health challenges in the Philippines while also providing an avenue to improve socio-economic conditions in key areas in the country. The contributions arising from this project in different aspects of development are detailed as follows.

A. Handbook of phytochemical properties of seaweeds

The major output expected from this project is the publication of a handbook of seaweeds outlining their phytochemical properties and locations where they are found. To maximize the utilization of seaweed as a source of food, food additive, and nutraceuticals, there is a need for systematic and comprehensive assessment of phytochemical properties of seaweeds available in the country. The wealth of information will pave the way for utilization of seaweeds for various applications. There is also a need to understand the influence of the symbiotic epiphytes to the properties of the seaweed raw materials and processed products. On the use of locally available seaweeds for screening of bioactive compounds and drug development, this remains unexplored and may potentially contribute novel compounds that may target different health problems. **The publication of a handbook of seaweeds outlining its phytochemical properties will serve as the starting point of future efforts toward product development for various seaweed applications including food and food additives and high-value compounds such as pigments, nutraceuticals, and pharmaceuticals.**

B. Processes

The processes that will be optimized and validated during the implementation of this project comprise different areas as indicated below:

1. Physico-chemical characterization and analysis, which includes at least 11 different analytical procedures for determination of amino acid, lipid, fatty acid content, and others.
2. Seaweed extraction and subsequent characterization and bioassay of extracts for drug development.
3. Preservation techniques for seaweeds and seaweed extracts.

These processes will be optimized to enable phytochemical analyses of the different seaweed samples. After the completion of the project, these processes will constitute the protocols that will be utilized for routine monitoring of raw materials and processed seaweed samples. In addition, the procedures may also be utilized for other researches targeting similar analytes or scientific questions.

C. Scientific outputs

The scientific outputs expected from this project include at least 4 publications in high impact peer-reviewed journals. In addition, the results are also envisaged to be presented in two (2) international and two (2) national conferences and shared with the Seaweed Industry Association of the Philippines (SIAP) to aid in the association's policy development. In each of the international conferences, at least two (2) members of the research staff are expected to participate. In each of the national conferences, at least four (4) members of the research staff are expected to participate.

D. Professional development

The project will support the career development of all research staff, as well as train seaweed farmers, towards contributing to relieve the deficit in the number of scientists and farmers dedicated on improving the seaweed industry status in the country.

E. Socio-economic impacts

One of the goals of this project is to identify the farming areas where a specific seaweed species with the optimum biochemical properties grows best. The seaweed farmers in the area will be trained on proper and sustainable farming practices to improve the quality of seaweed harvests and products resulting in a better income potential.

The activities that will be undertaken to deliver these project outputs are outlined in the Work Plan Matrix.

III. Attachments

Curriculum vitae of Drs. M. Roleda, W. Gerwick, I. Rodriguez, and L. Gerwick.

Attachments are in the file "8_IHITM_Roleda_Gerwick_Attachments_PCARI2018.pdf

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PCARI Project Form No. 3c: Line-Item Budget Summary

Project Title: **Phytochemical characterization of macroalgae for food and high value products (PhycoPRO)**

Philippine-California Advanced Research Institutes

PROPOSED BUDGET - SUMMARY

BUDGET ITEM		PH			US						TOTAL BUDGET REQUEST		
		YR 1	YR 2	Sub-Total	YR 1		YR 2		Sub-Total		YR 1	YR 2	
		in Php	in Php	in Php	in USD	in Php*	in USD	in Php*	in USD	in Php*	in Php	in Php	Total for 2 Years
I. PERSONNEL SERVICES													
1	**Personnel Salary: MSI**	4,774,152.00	4,774,152.00	9,548,304.00	154,423.20	8,493,276.27	163,543.42	8,994,888.31	317,966.63	17,488,164.58	13,267,428.27	13,769,040.31	**27,036,468.58**
	PS: MSU Naawan	795,120.00	795,120.00	1,590,240.00							795,120.00	795,120.00	**1,590,240.00**
	PS: Sorsogon State University	795,120.00	795,120.00	1,590,240.00							795,120.00	795,120.00	**1,590,240.00**
	PS: Univ Eastern Philippines	795,120.00	795,120.00	1,590,240.00							795,120.00	795,120.00	**1,590,240.00**
	PS: Partido State University	795,120.00	795,120.00	1,590,240.00							795,120.00	795,120.00	**1,590,240.00**
2	**Honoraria: MSI**	285,600.00	285,600.00	571,200.00	-	-	-	-	-	-	285,600.00	285,600.00	**571,200.00**
	Honoraria: MSU Naawan	90,000.00	90,000.00	180,000.00							90,000.00	90,000.00	**180,000.00**
	Honoraria: Sorsogon State University	90,000.00	90,000.00	180,000.00							90,000.00	90,000.00	**180,000.00**
	Honoraria: Univ Eastern Philippines	90,000.00	90,000.00	180,000.00							90,000.00	90,000.00	**180,000.00**
	Honoraria: Partido State University	90,000.00	90,000.00	180,000.00							90,000.00	90,000.00	**180,000.00**
	Sub-Total for PS	**8,600,232.00**	**8,600,232.00**	**17,200,464.00**	154,423.20	**8,493,276.27**	163,543.42	**8,994,888.31**	317,966.63	**17,488,164.58**	**17,093,508.27**	**17,595,120.31**	**34,688,628.58**
											-	-	-
II. MOOE											-	-	-
1	Travel Expenses	3,400,000.00	2,653,000.00	6,053,000.00	-	-	-	-	-	-	3,400,000.00	2,653,000.00	**6,053,000.00**
2	Supplies and Materials	1,400,000.00	1,100,000.00	2,500,000.00	37,700.00	2,073,500.00	33,044.00	1,817,420.00	70,744.00	3,890,920.00	3,473,500.00	2,917,420.00	**6,390,920.00**
3	Communication Expenses	20,000.00	20,000.00	40,000.00	2,673.00	147,015.00	2,870.00	157,850.00	5,543.00	304,865.00	167,015.00	177,850.00	**344,865.00**
4	Repairs and Maintenance of Facilities	520,000.00	30,000.00	550,000.00	-	-	-	-	-	-	520,000.00	30,000.00	**550,000.00**
5	Transportation and Delivery Expenses	150,000.00	150,000.00	300,000.00		-		-	-	-	150,000.00	150,000.00	**300,000.00**
6	Utilities Expenses	-	-	-		-		-	-	-	-	-	-
7	Training Expenses	14,000,000.00	365,000.00	14,365,000.00		-		-	-	-	14,000,000.00	365,000.00	**14,365,000.00**
8	Membership Dues for the Institution	-	-	-		-		-	-	-	-	-	-
9	Advertising Expenses	-	-	-		-		-	-	-	-	-	-
10	Rent Expenses	50,000.00	200,000.00	250,000.00		-		-	-	-	50,000.00	200,000.00	**250,000.00**
11	Representation Expenses	80,083.54	80,000.00	160,083.54		-		-	-	-	80,083.54	80,000.00	**160,083.54**
12	Subscription Expenses	20,000.00	50,000.00	70,000.00		-		-	-	-	20,000.00	50,000.00	**70,000.00**
13	Survey Expenses	-	-	-		-		-	-	-	-	-	-
14	Professional Services	400,000.00	400,000.00	800,000.00		-		-	-	-	400,000.00	400,000.00	**800,000.00**
15	Taxes, Insurance Premiums and Other Fees	100,000.00	150,000.00	250,000.00		-		-	-	-	100,000.00	150,000.00	**250,000.00**
16	Other Maintenance and Operating Expenses	310,000.00	460,000.00	770,000.00	8,869.52	487,823.72	13,000.00	715,000.00	21,869.52	1,202,823.72	797,823.72	1,175,000.00	**1,972,823.72**
				-							-	-	-
	MOOE: MSU Naawan	450,000.00	450,000.00	900,000.00							450,000.00	450,000.00	**900,000.00**
	MOOE: Sorsogon State University	1,000,000.00	500,000.00	1,500,000.00							1,000,000.00	500,000.00	**1,500,000.00**
	MOOE: Univ Eastern Philippines	800,000.00	300,000.00	1,100,000.00							800,000.00	300,000.00	**1,100,000.00**

	MOOE: Partido State University	400,000.00	400,000.00	800,000.00							400,000.00	400,000.00	**800,000.00**
	Sub-Total for MOOE	**23,100,083.54**	**7,308,000.00**	**30,408,083.54**	49,242.52	**2,708,338.72**	48,914.00	**2,690,270.00**	98,156.52	**5,398,608.72**	**25,808,422.26**	**9,998,270.00**	**35,806,692.26**
											-	-	-
III. Indirect Cost											-	-	-
	Indirect for PS	1,720,046.40	1,720,046.40	3,440,092.80	30,884.30	1,698,636.50	32,708.00	1,798,940.00	63,592.30	3,497,576.50	3,418,682.90	3,518,986.40	**6,937,669.30**
	Indirect for MOOE	4,620,016.71	1,461,600.00	6,081,616.71	9,848.50	541,667.72	9,782.00	538,010.00	19,630.50	1,079,677.72	5,161,684.43	1,999,610.00	**7,161,294.43**
	Sub-Total Indirect Cost	**6,340,063.11**	**3,181,646.40**	**9,521,709.51**	40,732.80	**2,240,304.22**	42,490.00	**2,336,950.00**	83,222.80	**4,577,254.22**	**8,580,367.33**	**5,518,596.40**	**14,098,963.73**
											-	-	-
IV. Equipment Outlay													
	UP-MSI	**66,750,000.00**	-	**66,750,000.00**							**66,750,000.00**	-	**66,750,000.00**
	MSU Naawan	**8,450,000.00**	-	**8,450,000.00**							**8,450,000.00**		**8,450,000.00**
	Sorsogon State Univ	**8,750,000.00**	-	**8,750,000.00**							**8,750,000.00**		**8,750,000.00**
	Eniv Eastern Philippines	**5,540,000.00**	-	**5,540,000.00**							**5,540,000.00**		**5,540,000.00**
	Partido State University	**8,970,000.00**		**8,970,000.00**							**8,970,000.00**	-	**8,970,000.00**
	Sub-Total for Equipment Outlay	**98,460,000.00**	-	**98,460,000.00**							**98,460,000.00**	-	**98,460,000.00**
TOTAL PROPOSED BUDGET		**136,500,378.65**	**19,089,878.40**	**155,590,257.05**	244,398.53	**13,441,919.19**	254,947.42	**14,022,108.31**	499,345.95	**27,464,027.52**	**149,942,297.84**	**33,111,986.71**	**183,054,284.54**

Note:

*Foreign Exchange Rate (USD to PHP)

PCARI Project Form No. 3c.1: PH Line-Item Budget

Philippine-California Advanced Research Institutes

PROPOSED BUDGET - BREAKDOWN FOR PH COMPONENT

Project Title:	**Phytochemical characterization of macroalgae for food and high value products (PhycoPRO)**
Implementing Agency/HEI:	**Marine Science Institute, University of the Philippines Diliman**
Proponents:	**Michael Y. Roleda, William H. Gerwick, Irene B. Rodriguez, Lena S. Gerwick**

I. PERSONNEL SERVICES (please see "Research Fellow Positions" sheet): Direct Cost

	Personnel Requirement		Year 1		Year 2		Total for 2 Years
	Role in Project	Name	Monthly Salary	Annual Salary	Monthly Salary	Annual Salary	
A. Personnel Salary							
1	Six (6) SRS II - SG 16		45,780.00	3,296,160.00	45,780.00	3,296,160.00	6,592,320.00
3	Two (2) URA II - SG 14		38,785.00	930,840.00	38,785.00	930,840.00	1,861,680.00
5	Two (2) Project Assistant 1 -SG 8		22,798.00	547,152.00	22,798.00	547,152.00	1,094,304.00
	Sub-Total for Salaries			*4,774,152.00*		4,774,152.00	9,548,304.00
B. HONORARIA							
1	Project Leader - MYRoleda		8,800.00	105,600.00	8,800.00	105,600.00	211,200.00
2	Co-Project Leader - IBRodriguez		7,500.00	90,000.00	7,500.00	90,000.00	180,000.00
3	Project Staff - LSReyes		7,500.00	90,000.00	7,500.00	90,000.00	180,000.00
	Sub-Total for Honoraria			*285,600.00*		*285,600.00*	571,200.00
	Sub-Total PS Direct Cost			*5,059,752.00*		*5,059,752.00*	*10,119,504.00*

II. MAINTENANCE, OPERATING AND OTHER EXPENSES (MOOE): Direct Cost (please see Note)

	MOOE Requirement	Year 1		Year 2		Total for 2 Years
1	**Travel Expenses**		3,400,000.00		2,653,000.00	6,053,000.00
	Sampling	1,755,000.00		1,298,000.00		
	Travel expenses for establishing network with local government units	785,000.00		640,000.00		
	Travel expenses for workshops	785,000.00		640,000.00		
	Travel insurance for staff	75,000.00		75,000.00		
2	**Supplies and Materials**		1,400,000.00		1,100,000.00	2,500,000.00
	Extraction solvents	100,000.00		200,000.00		
	HPLC grade solvents	150,000.00		150,000.00		
	Certified reference materials	250,000.00		100,000.00		
	ICPMS standards (single element)	250,000.00		50,000.00		
	High purity acids and high purity gases	150,000.00		200,000.00		
	General laboratory chemicals	150,000.00		150,000.00		
	Chemicals for preparation of artificial seawater and other required reagents	100,000.00		150,000.00		
	Laboratory glassware/plastic ware and other materials	250,000.00		100,000.00		
3	**Communication Expenses**		20,000.00		20,000.00	40,000.00
	Communication expenses during sampling	20,000.00		20,000.00		
4	**Repairs and Maintenance of Facilities**		520,000.00		30,000.00	550,000.00
	Maintenance of clean room	20,000.00		30,000.00		
	Renovation cost	500,000.00				
5	**Transportation and Delivery Expenses**		150,000.00		150,000.00	300,000.00
	Freight costs	150,000.00		150,000.00		
6	**Utilities Expenses**		-		-	-
7	**Training Expenses**		14,000,000.00		365,000.00	14,365,000.00
	Attendance to local conferences					
	Workshop in collaboration with Sorsogon State University			95,000.00		
	Workshop in collaboration with MSU Naawan			90,000.00		
	Workshop in collaboration with UEP			90,000.00		
	Workshop in collaboration with ParSU			90,000.00		
	Seaweed Farmers Training and workshops	14,000,000.00				
	Seaweed farmer training (5 barangays per province x 7 provinces) @950k per province	6,650,000.00				
	Multistakeholder provincial training (1 per province x 7 provinces) @300k per province	2,100,000.00				
	Model nursery development (1 per barangay = 5 per province x 7 provinces) @ 750k per provinc	5,250,000.00				
8	**Membership Dues for the Institution**		-		-	-
9	**Advertising Expenses**		-		-	-
10	**Rent Expenses**		50,000.00		200,000.00	250,000.00
	Boats for sampling	50,000.00		200,000.00		
11	**Representation Expenses (e.g. food for meetings, etc.)**		80,083.54		80,000.00	160,083.54

	Annual meetings (food expenses)	15,000.00		40,000.00		
	Kickoff meeting (food expenses)	15,000.00				
	Project evaluation meeting (food expenses)	50,083.54		40,000.00		
12	**Subscription Expenses**		20,000.00		50,000.00	70,000.00
	Purchase of access to relevant scientific articles	20,000.00		50,000.00		
13	**Survey Expenses**		-		-	-
14	**Professional Services**		400,000.00		400,000.00	800,000.00
	Bookkeeping	366,324.00		366,324.00		
	Daily Labor Support	33,676		33,676		
15	**Taxes, Insurance Premiums and Other Fees**		100,000.00		150,000.00	250,000.00
	Environmental fees for sampling sites	50,000.00		50,000.00		
	Chemical waste treatment	50,000.00		100,000.00		
16	**Other Maintenance and Operating Expenses (Please Itemize)**		310,000.00		460,000.00	770,000.00
	Instrument fee for inter-calibration purposes	60,000.00		60,000.00		
	Publication expenses	250,000.00		400,000.00		
	Sub-Total MOOE Direct Cost		***20,450,083.54***		***5,658,000.00***	***26,108,083.54***

III. INDIRECT COST (20%)

	Budget Item	Year 1		Year 2		Total for 2 Years
		Direct Cost	Indirect Cost	Direct Cost	Indirect Cost	
1	Personnel Services	5,059,752.00	1,011,950.40	5,059,752.00	1,011,950.40	2,023,900.80
2	MOOE	20,450,083.54	4,090,016.71	5,658,000.00	1,131,600.00	5,221,616.71
	Sub-Total Indirect Cost	***25,509,835.54***	***5,101,967.11***	***10,717,752.00***	***2,143,550.40***	***7,245,517.51***

IV. EQUIPMENT OUTLAY

	Equipment	Year 1			Year 2			Total for 2 Years
		Cost/ Unit	No. of Units	Sub-Total Cost	Cost/ Unit	No. of Units	Sub-Total Cost	
1	graphite furnace atomic absoprtion spectrophotometer with complete parts and	4,500,000.00	1	4,500,000.00			-	4,500,000.00
2	CHNS/O elemental analyzer with complete parts and accessories	4,500,000.00	1	4,500,000.00				4,500,000.00
3	microwave digestion system with complete parts and accessories	4,000,000.00	1	4,000,000.00				4,000,000.00
4	fumehoods with complete parts and accessories	1,000,000.00	1	1,000,000.00				1,000,000.00
5	GC-MS and accessories	11,000,000.00	1	11,000,000.00			-	11,000,000.00
6	HPLC with DAD detector and accessories	3,500,000.00	1	3,500,000.00			-	3,500,000.00
7	HPLC-MS and accessories	20,000,000.00	1	20,000,000.00			-	20,000,000.00
8	lyophilizer and accessories	1,200,000.00	1	1,200,000.00			-	1,200,000.00
9	ultralow freezer and accessories	1,300,000.00	2	2,600,000.00			-	2,600,000.00
10	UV-Vis spectrophotometer and accessories	700,000.00	1	700,000.00			-	700,000.00
11	Multi-channel peristaltic pump and accessories	300,000.00	2	600,000.00			-	600,000.00
12	rotary evaporator and accessories	500,000.00	2	1,000,000.00			-	1,000,000.00
13	Oven drier and accessories	200,000.00	1	200,000.00			-	200,000.00
14	analytical balance and accessories	100,000.00	2	200,000.00			-	200,000.00
15	toploading balance and accessories	50,000.00	2	100,000.00			-	100,000.00
16	High resolution camera and accessories for seaweed documentation	100,000.00	1	100,000.00			-	100,000.00
17	Temperature and humidity data logger and accessories	70,000.00	4	280,000.00			-	280,000.00
18	hot plate/magnetic stirrer and accessories	50,000.00	4	200,000.00			-	200,000.00
19	Desktops/laptops and accessories	50,000.00	6	300,000.00			-	300,000.00
20	furnace and accessories	450,000.00	1	450,000.00			-	450,000.00
21	sonicator and accessories	500,000.00	2	1,000,000.00			-	1,000,000.00
22	vacuum pump and accessories	200,000.00	2	400,000.00			-	400,000.00
23	pH meter, microwave oven, micro-centrifuge, vortex mixer, and accessories	170,000.00	1	170,000.00			-	170,000.00
24	qPCR, gradient PCR, and accessories	3,500,000.00	1	3,500,000.00			-	3,500,000.00
25	refrigerated centrifuge with different rotors and accessories	1,500,000.00	1	1,500,000.00			-	1,500,000.00
26	water bath with temperature control and accessories	475,000.00	2	950,000.00			-	950,000.00
27	multiplate reader and accessories	2,000,000.00	1	2,000,000.00			-	2,000,000.00
28	multiparameter and accessories for water analysis	400,000.00	2	800,000.00			-	800,000.00
	Sub-Total Equipmeny Outlay Cost			***66,750,000.00***	-		-	***66,750,000.00***

V. G	Budget Item	Year 1	Year 2	Total for 2 Years
	Sub-Total for PS Direct Cost	5,059,752.00	5,059,752.00	10,119,504.00
1	Sub-Total for MOOE Direct Cost	20,450,083.54	5,658,000.00	26,108,083.54
2	Sub-Total for Indirect Cost (PS + MOOE Indirect Cost)	5,101,967.11	2,143,550.40	7,245,517.51
3	Sub-Total for for Equipment Outlay	66,750,000.00	-	66,750,000.00
4	***TOTAL BUDGET REQUIREMENT***	***97,361,802.65***	***12,861,302.40***	***110,223,105.05***

PCARI Project Form No. 3c.1: PH Line-Item Budget

Philippine-California Advanced Research Institutes

PROPOSED BUDGET - BREAKDOWN FOR PH COMPONENT

Project Title:	Phytochemical characterization of macroalgae for food and high value products (PhycoPRO)
Implementing Agency/HEI:	Mindanao State University - Naawan
Proponents:	Michael Y. Roleda, William H. Gerwick, Irene B. Rodriguez, Lena Gerwick

I. PERSONNEL SERVICES (please see "Research Fellow Positions" sheet): Direct Cost

	Personnel Requirement		Year 1		Year 2		Total for 2 Years
	Role in Project	Name	Monthly Salary	Annual Salary	Monthly Salary	Annual Salary	
1	URA I - SG 12		33,130.00	397,560.00	33,130.00	397,560.00	795,120.00
2	URA I - SG 12		33,130.00	397,560.00	33,130.00	397,560.00	795,120.00
	Sub-Total for Salaries		66,260.00	*795,120.00*	66,260.00	795,120.00	1,590,240.00
B. HONORARIA							
1	Dr. Rey Capangpangan		7,500.00	90,000.00	7,500.00	90,000.00	180,000.00
	Sub-Total for Honoraria		*7,500.00*	*90,000.00*	*7,500.00*	*90,000.00*	180,000.00
	Sub-Total PS Direct Cost			*885,120.00*		*885,120.00*	*1,770,240.00*

II. MAINTENANCE, OPERATING AND OTHER EXPENSES (MOOE): Direct Cost (please see Note)

	MOOE Requirement	Year 1		Year 2		Total for 2 Years
1	Travel Expenses					-
2	Supplies and Materials		450,000.00		450,000.00	900,000.00
	Extraction solvents	100,000.00		100,000.00		
	HPLC grade solvents	150,000.00		150,000.00		
	General laboratory chemicals	100,000.00		100,000.00		
	Laboratory glassware/plastic ware and other materials	100,000.00		100,000.00		
3	Communication Expenses					-
4	Repairs and Maintenance of Facilities					-
5	Transportation and Delivery Expenses					-
6	Utilities Expenses					-
7	Training Expenses					-
8	Membership Dues for the Institution					-
9	Advertising Expenses					-
10	Rent Expenses					-
11	Representation Expenses (e.g. food for meetings, etc.)					-
12	Subscription Expenses					-
13	Survey Expenses					-
14	Professional Services					-
15	Taxes, Insurance Premiums and Other Fees					-
16	Other Maintenance and Operating Expenses (Please Itemize)		-		-	-
	Sub-Total MOOE Direct Cost		*450,000.00*		*450,000.00*	*900,000.00*

III. INDIRECT COST (20%)

	Budget Item	Year 1		Year 2		Total for 2 Years
		Direct Cost	Indirect Cost	Direct Cost	Indirect Cost	
1	Personnel Services	885,120.00	177,024.00	885,120.00	177,024.00	354,048.00
2	MOOE	450,000.00	90,000.00	450,000.00	90,000.00	180,000.00
	Sub-Total Indirect Cost	*1,335,120.00*	*267,024.00*	*1,335,120.00*	*267,024.00*	*534,048.00*

IV. EQUIPMENT OUTLAY

	Equipment	Year 1			Year 2			Total for 2 Years
		Cost/ Unit	No. of Units	Sub-Total Cost	Cost/ Unit	No. of Units	Sub-Total Cost	
1	FT-IR and accessories	2,500,000.00	1	2,500,000.00			-	2,500,000.00
2	HPLC with UV/RI detector and accessories	3,500,000.00	1	3,500,000.00			-	3,500,000.00
3	UV-Vis spectrophotometer and accessories	700,000.00	1	700,000.00			-	700,000.00
4	ultralow freezer and accessories	1,000,000.00	1	1,000,000.00			-	1,000,000.00
5	furnace and accessories	450,000.00	1	450,000.00			-	450,000.00
6	Rotary evaporator with accessories	300,000.00	1	300,000.00				300,000.00
21								
22				-			-	-
	Sub-Total Equipmeny Outlay Cost			*8,450,000.00*	-		-	*8,450,000.00*

V. GRAND TOTAL

	Budget Item	Year 1	Year 2	Total for 2 Years
1	Sub-Total for PS Direct Cost	885,120.00	885,120.00	1,770,240.00
2	Sub-Total for MOOE Direct Cost	450,000.00	450,000.00	900,000.00
3	Sub-Total for Indirect Cost (PS + MOOE Indirect Cost)	267,024.00	267,024.00	534,048.00
4	Sub-Total for for Equipment Outlay	8,450,000.00	-	8,450,000.00
	TOTAL BUDGET REQUIREMENT	***10,052,144.00***	***1,602,144.00***	***11,654,288.00***

Philippine-California Advanced Research Institutes

PROPOSED BUDGET - BREAKDOWN FOR PH COMPONENT

Project Title:	Phytochemical characterization of macroalgae for food and high value products (PhycoPRO)
Implementing Agency/HEI:	Partido State University
Proponents:	Michael Y. Roleda, William H. Gerwick, Irene B. Rodriguez, Lena Gerwick

I. PERSONNEL SERVICES (please see "Research Fellow Positions" sheet): Direct Cost

	Personnel Requirement		Year 1		Year 2		Total for 2 Years
	Role in Project	Name	Monthly Salary	Annual Salary	Monthly Salary	Annual Salary	
1	URA I - SG 12		33,130.00	397,560.00	33,130.00	397,560.00	795,120.00
2	URA I - SG 12		33,130.00	397,560.00	33,130.00	397,560.00	795,120.00
	Sub-Total for Salaries		66,260.00	*795,120.00*	66,260.00	795,120.00	1,590,240.00
B. HONORARIA							
1	Dr. Michael Clores		7,500.00	90,000.00	7,500.00	90,000.00	180,000.00
	Sub-Total for Honoraria		*7,500.00*	*90,000.00*	*7,500.00*	*90,000.00*	180,000.00
	Sub-Total PS Direct Cost			*885,120.00*		*885,120.00*	*1,770,240.00*

II. MAINTENANCE, OPERATING AND OTHER EXPENSES (MOOE): Direct Cost (please see Note)

	MOOE Requirement	Year 1		Year 2		Total for 2 Years
1	Travel Expenses					-
2	Supplies and Materials		400,000.00		400,000.00	800,000.00
	Field and sample preservation materials	200,000.00		200,000.00		
	General laboratory chemicals	100,000.00		100,000.00		
	Laboratory glassware/plastic ware and other materials	100,000.00		100,000.00		
3	Communication Expenses					-
4	Repairs and Maintenance of Facilities					-
5	Transportation and Delivery Expenses					-
6	Utilities Expenses					-
7	Training Expenses					-
8	Membership Dues for the Institution					-
9	Advertising Expenses					-
10	Rent Expenses					-
11	Representation Expenses (e.g. food for meetings, etc.)					-
12	Subscription Expenses					-
13	Survey Expenses					-
14	Professional Services					-
15	Taxes, Insurance Premiums and Other Fees					-
16	Other Maintenance and Operating Expenses (Please Itemize)		-		-	-
	Sub-Total MOOE Direct Cost		*400,000.00*		*400,000.00*	*800,000.00*

III. INDIRECT COST (20%)

	Budget Item	Year 1		Year 2		Total for 2 Years
		Direct Cost	Indirect Cost	Direct Cost	Indirect Cost	
1	Personnel Services	885,120.00	177,024.00	885,120.00	177,024.00	354,048.00
2	MOOE	400,000.00	80,000.00	400,000.00	80,000.00	160,000.00
	Sub-Total Indirect Cost	*1,285,120.00*	*257,024.00*	*1,285,120.00*	*257,024.00*	*514,048.00*

IV. EQUIPMENT OUTLAY

	Equipment	Year 1			Year 2			Total for 2 Years
		Cost/ Unit	No. of	Sub-Total Cost	Cost/ Unit	No. of Units	Sub-Total Cost	
1	Oven drier and accessories	200,000.00	2	400,000.00			-	400,000.00
2	analytical balance and accessories	100,000.00	1	100,000.00			-	100,000.00
3	toploading balance and accessories	50,000.00	2	100,000.00			-	100,000.00
4	furnace and accessories	450,000.00	1	450,000.00				450,000.00
5	filtration system (vacuum pump, manifold, and accessories)	200,000.00	1	200,000.00				200,000.00
6	UV-Vis spectrophotometer and accessories	700,000.00	1	700,000.00				700,000.00
7	Spectrofluorometer and accessories	1,000,000.00	1	1,000,000.00				1,000,000.00
8	pump, motor with accessories	350,000.00	1	350,000.00				350,000.00
9	data loggers and accessories	70,000.00	3	210,000.00				210,000.00
10	scuba diving sets and accessories	80,000.00	2	160,000.00				160,000.00
11	Compound microscope and accessories	300,000.00	2	600,000.00				600,000.00
12	inverted microscope and accessories	800,000.00	1	800,000.00				800,000.00
13	Stereomicroscope and accessories	300,000.00	1	300,000.00				300,000.00
14	Mechanized rotating drum mixer and accessories	200,000.00	1	200,000.00				200,000.00
15	Gopro and underwater camera and accessories	100,000.00	2	200,000.00				200,000.00
16	tabletop centrifuge and accessories	150,000.00	1	150,000.00				150,000.00
17	pH meter, vortex mixer, hotplate, and accessories	150,000.00	1	150,000.00				150,000.00
18	Refractometer and accessories	100,000.00	1	100,000.00				100,000.00
19	Dessicator and accessories	250,000.00	1	250,000.00				250,000.00
20	electronic image analysis system and accessories	150,000.00	1	150,000.00				150,000.00
21	shaker and accessories	500,000.00	1	500,000.00				500,000.00
22	multiparameter and accessories for water sampling	400,000.00	1	400,000.00			-	400,000.00
23	refrigerated centrifuge with different rotors and accessories	1,500,000.00	1	1,500,000.00				1,500,000.00

	Sub-Total Equipmeny Outlay Cost			*8,970,000.00*	-		-	*8,970,000.00*

V. GRAND TOTAL					-
	Budget Item	**Year 1**	**Year 2**	**Total for 2 Years**	
1	Sub-Total for PS Direct Cost	885,120.00	885,120.00	**1,770,240.00**	
2	Sub-Total for MOOE Direct Cost	400,000.00	400,000.00	**800,000.00**	
3	Sub-Total for Indirect Cost (PS + MOOE Indirect Cost)	257,024.00	257,024.00	**514,048.00**	
4	Sub-Total for for Equipment Outlay	8,970,000.00	-	**8,970,000.00**	
	TOTAL BUDGET REQUIREMENT	***10,512,144.00***	***1,542,144.00***	***12,054,288.00***	

PCARI Project Form No. 3c.1: PH Line-Item Budget

Philippine-California Advanced Research Institutes

PROPOSED BUDGET - BREAKDOWN FOR PH COMPONENT

Project Title:	Phytochemical characterization of macroalgae for food and high value products (PhycoPRO)
Implementing Agency/HEI:	Sorsogon State University
Proponents:	Michael Y. Roleda, William H. Gerwick, Irene B. Rodriguez, Lena Gerwick

I. PERSONNEL SERVICES (please see "Research Fellow Positions" sheet): Direct Cost

	Personnel Requirement		Year 1		Year 2		Total for 2 Years
	Role in Project	Name	Monthly Salary	Annual Salary	Monthly Salary	Annual Salary	
1	URA I - SG 12		33,130.00	397,560.00	33,130.00	397,560.00	795,120.00
2	URA I - SG 12		33,130.00	397,560.00	33,130.00	397,560.00	795,120.00
	Sub-Total for Salaries		66,260.00	*795,120.00*	66,260.00	795,120.00	1,590,240.00
B. HONORARIA							
1	Dr. Richrad Dumilag		7,500.00	90,000.00	7,500.00	90,000.00	180,000.00
	Sub-Total for Honoraria		*7,500.00*	*90,000.00*	*7,500.00*	*90,000.00*	180,000.00
	Sub-Total PS Direct Cost			*885,120.00*		*885,120.00*	*1,770,240.00*

II. MAINTENANCE, OPERATING AND OTHER EXPENSES (MOOE): Direct Cost (please see Note)

	MOOE Requirement	Year 1		Year 2		Total for 2 Years
1	Travel Expenses					-
2	Supplies and Materials		500,000.00		500,000.00	1,000,000.00
	RNA extraction kits and other chemicals	300,000.00		300,000.00		
	General laboratory chemicals	100,000.00		100,000.00		
	Laboratory glassware/plastic ware and other materials	100,000.00		100,000.00		
3	Communication Expenses					-
4	Repairs and Maintenance of Facilities					-
5	Transportation and Delivery Expenses					-
6	Utilities Expenses					-
7	Seaweed Farmers Training/Workshop Expenses	500,000.00	500,000.00			500,000.00
	Seaweed farmer training (5 barangays)					
8	Membership Dues for the Institution					-
9	Advertising Expenses					-
10	Rent Expenses					-
11	Representation Expenses (e.g. food for meetings, etc.)					-
12	Subscription Expenses					-
13	Survey Expenses					-
14	Professional Services					-
15	Taxes, Insurance Premiums and Other Fees					-
16	Other Maintenance and Operating Expenses (Please Itemize)		-		-	-
	Sub-Total MOOE Direct Cost		*1,000,000.00*		*500,000.00*	*1,500,000.00*

III. INDIRECT COST (20%)

	Budget Item	Year 1		Year 2		Total for 2 Years
		Direct Cost	Indirect Cost	Direct Cost	Indirect Cost	
1	Personnel Services	885,120.00	177,024.00	885,120.00	177,024.00	354,048.00
2	MOOE	1,000,000.00	200,000.00	500,000.00	100,000.00	300,000.00
	Sub-Total Indirect Cost	*1,885,120.00*	*377,024.00*	*1,385,120.00*	*277,024.00*	*654,048.00*

IV. EQUIPMENT OUTLAY

	Equipment	Year 1			Year 2			Total for 2 Years
		Cost/ Unit	No. of	Sub-Total Cost	Cost/ Unit	No. of	Sub-Total Cost	
1	Inverted microscope and accessories	1,000,000.00	1	1,000,000.00			-	1,000,000.00
2	gradient PCR and accessories	600,000.00	1	600,000.00			-	600,000.00
3	Compound microscope and accessories	300,000.00	1	300,000.00			-	300,000.00
4	Stereomicroscope and accessories	300,000.00	1	300,000.00			-	300,000.00
5	gel documentation system with complete parts and accessories	1,450,000.00	1	1,450,000.00			-	1,450,000.00
6	bead ruptor tissue lyser and accessories	500,000.00	1	500,000.00			-	500,000.00
7	liquid nitrogen containers and accessories	80,000.00	1	80,000.00			-	80,000.00
8	refrigerator, freezer and accessories	100,000.00	2	200,000.00			-	200,000.00
9	multiparameter for water sampling (data loggers)	1,250,000.00	1	1,250,000.00			-	1,250,000.00
10	refrigerated centrifuge with accessories	1,200,000.00	1	1,200,000.00			-	1,200,000.00
11	analytical balance and accessories	100,000.00	1	100,000.00			-	100,000.00
12	toploading balance and accessories	50,000.00	1	50,000.00			-	50,000.00
13	digital dry bath with complete parts and accessories	90,000.00	1	90,000.00			-	90,000.00
14	pipettors with complete parts and accessories	95,000.00	4	380,000.00			-	380,000.00
15	vortex mixer with complete parts and accessories	80,000.00	1	80,000.00			-	80,000.00
16	dead air cabinet with complete parts and accessories	450,000.00	1	450,000.00			-	450,000.00
17	water purification system with complete parts and accessories	500,000.00	1	500,000.00			-	500,000.00
18	mini-centrifuge with complete parts and accessories	60,000.00	1	60,000.00			-	60,000.00
19	DSLR camera and accessories	90,000.00	1	90,000.00			-	90,000.00

20	Gel electrophoresis system with complete parts and accessories	70,000.00	1	70,000.00			-	70,000.00
21				-			-	-
22				-			-	-
	Sub-Total Equipmeny Outlay Cost			***8,750,000.00***	-		-	***8,750,000.00***

V. GRAND TOTAL					-
	Budget Item	**Year 1**	**Year 2**	**Total for 2 Years**	
1	Sub-Total for PS Direct Cost	885,120.00	885,120.00	**1,770,240.00**	
2	Sub-Total for MOOE Direct Cost	1,000,000.00	500,000.00	**1,500,000.00**	
3	Sub-Total for Indirect Cost (PS + MOOE Indirect Cost)	377,024.00	277,024.00	**654,048.00**	
4	Sub-Total for for Equipment Outlay	8,750,000.00	-	**8,750,000.00**	
	TOTAL BUDGET REQUIREMENT	***11,012,144.00***	***1,662,144.00***	***12,674,288.00***	

PCARI Project Form No. 3c.1: PH Line-Item Budget

Philippine-California Advanced Research Institutes

PROPOSED BUDGET - BREAKDOWN FOR PH COMPONENT

Project Title:	Phytochemical characterization of macroalgae for food and high value products (PhycoPRO)
Implementing Agency/HEI:	University of Eastern Philippines
Proponents:	Michael Y. Roleda, William H. Gerwick, Irene B. Rodriguez, Lena Gerwick

I. PERSONNEL SERVICES (please see "Research Fellow Positions" sheet): Direct Cost

	Personnel Requirement		Year 1		Year 2		Total for 2 Years
	Role in Project	Name	Monthly Salary	Annual Salary	Monthly Salary	Annual Salary	
1	URA I - SG 12		33,130.00	397,560.00	33,130.00	397,560.00	795,120.00
2	URA I - SG 12		33,130.00	397,560.00	33,130.00	397,560.00	795,120.00
	Sub-Total for Salaries		66,260.00	*795,120.00*	66,260.00	795,120.00	1,590,240.00
B. HONORARIA							
1	Dr. Ronelie Salvador		7,500.00	90,000.00	7,500.00	90,000.00	180,000.00
	Sub-Total for Honoraria		*7,500.00*	*90,000.00*	*7,500.00*	*90,000.00*	180,000.00
	Sub-Total PS Direct Cost			*885,120.00*		*885,120.00*	*1,770,240.00*

II. MAINTENANCE, OPERATING AND OTHER EXPENSES (MOOE): Direct Cost (please see Note)

	MOOE Requirement	Year 1		Year 2		Total for 2 Years
1	Travel Expenses					-
2	Supplies and Materials		300,000.00		300,000.00	600,000.00
	Field and other materials	100,000.00		100,000.00		
	General laboratory chemicals	100,000.00		100,000.00		
	Laboratory glassware/plastic ware and other materials	100,000.00		100,000.00		
3	Communication Expenses					-
4	Repairs and Maintenance of Facilities					-
5	Transportation and Delivery Expenses					-
6	Utilities Expenses					-
7	Seaweed Farmers Training/Workshop Expenses	500,000.00	500,000.00			500,000.00
	Seaweed farmer training (5 barangays)					
8	Membership Dues for the Institution					-
9	Advertising Expenses					-
10	Rent Expenses					-
11	Representation Expenses (e.g. food for meetings, etc.)					-
12	Subscription Expenses					-
13	Survey Expenses					-
14	Professional Services					-
15	Taxes, Insurance Premiums and Other Fees					-
16	Other Maintenance and Operating Expenses (Please Itemize)		-		-	-
	Sub-Total MOOE Direct Cost		*800,000.00*		*300,000.00*	*1,100,000.00*

III. INDIRECT COST (20%)

	Budget Item	Year 1		Year 2		Total for 2 Years
		Direct Cost	Indirect Cost	Direct Cost	Indirect Cost	
1	Personnel Services	885,120.00	177,024.00	885,120.00	177,024.00	354,048.00
2	MOOE	800,000.00	160,000.00	300,000.00	60,000.00	220,000.00
	Sub-Total Indirect Cost	*1,685,120.00*	*337,024.00*	*1,185,120.00*	*237,024.00*	*574,048.00*

IV. EQUIPMENT OUTLAY

	Equipment	Year 1			Year 2			Total for 2 Years
		Cost/ Unit	No. of	Sub-Total Cost	Cost/ Unit	No. of Units	Sub-Total Cost	
1	Inverted microscope and accessories	800,000.00	1	800,000.00			-	800,000.00
2	stereomicroscope and accessories	300,000.00	1	300,000.00			-	300,000.00
3	Oven and accessories	300,000.00	1	300,000.00			-	300,000.00
4	Furnace and accessories	450,000.00	1	450,000.00				450,000.00
5	analytical balance and accessories	100,000.00	1	100,000.00				100,000.00
6	toploading balance and accessories	50,000.00	1	50,000.00				50,000.00
7	UV-Vis spectrophotometer and accessories	700,000.00	1	700,000.00				700,000.00
8	Water bath with temperature control and accessories	450,000.00	1	450,000.00				450,000.00
9	Centrifuge with different rotors and accessories	1,200,000.00	1	1,200,000.00				1,200,000.00
10	Multiparameter and accessories for water sampling	400,000.00	1	400,000.00				400,000.00
11	pH meter, vortex mixer, hotplate, and accessories	150,000.00	1	150,000.00				150,000.00
12	freezer and refrigerator and accessories for sample storage	100,000.00	1	100,000.00				100,000.00
13	compound microscope and accessories	300,000.00	1	300,000.00				300,000.00
14	Gopro and underwater camera and accessories	100,000.00	1	100,000.00				100,000.00
15	data loggers and accessories	70,000.00	2	140,000.00				140,000.00

21				-			-	-
22				-			-	-
	Sub-Total Equipmeny Outlay Cost			***5,540,000.00***	-		-	***5,540,000.00***

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V. GRAND TOTAL

	Budget Item	Year 1	Year 2	Total for 2 Years
1	Sub-Total for PS Direct Cost	885,120.00	885,120.00	**1,770,240.00**
2	Sub-Total for MOOE Direct Cost	800,000.00	300,000.00	**1,100,000.00**
3	Sub-Total for Indirect Cost (PS + MOOE Indirect Cost)	337,024.00	237,024.00	**574,048.00**
4	Sub-Total for for Equipment Outlay	5,540,000.00	-	**5,540,000.00**
	TOTAL BUDGET REQUIREMENT	***7,562,144.00***	***1,422,144.00***	***8,984,288.00***

PCARI Project Form No. 3c.2: US Line-Item Budget

Philippine-California Advanced Research Institutes

PROPOSED BUDGET - BREAKDOWN FOR US COMPONENT

Project Title:	Phytochemical characterization of macroalgae for food and high value products (PhycoPRO)
Implementing Agency/HEI:	Marine Science Institute, University of the Philippines Diliman
Proponents:	Michael Y. Roleda, William H. Gerwick, Irene B. Rodriguez, Lena Gerwick

I. PERSONNEL SERVICES: Direct Cost

Personnel Requirement			Year 1				Year 2				Total for 2 Years
	Role in Project	Name	Salary (in	%Effort	Fringe Benefit Rate	Total Compensation (in USD)	Salary (in USD)	%Effort	Fringe Benefit Rate	Total Compensation (in USD)	
			(A)	(B)	(C)	(A x B) x (1 + C)	(A)	(B)	(C)	(A x B) x (1 + C)	
1	Principal Investigator	Gerwick, William	327	12.5%	29.80%	53,138	337,332.00	12.5%	30.690%	55,108	108,246
2	Principal Investigator	Gerwick, Lena	136	12.5%	9.6000	18,663	140,316.00	12.5%	9.89%	19,275	37,937
3	Postdoctoral Fellow	Harpreet Singh	55,	100%	19.90%	66,674	60,144.00	100.00%	20.50%	72,474	139,148
4	SRA	Evgenia Glukhov	61,	16.67%	56.30%	15,949	63,372.00	16.67%	57.99%	16,687	32,636
6						-				0.00	-
7						-				0.00	-
8						-				0.00	-
9						-				0.00	-
10						-				0.00	-
	Sub-Total PS Direct Cost					*154,423.20*				*163,543.42*	*317,966.63*

II. MAINTENANCE, OPERATING AND OTHER EXPENSES (MOOE): Direct Cost

	MOOE Requirement	Year 1			Year 2		Total for 2 Years
1	**Supplies and Materials**		37,700.00			33,044.00	70,744.00
	Reagents and materials for sample identification and preliminary testing	15,000.00					
	Chemicals for analysis and evaluation of extracts for activity and other parar	12,200.00					
	Certified reference materials for identification of isolated compounds with a	10,500.00					
	Reagents for sample analyses of various parameters				18,000.00		
	Chemicals and reagents for analysis of seaweed extracts				15,044.00		
2	**Communication Expenses**		2,673.00			2,870.00	5,543.00
	Project Specifc Costs	2,673.00			2,870.00		
11	**Other Maintenance and Operating Expenses (Please Itemize and describe as these are subject to further consideration)**		8,869.52			13,000.00	21,869.52
	E.3 Write research papaes 6.2 Project Evaluation	8,869.52					
	C.1 Write research papers				8000		
	D.1 Project Evalulation				5,000.00		
	Sub-Total MOOE Direct Cost		49,242.52			48,914.00	98,156.52

III. INDIRECT COST (20%)

	Budget Item	Year 1			Year 2		Total for 2 Years
		Direct Cost	Indirect Cost		Direct Cost	Indirect Cost	
1	Personnel Services	154,423.20	30,884.30		163,543.42	32,708.00	63,592.30
2	MOOE	49,242.52	9,848.50		48,914.00	9,782.00	19,630.50
	Sub-Total	*203,665.72*	*40,732.80*		*212,457.42*	*42,490.00*	*83,222.80*

IV. GRAND TOTAL

	Budget Item	Year 1		Year 2	Total for 2 Years
1	Total Direct Cost	203,665.72		212,457.42	416,123.15
2	Total Indirect Cost	40,732.80		42,490.00	83,222.80
	TOTAL BUDGET REQUIREMENT	*244,398.53*		*254,947.42*	*499,345.95*

Budget and Justification

Project Title: Phytochemical characterization of macroalgae for food and high value products (PhycoPRO)

LINE ITEM BUDGET FOR UP-MSI
TOTAL PROPOSED BUDGET FOR 2 YEARS: 110,223,004.80

A. Personnel: List employee(s) (including names for each identified position) of the applicant/recipient organization, including in-kind costs for those positions whose work is tied to the grant project.+

Personnel for PH Component: UP-MSI

Position	Name	Annual Salary/Rate	Level of Effort	Cost for YR 1	Cost for YR 2
(1) Project Leader	Michael Y. Roleda	105,600.00	30%	105,600.00	105,600.00
(2) Co-Project Leader	Irene B. Rodriguez	90,000.00	30%	90,000.00	90,000.00
(3) Project Staff	Lilibeth Salvador-Reyes	90,000.00	10%	90,000.00	90,000.00
(4-9) SRS II	To be selected	3,926,160.00	100%	3,926,160.00	3,926,160.00
(10-11) URA II	To be selected	930,840.00	100%	930,840.00	930,840.00
(12-13) Project Assistant 1	To be selected	547,152.00	100%	547,152.00	547,152.00
		TOTAL	**PhP**	**5,059,752.00**	**5,059,752.00**

JUSTIFICATION: Describe the role and responsibilities of each position.

(1) and (2): The Project Leader and Co-Project Leader will provide general research directions, oversight of the project, supervision of research staff, and will be considered key staff. The Project Leaders will be paid honoraria.

(3) : The Project Staff will provide guidance and suggestions for successful implementation of the grant and will be considered key staff. The Project Staff will be paid honoraria.

(4-9) : Six (6) Science Research Specialist II (SRS II) are MS graduates and will carry out majority of the experiments and will be considered key staff. Each SRS II will be in-charge of specific task or research target including method optimization, seaweed culturing, bio-assays, chemical extraction, and other research tasks.

(10-11): Two University Research Associate II (URA II) are MS students and will carry out majority of the experiments and will be considered key staff. Each URA II will be in-charge of specific task or research target including method optimization, seaweed culturing, bio-assays, chemical extraction, and other research tasks.

(12-13): Two Project Assistants will assist in document processing including licenses, bids and awards, assist in sampling activities, hatchery cultures, preparation of seaweed voucher specimens, and laboratory activities.

C. Travel (for Philippines side only): UP-MSI

Travel expenses for PH counterpart

Item(s)	Cost for YR 1	Cost for YR 2
Travel Expenses		
Sampling	1,755,000.00	1,298,000.00
Travel expenses for establishing network with local government units	785,000.00	640,000.00
Travel expenses for workshops	785,000.00	640,000.00
Travel insurance for staff	75,000.00	75,000.00
TOTAL	**3,400,000.00**	**2,653,000.00**

Sampling costs: The sampling cost will cover airfare and per diem costs. The sampling will involve number of participants and number of days that will be unique to the requirements of the study site.

	YEAR 1	YEAR 2
Caramoan, Camarines Sur:	199,000.00	
Dinagat, Surigao Del Norte:		189,000.00
Tandubas, Tawi Tawi:		254,000.00
Batan, Batanes:	464,000.00	
Bolinao, Pangasinan:	100,000.00	100,000.00
Currimao, Ilocos Norte:	100,000.00	100,000.00
Guiuan, Eastern Samar:	169,000.00	
Biri, Northern Samar:		219,000.00
Iba, Zambales:		112,000.00
Busuanga, Palawan:	212,000.00	
Sorsogon, Sorsogon:		162,000.00
Legazpi City, Albay:	169,000.00	
Bien Unido, Bohol:	180,000.00	
Lian, Batangas:	162,000.00	162,000.00
TOTAL	1,755,000.00	1,298,000.00

Travel Expenses for Sampling:

To Caramoan, Camarines Sur: 199,000.00 [airfare and per diem during travel = 90,000; per diem during sampling = 1500/day; 1,500 x 10 people x 7 days = 105,000.00; incidentals = 4,000.00]

To Dinagat, Surigao Del Norte: 189,000.00 [airfare and per diem during travel = 80,000; per diem during sampling = 1500/day; 1,500 x 10 people x 7 days = 105,000.00; incidentals = 4,000.00]

To Tandubas, Tawi Tawi: 254,000.00 [airfare = 100,000; per diem = 1500/day; 1,500 x 10 x people x 10 days = 150,000.00; incidentals = 4,000.00]

To Batan, Batanes: 464,000.00 [airfare = 310,000; per diem = 1500/day; 1,500 x 10 x people x 10 days = 150,000.00; incidentals = 4,000.00]

To Bolinao, Pangasinan: 100,000.00 [per diem = 1500/day; 1500 x 8 people x 8 days = 96,000.00; incidentals = 4,000.00]

To Currimao, Ilocos Norte: 100,000.00 [per diem = 1500/day; 1500 x 8 people x 8 days = 96,000.00; incidentals = 4,000.00]

To Guiuan, Eastern Samar: 169,000.00 [airfare and per diem during travel = 60,000; per diem during sampling = 1500/day; 1,500 x 10 people x 7 days = 105,000.00; incidentals = 4,000.00]

To Biri, Northern Samar: 219,000.00 [airfare and per diem during travel = 110,000; per diem during sampling = 1500/day; 1,500 x 10 people x 7 days = 105,000.00; incidentals = 4,000.00]

To Iba, Zambales: 112,000.00 [per diem = 1500/day; 1500 x 8 people x 9 days = 108,000.00; incidentals = 4,000.00]

To Busuanga, Palawan: 212,000.00 [airfare and per diem during travel = 54,000; per diem during sampling = 2200/day; 2200 x 10 people x 7 days = 154,000.00; incidentals = 4,000.00]

To Sorsogon, Sorsogon: 162,000.00 [airfare and per diem during travel = 50,000; per diem = 1500/day; 1500 x 8 people x 9 days = 108,000.00; incidentals = 4,000.00]

To Legazpi, Albay: 169,000.00 [airfare and per diem during travel = 60,000; per diem during sampling = 1500/day; 1,500 x 10 people x 7 days = 105,000.00; incidentals = 4,000.00]

To Bien Unido, Bohol: 180,000.00 [airfare and per diem during travel = 50,000; per diem during sampling = 1800/day; 1,800 x 10 people x 7 days = 126,000.00; incidentals = 4,000.00]

To Lian, Batangas: 162,000.00 [per diem = 2200/day; 2200 x 8 people x 9 days = 158,400.00; incidentals = 3,600.00]

NOTE: In most of the sampling trips, the researchers will be availing of the MSI vehicle and will entail services of an MSI driver, which will make the number of people equal to 10.

Travel Expenses for Establishing LGU network/Workshops:

Year 1

To Caramoan, Camarines Sur: 169,000.00 [airfare and per diem during travel = 90,000; per diem during workshop = 1500/day; 1,500 x 10 people x 5 days = 75,000.00; incidentals = 4,000.00]

To Dinagat, Surigao Del Norte: 159,000.00 [airfare and per diem during travel = 80,000; per diem during workshop = 1500/day; 1,500 x 10 people x 5 days = 75,000.00; incidentals = 4,000.00]

To Tandubas, Tawi Tawi: 179,000.00 [airfare = 100,000; per diem = 1500/day; 1,500 x 10 x people x 5 days = 75,000.00; incidentals = 4,000.00]

To Guiuan, Eastern Samar: 139,000.00 [airfare and per diem during travel = 60,000; per diem during workshop = 1500/day; 1,500 x 10 people x 5 days = 75,000.00; incidentals = 4,000.00]

To Legazpi, Albay: 139,000.00 [airfare and per diem during travel = 60,000; per diem during workshop = 1500/day; 1,500 x 10 people x 5 days = 75,000.00; incidentals = 4,000.00]

Year 2

To Caramoan, Camarines Sur: 169,000.00 [airfare and per diem during travel = 90,000; per diem during workshop = 1500/day; 1,500 x 10 people x 5 days = 75,000.00; incidentals = 4,000.00]

To Biri, Northern Samar: 189,000.00 [airfare and per diem during travel = 110,000; per diem during workshop = 1500/day; 1,500 x 10 people x 5 days = 75,000.00; incidentals = 4,000.00]

To Busuanga, Palawan: 168,000.00 [airfare and per diem during travel = 54,000; per diem during workshop = 2200/day; 2200 x 10 people x 5 days = 110,000.00; incidentals = 4,000.00]

To Sorsogon, Sorsogon: 114,000.00 [airfare and per diem during travel = 50,000; per diem = 1500/day; 1500 x 8 people x 5 days = 60,000.00; incidentals = 4,000.00]

Travel insurance for staff

Insurance for research staff participating in sampling trips will be applied for. With the current situation of the COVID-19 pandemic, insurance covering COVID-19 and dengue will also be availed of aside from the travel insurance. This is estimated to be around PhP 7,500 per year for each research staff.

D.Equipment (for Philippines-side only): UP-MSI

Item(s)	Cost/unit (PhP)	YR 1		YR 2	
		No. of Units	Sub-Total for YR 1	No. of Units	Sub-Total for YR 2
graphite furnace atomic absoprtion spectrophotometer with complete parts and accessories	4,500,000.00	1	4,500,000.00		
CHNS/O elemental analyzer with complete parts and accessories	4,500,000.00	1	4,500,000.00		
microwave digestion system with complete parts and accessories	4,000,000.00	1	4,000,000.00		
fumehoods with complete parts and accessories	1,000,000.00	1	1,000,000.00		
GCMS and accessories	11,000,000.00	1	11,000,000.00		

HPLC with DAD detector and accessories	3,500,000.00	1	3,500,000.00		
HPLC_MS and accessories	20,000,000.00	1	20,000,000.00		
lyophilizer and accessories	1,200,000.00	1	1,200,000.00		
ultralow freezer and accessories	1,300,000.00	2	2,600,000.00		
UV-Vis spectrophotometer and accessories	700,000.00	1	700,000.00		
Multi-channel peristaltic pump and accessories	300,000.00	2	600,000.00		
rotary evaporator and accessories	500,000.00	2	1,000,000.00		
Oven drier and accessories	200,000.00	1	200,000.00		
analytical balance and accessories	100,000.00	2	200,000.00		
toploading balance and accessories	50,000.00	2	100,000.00		
High resolution camera and accessories for seaweed documentation	100,000.00	1	100,000.00		
Temperature and humidity data logger and accessories	70,000.00	4	280,000.00		
hot plate/magnetic stirrer and accessories	50,000.00	4	200,000.00		
Desktops/laptops and accessories	50,000.00	6	300,000.00		
Furnace and accessories	450,000.00	1	450,000.00		
Sonicator and accessories	500,000.00	2	1,000,000.00		
vacuum pump and accessories	200,000.00	2	400,000.00		
pH meter, microwave oven, micro-centrifuge, vortex mixer, and accessories	170,000.00	1	170,000.00		
qPCR, gradient PCR, and accessories	3,500,000.00	1	3,500,000.00		
refrigerated centrifuge with different rotors and accessories	1,500,000.00	1	1,500,000.00		
water bath with temperature control and accessories	475,000.00	2	950,000.00		
multiplate reader and accessories	2,000,000.00	1	2,000,000.00		
multiparameter and accessories for water analysis	400,000.00	2	800,000.00		
TOTAL (PhP)			**66,750,000.00**		**0.00**

JUSTIFICATION: Describe the need or purpose of the equipment (e.g. how each unit will be used). UP-MSI

Equipment	Purpose	User	Location of the equipment
graphite furnace atomic absorption spectrophotometer with complete parts and accessories	For multi-elemental analysis of trace metals in seaweed samples and seaweed extracts.	For use of the research staff during the execution of the grant and will become part of the *Seaweed Chemistry facility*. Will be accessed by collaborating HEIS through a MOA that will be executed.	Will be housed and maintained at UP-MSI.
CHNS/O elemental analyzer with complete parts and accessories	For the determination of particulate organic carbon and nitrogen content of seaweed samples and extracts.	For use of the research staff during the execution of the grant and will become part of the *Seaweed Chemistry facility*.	Will be housed and maintained at UP-MSI.

microwave digestion system with complete parts and accessories	Will be used for the sample processing of seaweed samples and extracts to make them suitable for trace metal determination.	For use of the research staff during the execution of the grant and will become part of the *Seaweed Chemistry facility.*	Will be housed and maintained at UP-MSI.
fumehoods with complete parts and accessories	Will be used for the safe processing of the samples to ensure protection of research staff.	For use of the research staff during the execution of the grant and will become part of the *Seaweed Chemistry facility.*	Will be housed and maintained at UP-MSI.
GCMS and accessories	For analysis of fatty acid methyl esters (FAMEs), derivatized iodine-containing compounds, and thermally-labile compounds of interest.	For use of the research staff during the execution of the grant and will become part of the *Seaweed Chemistry facility.* Will be accessed by collaborating HEIS through a MOA that will be executed.	Will be housed and maintained at UP-MSI.
HPLC-DAD detector and accessories	For analysis of amino acids, fucoxanthin and fucosterol, β-carotene, and other thermally-stable compounds of interest.	For use of the research staff during the execution of the grant and will become part of the *Seaweed Chemistry facility.*	Will be housed and maintained at UP-MSI.
HPLC-MS and accessories	For analysis of small molecules of interest and other thermally-unstable compounds of interest.	For use of the research staff during the execution of the grant and will become part of the *Seaweed Chemistry facility.* Will be accessed by collaborating HEIS through a MOA that will be executed.	Will be housed and maintained at UP-MSI.
Lyophilizer and accessories	Required for sample preparation and pre-treatment.	For use of the research staff during the execution of the grant and will become part of the *Seaweed Chemistry facility*	Will be housed and maintained at UP-MSI.
Ultralow freezer and accessories	For use in preservation and storage of samples and extracts.	For use of the research staff during the execution of the grant and will become part of the *Seaweed Chemistry facility*	Will be housed and maintained at UP-MSI.
UV-Vis spectrophotometer and accessories	For analysis of phenolic content, determination of absorption spectra, and relevant antioxidant assays.	For use of the research staff during the execution of the grant and will become part of the *Seaweed Chemistry facility*	Will be housed and maintained at UP-MSI.
Multi-channel peristaltic pump and accessories	Required for volume-controlled and automated	For use of the research staff during the execution of the grant and will be utilized by	Will be housed and

	mixing of reagents in antioxidant assays	the *Seaweed Chemistry facility*.	maintained at UP-MSI.
Rotary evaporator and accessories	Required for extraction of target compounds from samples.	For use of the research staff during the execution of the grant and will be utilized by the *Seaweed Chemistry facility*.	Will be housed and maintained at UP-MSI.
Oven drier and accessories	Required for sample preparation and pre-treatment.	For use of the research staff during the execution of the grant and will become part of the *Seaweed Chemistry facility*.	Will be housed and maintained at UP-MSI.
Analytical balance and accessories	Required for sample preparation and pre-treatment.	For use of the research staff during the execution of the grant and will be utilized by the *Seaweed Chemistry facility*.	Will be housed and maintained at UP-MSI.
Toploading balance and accessories	Required for sample preparation and pre-treatment.	For use of the research staff during the execution of the grant and will be utilized by the *Seaweed Chemistry facility*.	Will be housed and maintained at UP-MSI.
High resolution camera and accessories	Required for sample documentation.	For use of the research staff during the execution of the grant and will be utilized by the *Seaweed Chemistry facility*.	Will be housed and maintained at UP-MSI.
Temperature and humidity data logger and accessories	Required for recording of temperature and humidity.	For use of the research staff during the execution of the grant and will be utilized by the *Seaweed Chemistry facility*.	Will be housed and maintained at UP-MSI.
Hot plate and magnetic stirrer and accessories	Required for sample preparation and pre-treatment.	For use of the research staff during the execution of the grant and will be utilized by the *Seaweed Chemistry facility*.	Will be housed and maintained at UP-MSI.
Desktops/laptops and accessories	Required for documentation, data analysis, data processing, and writing.	For use of the research staff during the execution of the grant and will be utilized by the *Seaweed Chemistry facility*.	Will be housed and maintained at UP-MSI.
Furnace and accessories	Required for ashing of seaweed raw materials and products.	For use of the research staff during the execution of the grant and will be utilized by the *Seaweed Chemistry facility*.	Will be housed and maintained at UP-MSI.
Sonicator and accessories	Required for cleaning and extraction purposes.	For use of the research staff during the execution of the grant and will be utilized by the *Seaweed Chemistry facility*.	Will be housed and maintained at UP-MSI.
Vacuum pump and accessories	Required for filtration and extraction processes.	For use of the research staff during the execution of the grant and will be utilized by the *Seaweed Chemistry facility*.	Will be housed and maintained at UP-MSI.
pH meter, microwave oven, microcentrifuge,	Required for routine pH determination and other laboratory procedures.	For use of the research staff during the execution of the grant and will be utilized by	Will be housed and maintained at UP-MSI.

vortex mixer, and accessories		the *Seaweed Chemistry facility*.	
qPCR, gradient PCR, and accessories	Will be used for targeted transcriptomics analyses.	For use of the research staff during the execution of the grant and will be utilized by the *Seaweed Chemistry facility*. Will be accessed by collaborating HEIS through a MOA that will be executed.	Will be housed and maintained at UP-MSI.
refrigerated centrifuge with different rotors and accessories	Will be used for samples processing for pre-concentration or separation purposes.	For use of the research staff during the execution of the grant and will be utilized by the *Seaweed Chemistry facility*.	Will be housed and maintained at UP-MSI.
water bath with temperature control and accessories	Will be used for experiments requiring precise control of temperature during processing.	For use of the research staff during the execution of the grant and will be utilized by the *Seaweed Chemistry facility*.	Will be housed and maintained at UP-MSI.
multiplate reader and accessories	Will be used for biochemical assays with samples that are available at minimum volumes and require almost simultaneous measurements.	For use of the research staff during the execution of the grant and will be utilized by the *Seaweed Chemistry facility*.	Will be housed and maintained at UP-MSI.
multiparameter and accessories for water analysis	Will be used for routine measurements of physico-chemical parameters during sampling.	For use of the research staff during the execution of the grant and will be utilized by the *Seaweed Chemistry facility*.	Will be housed and maintained at UP-MSI.

After the end of the grant, most of the equipment acquired through this project will constitute the *Seaweed Chemistry facility* housing sophisticated equipment. A close coordination with the UP-MSI will be sought to address maintenance and operation costs for the equipment. For common use equipment, all MSI research staff may use the equipment provided that operators will undergo sufficient training and agree to shoulder technical fees that may be incurred if their usage of specific equipment led to equipment malfunction. For sophisticated equipment, the use may be made accessible to other researchers after training and fees will be collected to cover cost of chemicals and other requirements. The core equipment facility will be supervised by the Project Leaders of this project with the assistance of other UP-MSI faculty with relevant background.

Is the host institution willing to maintain the equipment/facility?

_✓_ YES.

If YES, provide means by which the equipment/facility will be maintained.

The equipment will be maintained in the Seaweed Chemistry Facility at the UP-MSI.

__ NO

E. Supplies and Materials:
Supplies and Materials for PH side: UP-MSI

Item(s)	Cost/unit (PhP)	YR 1		YR 2	
		No. of units	Sub-Total for YR 1	No. of units	Sub-Total for YR 1
Extraction solvents	~6,000.00	~20	100,000.00	~40	200,000.00
HPLC grade solvents	~16,500.00	~9	150,000.00	~9	150,000.00
Certified reference materials	~30,000.00	~9	250,000.00	~4	100,000.00
ICPMS single element standards	~8,000.00	~32	250,000.00	~6	50,000.00
High purity acids and high purity gases (He, Ar, N_2, air)	~10,000.00	~15	150,000.00	~20	200,000.00
General laboratory chemicals	~9,000.00	~18	150,000.00	~18	150,000.00
Chemicals for preparation of artificial seawater and other required reagents	~1,000.00 to 5,000.00	~15	100,000.00	~25	150,000.00
Laboratory glassware/plastic ware and other materials	~500.00 to 10,000.00	1 to 50 (various)	250,000.00	1 to 20 (various)	100,000.00
TOTAL (PhP)			**1,400,000.00**		**1,100,000.00**

JUSTIFICATION:

Extraction solvents: the solvents that will be used in this work include hexane, toluene, methanol, ethanol, diethyl ether, and other solvents that may be needed for harsher extraction conditions. The average cost of hexane, toluene, and diethyl ether is about PhP 6,000.00 per 2L while the average cost of methanol, ethanol, and other alcohol-containing solvent is about PhP 6,500.00 per 4L. It is expected that significant volumes of extraction solvents will be utilized for the duration of this project.

HPLC-grade solvents: elution solvents for HPLC analysis will include methanol, acetonitrile, dichloromethane, and other solvents that may be necessary. The average cost of HPLC-grade solvents is PhP 16,500.00 per 8L and significant volumes will be utilized for optimization and validation of methods and analysis of seaweed samples.

Certified reference materials: optimization and validation of standard methods will require the use of certified reference materials with matrices that are closest to the real samples. The average cost of reference materials is about PhP 30,000.00 including freight costs.

ICPMS single element standards: single element standards will be used for preparation of standard calibration standard solution of target trace metal analytes and will be continually used for evaluation of the standard procedures and in the actual sample analyses.

High purity acids: high purity nitric acid, hydrochloric acid, hydrofluoric acid, acetic acid, and sulfuric acid will be required in this project for the complete mineralization of seaweed and seaweed extracts through acid-assisted hot plate digestion. In addition, high purity acids will also be used in the preparation of ICPMS standards and samples for analysis.

High purity gases: high purity gases such as argon, nitrogen, helium, and oxygen will be required for the operation of the ICPMS (argon, helium, nitrogen, and oxygen) and GC-MS (helium).

Chemicals for preparation of artificial seawater and other reagents will be required in keeping seaweed species alive in culture when samples are required for subsequent validation of physicochemical parameters.

Laboratory glassware/plastic ware and other materials: this project will require use of both glass and plastic ware for solution and sample preparation. For all work focusing on trace metal analysis and speciation, plastic ware will be used while glass will be used for all other experiments.

F. Professional Services: Services of a consultant in bookkeeping will be sought whenever necessary and services of a lawyer will also be sought when the need arises (e.g., when purchase of equipment is problematic due to unwanted practices of suppliers).

COSTS FOR CONTRACTS MUST BE BROKEN DOWN IN DETAIL AND NARRATIVE JUSTIFICATION. IF APPLICABLE, NUMBERS OF CLIENTS SHOULD BE INCLUDED IN THE COSTS.

Name	Service	Rate	Cost for YR —	Cost for YR —
To be identified	Bookkeeping	SG 11 (30,527/mo)	366,324.00	366,324.00
To be identified	Daily labor support	400/day	33,676.00	33,676.00
		TOTAL	400,000.00	400,000.00

JUSTIFICATION: Explain the need for each contractual agreement and how they relate to the overall project.

1. The services of a bookkeeper is needed because this is a big grant that will require timely submission of financial and technical reports. The processing of vouchers for procurement, payment or travel of research staff will require the services of the bookkeeper.
2. The remaining funds will be allotted for minimum wages (daily labor costs) for daily wage earners that will be hired for services rendered like assistance in handling heavy sampling containers, cleaning of hatchery spaces/tanks, and other manual tasks that the research staff will need assistance for.

G. Construction: NOT ALLOWED

H. Other: expenses not covered in any of the previous budget categories

OTHER EXPENSES FOR THE PH COUNTERPART: UP-MSI

Item(s)	Cost for YR 1	Cost for YR 2
Communication expenses		
Communication expenses during sampling	20,000.00	20,000.00
Repairs and Maintenance of Facilities		
Maintenance of clean room	20,000.00	30,000.00
Renovation cost	500,000.00	
Transportation and Delivery Expenses		
Freight costs	150,000.00	150,000.00
Training Expenses		
Workshop in collaboration with Sorsogon State University		95,000.00

Workshop in collaboration with MSU Naawan		90,000.00
Workshop in collaboration with University of Eastern Philippines		90,000.00
Workshop in collaboration with Partido State University		90,000.00
Seaweed Farmers Training/Workshops	14,000,000.00	
Membership Dues for the Institution		
		-
Rent Expenses		
Boats for sampling	50,000.00	200,000.00
Representation Expenses		
Annual meetings (food expenses)	15,000.00	15,000.00
Kickoff meeting (food expenses)	15,000.00	
Project evaluation meeting (food expenses)		15,000.00
Subscription expenses		
Purchase of access to relevant scientific articles	20,000.00	50,000.00
Professional fees		
Consultant	400,000.00	400,000.00
Taxes, Insurance Premiums, and Other Fees		
Environmental fees for sampling	50,000.00	50,000.00
Chemical waste treatment	50,000.00	100,000.00
Other Maintenance and Operating Expenses		
Instrument fee for inter-calibration purposes	60,000.00	60,000.00
Publication expenses	250,000.00	400,000.00
TOTAL	**15,600,000.00**	**1,855,000.00**

JUSTIFICATION:

Communication expenses will cover cost for purchasing of SIM and prepaid cards for use of project staff in coordinating sampling activities.

Maintenance of clean room will include costs of changing eap filters every quarter during the operation of a clean room. Renovation cost will cover costs of room preparation for the sophisticated equipment that will be purchased through this project. The three MS equipment (ICPMS, HPLC-MS, GC-MS) will require specific conditions for the optimum performance of the equipment. HPLC-MS and GC-MS will need dedicated rooms and the ICPMS will require a class 1000 or better clean conditions. In addition to this, electrical supply, water and gas lines, and provisions for fume hoods in rooms housing specific equipment will also need to be properly installed.

Freight costs will cover fees for sending samples to the CA counterpart and costs for sending samples from within the country whenever the need arises, e.g., from Tawi Tawi to QC.

Water and electricity costs will cover fees that may be incurred during sampling especially in remote areas where use of generators may be required.

Training Expenses – the funds will cover the cost of workshop materials and food during the duration of the workshops in the partner HEIs and in UP-MSI. Th

Seaweed Farmers Training/Workshops– the funds will cover (1) Farmer training (20-30 farmers per village in 5 villages per hub) focusing on practices that can minimise losses from disease and epiphytes and reduces marine plastic pollution. Costs will cover community organising/preparation for the workshops, seaweed farmer mobilisation, resource personnel to deliver training, training venue, materials, snacks/refreshments for participants; (2) Multi-stakeholder provincial training by bringing together relevant members from DA, LGU, GAs, NGOs, Industry and academia to e.g., increase understanding of the need for new strains of seaweed from sexual reproduction, and progress to date; (3) establish community nursery that will act as model farms demonstrating the implementation of best practices because seeing is believing for small-scale fishers. Together with training this will help drive adoption. Seaweed-farmer scientists will be trained who can provide growth data on a monthly basis to help build up a clearer picture of seaweed growth rates for seedstock selection that will provide reliable and continuous source of high-quality seedlings for local community.

Rent expenses will cover costs that will be incurred during the sampling activities including accommodation, and boat and car rental during the sampling activities. Airfare costs are listed in the item for travel expenses.

Representation expenses will cover cost for food (lunch and snacks) in the annual meetings, and planning and evaluation meetings.

Subscription expenses will cover fees for access to scientific publications.

Services of a consultant in bookkeeping will be sought whenever necessary and services of a lawyer will also be sought when the need arises (e.g., when purchase of equipment is problematic due to unwanted practices of suppliers).

Other expenses include fees for inter-calibration purposes that may be required for some analytical methods and a budget for incidentals. Publication expenses will cover cost for page charges in journals with publication charges.

Indirect Cost Rate: UP-MSI

	BUDGET ITEM	Budget Requirement YR 1	YR 2
I. PERSONNEL SERVICES			
1	Personnel Salary	4,774,152.00	4,774,152.00
2	Honoraria	285,600.00	285,600.00
	Sub-Total for PS	**5,059,752.00**	**5,059,752.00**
II. MOOE			
1	Travel Expenses	3,400,000.00	2,653,000.00
2	Supplies and Materials	1,400,000.00	1,100,000.00
3	Communication Expenses	20,000.00	20,000.00
4	Repairs and Maintenance of Facilities	520,000.00	30,000.00
5	Transportation and Delivery Expenses	150,000.00	150,000.00
6	Utilities Expenses	-	-
7	Training Expenses	14,000,000.00	365,000.00
8	Membership Dues for the Institution	-	-
9	Advertising Expenses	-	-
10	Rent Expenses	50,000.00	200,000.00
11	Representation Expenses	30,000.00	30,000.00
12	Subscription Expenses	20,000.00	50,000.00
13	Survey Expenses	-	-
14	Professional Services	400,000.00	400,000.00
15	Taxes, Insurance Premiums and Other Fees	100,000.00	150,000.00
16	Other Maintenance and Operating Expenses	310,000.00	460,000.00
	Sub-Total for MOOE	**20,400,000.00**	**5,608,000.00**
III. Indirect Cost			
	Indirect for PS	1,011,950.40	1,011,950.40
	Indirect for MOOE	4,090,016.71	1,131,600.00
	Sub-Total Indirect Cost	**5,109,967.11**	**2,143.550.40**
IV. Equipment Outlay		**66,750,000.00**	
TOTAL PROPOSED BUDGET (UP-MSI)		**97,361,802.65**	**12,861,302.40**
TOTAL PROPOSED BUDGET FOR TWO YEARS (UP-MSI)			**110,223,105.05**

Equipment, MOOE and Justification for collaborating HEIs:

LINE ITEM BUDGET FOR MINDANAO STATE UNIVERSITY NAAWAN:
TOTAL PROPOSED BUDGET FOR 2 YEARS: 11,654,288.00

MSU Naawan will participate in research activities including sample collection, preservation for herbarium and chemical analyses, sample pre-treatment and analyses including anti-oxidative assays and basic biochemical profiling, which dictates the equipment being requested for this institution.

Personnel for PH Component: MSU Naawan

Position	Name	Annual Salary/Rate	Level of Effort	Cost for YR 1	Cost for YR 2
(1) HEI Collaborator	Rey Capangpangan	90,000.00	30%	90,000.00	90,000.00
(2-3) URA I	To be selected	795,120.00	100%	795,120.00	795,120.00
		TOTAL	**PhP**	**885,120.00**	**885,120.00**

JUSTIFICATION: Describe the role and responsibilities of each position.

(1): Dr. Rey Capangpangan of MSU Naawan will coordinate all research efforts at the partner HEIs including supervision of research staff based at the MSU Naawan.
(2) and (3) Two URAI: the two university research associates that will be identified from the pool of applicants (preferably a graduate student and hailing from nearby towns/provinces of the partner institutions) will assist in project activities from sampling to sample processing, data analysis, manuscript and report writing, and administrative tasks that will be necessary for smooth implementation of the project.

Equipment for MSU NAAWAN:

Item(s)	Cost/unit (PhP)	YR 1		YR 2	
		No. of Units	Sub-Total for YR 1	No. of Units	Sub-Total for YR 2
FT-IR and accessories	2,500,000.00	1	2,500,000.00		
HPLC with UV/RI detector and accessories	3,500,000.00	1	3,500,000.00		
UV-Vis spectrophotometer and accessories	700,000.00	1	700,000.00		
ultralow freezer and accessories	1,000,000.00	1	1,000,000.00		
furnace and accessories	450,000.00	1	450,000.00		
Rotary Evaporator with accessories	300,000.00		300,000.00		
TOTAL (PhP)			**8,450,000.00**		**0.00**

JUSTIFICATION: Describe the need or purpose of the equipment (e.g. how each unit will be used).

Equipment	Purpose	User	Location of the equipment
FT-IR and accessories	Will be used in the analyses of major organic groups present in an unknown compound to guide further purification steps.	For use of the research staff in MSU Naawan during the execution of the grant.	Will be housed and maintained at MSU NAAWAN.
HPLC with UV/RI detector and accessories	Will be used for qualitative and quantitative measurements of target analytes from samples.	For use of the research staff in MSU Naawan during the execution of the grant.	Will be housed and maintained at MSU NAAWAN.
UV-Vis spectrophotometer and accessories	Will be used for analyses of target compounds in samples.	For use of the research staff in MSU Naawan during the execution of the grant.	Will be housed and maintained at MSU NAAWAN.

Ultralow freezer and accessories	Will be used for sample storage especially samples that are thermally sensitive.	For use of the research staff in MSU Naawan during the execution of the grant.	Will be housed and maintained at MSU NAAWAN.
Furnace and accessories	Will be used for sample processing with focus on total carbon and nitrogen content.	For use of the research staff in MSU Naawan during the execution of the grant.	Will be housed and maintained at MSU NAAWAN.
Rotary Evaporator with accessories	Will be used for the removal of solvent under pressure during the extraction of target compounds from natural products.	For use of the research staff in MSU Naawan during the execution of the grant	Will be housed and maintained at MSU Naawan

Supplies and Materials for PH side: MSU NAAWAN

Item(s)	Cost/unit (PhP)	YR 1		YR 2	
		No. of units	Sub-Total for YR 1	No. of units	Sub-Total for YR 1
Extraction solvents	~2,000.00 to 5,000.00	~50	100,000.00	~50	100,000.00
HPLC grade solvents	~5,000.00 to 8,000.00	~25	150,000.00	~25	150,000.00
General laboratory chemicals	~500.00 to 2,500.00	~50	100,000.00	~50	100,000.00
Laboratory glassware/plastic ware and other materials	~500.00 to 5,000.00	~60	100,000.00	~60	100,000.00
TOTAL (PhP)			**450,000.00**		**450,000.00**

JUSTIFICATION:

Extraction solvents: Various extraction solvents (organic solvents or acidic/basic solvents will be used for extraction of desirable compounds from seaweeds. These solvents will be used in varying amounts and concentrations depending on the chemical properties of target compounds for extraction. Thus, the projected amount and cost are rough estimates only.

HPLC grade solvents: These solvents (acetonitrile, methanol, ultrapure water, acetone, and others) will be used in operating the HPLC for the precise separation and collection of compounds eluting from the HPLC. The costs and consumption of these solvents will be dependent on the nature of the target compounds for isolation.

General laboratory chemicals: these chemicals are required in sample pretreatment, clean-up, preconcentration, or cleaning steps in the laboratory.

Laboratory glassware/plastic ware and other materials: these are items needed as sample containers, reaction vessels, transfer and storage containers, and other routine laboratory procedures.

Indirect Cost Rate: MSU NAAWAN

	BUDGET ITEM	Budget Requirement YR 1	YR 2
I. PERSONNEL SERVICES			
1	Personnel Salary	795,120.00	795,120.00
2	Honoraria	90,000.00	90,000.00
	Sub-Total for PS	**885,120.00**	**885,120.00**
II. MOOE			
1	Travel Expenses		
2	Supplies and Materials	450,000.00	450,000.00
3	Communication Expenses		
4	Repairs and Maintenance of Facilities		
5	Transportation and Delivery Expenses		
6	Utilities Expenses		
7	Training Expenses		
8	Membership Dues for the Institution		
9	Advertising Expenses		

10	Rent Expenses		
11	Representation Expenses		
12	Subscription Expenses		
13	Survey Expenses		
14	Professional Services		
15	Taxes, Insurance Premiums and Other Fees		
16	Other Maintenance and Operating Expenses		
	Sub-Total for MOOE	**450,000.00**	**450,000.00**
	Indirect for PS	177,024.00	177,024.00
	Indirect for MOOE	90,000.00	90,000.00
	Sub-Total Indirect Cost	**267,024.00**	**267,024.00**
IV. Equipment Outlay		**8,450,000.00**	
TOTAL PROPOSED BUDGET: MSU NAAWAN		**10,052,144.00**	**1,602,144.00**
TOTAL PROPOSED BUDGET FOR 2 YEARS			**11,654,288.00**

LINE ITEM BUDGET FOR SORSOGON STATE UNIVERSITY:
TOTAL PROPOSED BUDGET FOR 2 YEARS: 12,674,288.00

Sorsogon State University will participate in research activities including sample collection, preservation for herbarium and chemical analyses, sample pre-treatment and analyses including taxonomic identification, molecular analyses, and morphological characterization, which dictates the equipment being requested for this institution.

Personnel for PH Component: Sorsogon State University

Position	Name	Annual Salary/Rate	Level of Effort	Cost for YR 1	Cost for YR 2
(1) HEI Collaborator	Richard Dumilag	90,000.00	30%	90,000.00	90,000.00
(2-3) URA I	To be selected	795,120.00	100%	795,120.00	795,120.00
		TOTAL	**PhP**	**885,120.00**	**885,120.00**

JUSTIFICATION: Describe the role and responsibilities of each position.

(1): Dr. Richard Dumilag of Sorsogon State University will coordinate all research efforts at the partner HEIs including supervision of research staff based at the SSU.
(2) and (3) Two URAI: the two university research associates that will be identified from the pool of applicants (preferably a graduate student and hailing from nearby towns/provinces of the partner institutions) will assist in project activities from sampling to sample processing, data analysis, manuscript and report writing, and administrative tasks that will be necessary for smooth implementation of the project.

Equipment for SSU:

Item(s)	Cost/unit (PhP)	YR 1		YR 2	
		No. of Units	Sub-Total for YR 1	No. of Units	Sub-Total for YR 2
Inverted microscope and accessories	1,000,000.00	1	1,000,000.00		
gradient PCR and accessories	600,000.00	1	600,000.00		
Compound microscope and accessories	300,000.00	1	300,000.00		
Stereomicroscope and accessories	300,000.00	1	300,000.00		
Gel Documentation System with complete parts and accessories	1,450,000.00	1	1,450,000.00		
bead ruptor tissue lyser and accessories	500,000.00	1	500,000.00		
liquid nitrogen containers and accessories	80,000.00	1	80,000.00		
refrigerator, freezer and accessories	100,000.00	2	200,000.00		
multiparameter and accessories for water sampling (data loggers)	1,250,000.00	1	1,250,000.00		
refrigerated centrifuge with accessories	1,200,000.00	1	1,200,000.00		
analytical balance and accessories	100,000.00	1	100,000.00		
toploading balance and accessories	50,000.00	1	50,000.00		
Digital Dry Bath with complete parts and accessories	90,000.00	1	90,000.00		
Pipettors with complete parts and accessories	95,000.00	4	380,000.00		
Vortex Mixer with complete parts and accessories	80,000.00	1	80,000.00		
Dead Air Cabinet with complete parts and accessories	450,000.00	1	450,000.00		

water purification system with complete parts and accessories	500,000.00	1	500,000.00		
Mini Centrifuge with complete parts and accessories	60,000.00	1	60,000.00		
DSLR camera and accessories	90,000.00	1	90,000.00		
Gel electrophoresis system with complete parts and accessories	70,000.00	1	70,000.00		
TOTAL (PhP)			8,750,000.00		0.00

JUSTIFICATION: Describe the need or purpose of the equipment (e.g. how each unit will be used).

Equipment	Purpose	User	Location of the equipment
Inverted microscope and accessories	Will be used for documentation especially looking at details of samples.	For use of the research staff in Sorsogon State University during the execution of the grant.	Will be housed and maintained at SSU.
gradient PCR and accessories	Will be used to determine presence of target genes.	For use of the research staff in Sorsogon State University during the execution of the grant.	Will be housed and maintained at SSU.
Compound microscope and accessories	Will be used for documentation especially looking at details of samples.	For use of the research staff in Sorsogon State University during the execution of the grant.	Will be housed and maintained at SSU.
Stereomicroscope and accessories	Will be used for documentation especially looking at details of samples.	For use of the research staff in Sorsogon State University during the execution of the grant.	Will be housed and maintained at SSU.
Gel Documentation System with complete parts and accessories	Will be used to check/visualize amplicons after gel electrophoresis	For use of the research staff in Sorsogon State University during the execution of the grant.	Will be housed and maintained at SSU.
bead ruptor tissue lyser and accessories	Required for sample preparation and pre-treatment.	For use of the research staff in Sorsogon State University during the execution of the grant.	Will be housed and maintained at SSU.
liquid nitrogen containers and accessories	Will be used for sample storage and preservation during transport.	For use of the research staff in Sorsogon State University during the execution of the grant.	Will be housed and maintained at SSU.
refrigerator, freezer and accessories	Will be used for sample storage and preservation.	For use of the research staff in Sorsogon State University during the execution of the grant.	Will be housed and maintained at SSU.
multiparameter for water sampling (data loggers)	Will be used for routine measurements of physico-chemical parameters during sampling.	For use of the research staff in Sorsogon State University during the execution of the grant.	Will be housed and maintained at SSU.
Refrigerated centrifuge with different rotors and accessories	Will be used for samples processing for pre-concentration or separation purposes.	For use of the research staff in Sorsogon State University during the execution of the grant.	Will be housed and maintained at SSU.
Analytical balance and accessories	Will be used for determination of sample weight.	For use of the research staff in Sorsogon State University during the execution of the grant.	Will be housed and maintained at SSU.
Toploading balance and accessories	Will be used for determination of sample weight.	For use of the research staff in Sorsogon State University during the execution of the grant.	Will be housed and maintained at SSU.
Digital Dry Bath with complete parts and accessories	Will be used for DNA extraction process	For use of the research staff in Sorsogon State University during the execution of the grant.	Will be housed and maintained at SSU.
Pipettors with complete parts and accessories	Will be used for various steps for sample preservation and other molecular downstream analysis (e.g. DNA extraction process, AGE, PCR, etc.)	For use of the research staff in Sorsogon State University during the execution of the grant.	Will be housed and maintained at SSU.
Vortex Mixer with complete parts and accessories	Will be used to mix and agitate samples during DNA extraction	For use of the research staff in Sorsogon State University during the execution of the grant.	Will be housed and maintained at SSU.

Dead Air Cabinet with complete parts and accessories	Will be used as a an axenic working station during molecular work	For use of the research staff in Sorsogon State University during the execution of the grant.	Will be housed and maintained at SSU.
Water purification system with complete parts and accessories	Will be used for producing ultrapure water	For use of the research staff in Sorsogon State University during the execution of the grant.	Will be housed and maintained at SSU.
Mini Centrifuge with complete parts and accessories	Will be used for a quick agitation and separation process during DNA extraction	For use of the research staff in Sorsogon State University during the execution of the grant.	Will be housed and maintained at SSU.
DSLR camera and accessories	For documentation	For use of the research staff in Sorsogon State University during the execution of the grant.	Will be housed and maintained at SSU.
Gel Electrophoresis System with complete parts and accessories	Will be used to separate DNA fragments (bands) and visualize amplicons	For use of the research staff in Sorsogon State University during the execution of the grant.	Will be housed and maintained at SSU.

E. Supplies and Materials:

Supplies and Materials for PH side: SSU

Item(s)	Cost/unit (PhP)	YR 1		YR 2	
		No. of units	Sub-Total for YR 1	No. of units	Sub-Total for YR 1
RNA extraction kits and other chemicals	~50,000.00	~6	300,000.00	~6	300,000.00
General laboratory chemicals	~5,000.00	~20	100,000.00	~20	100,000.00
Laboratory glassware/plastic ware and other materials	~2,000.00 to 5,000.00	~15	100,000.00	~15	100,000.00
TOTAL (PhP)			**500,000.00**		**500,000.00**

JUSTIFICATION:

RNA extraction kits and other chemicals: These kits are necessary for collection, preservation and analyses of RNA from samples.

General laboratory chemicals: these chemicals are required in sample pretreatment, clean-up, preconcentration, or cleaning steps in the laboratory.

Laboratory glassware/plastic ware and other materials: these are items needed as sample containers, reaction vessels, transfer and storage containers, and other routine laboratory procedures.

Indirect Cost Rate: SSU

	BUDGET ITEM	Budget Requirement YR 1	YR 2
I. PERSONNEL SERVICES			
1	Personnel Salary	795,120.00	795,120.00
2	Honoraria	90,000.00	90,000.00
	Sub-Total for PS	**885,120.00**	**885,120.00**
II. MOOE			
1	Travel Expenses		
2	Supplies and Materials	500,000.00	500,000.00
3	Communication Expenses		
4	Repairs and Maintenance of Facilities		
5	Transportation and Delivery Expenses		
6	Utilities Expenses		
7	Seaweed Farmers Training/Workshops Expenses	500,000.00	
8	Membership Dues for the Institution		
9	Advertising Expenses		

10	Rent Expenses		
11	Representation Expenses		
12	Subscription Expenses		
13	Survey Expenses		
14	Professional Services		
15	Taxes, Insurance Premiums and Other Fees		
16	Other Maintenance and Operating Expenses		
	Sub-Total for MOOE	**1,000,000.00**	**500,000.00**
	Indirect for PS	177,024.00	177,024.00
	Indirect for MOOE	200,000.00	100,000.00
	Sub-Total Indirect Cost	**377,024.00**	**277,024.00**
IV. Equipment Outlay		**8,750,000.00**	
TOTAL PROPOSED BUDGET: SSU		**11,012,144.00**	**1,662,144.00**
TOTAL PROPOSED BUDGET FOR 2 YEARS			**12,674,288.00**

Justification for MOOE #7 Seaweed Farmers Trainings/Workshop:

Farmer training (20-30 farmers per village in 5 villages per hub) focusing on practices that can minimise losses from disease and epiphytes and reduces marine plastic pollution. Costs will cover community organising/preparation for the workshops, seaweed farmer mobilisation, resource personnel to deliver training, training venue, materials, snacks/refreshments for participants

LINE ITEM BUDGET FOR UNIVERSITY OF EASTERN PHILIPPINES:
TOTAL PROPOSED BUDGET FOR 2 YEARS: 8,984,288.00

University of Eastern Philippines will participate in research activities including sample collection, preservation for herbarium and chemical analyses, sample pre-treatment and analyses including taxonomic identification, molecular analyses, and morphological characterization, which dictates the equipment being requested for this institution.

Personnel for PH Component: University of Eastern Philippines

Position	Name	Annual Salary/Rate	Level of Effort	Cost for YR 1	Cost for YR 2
(1) HEI Collaborator	Ronelie Salvador	90,000.00	30%	90,000.00	90,000.00
(2-3) URA I	To be selected	795,120.00	100%	795,120.00	795,120.00
		TOTAL	**PhP**	**885,120.00**	**885,120.00**

JUSTIFICATION: Describe the role and responsibilities of each position.

(1): Dr. Ronelie Salvador of University of Eastern Philippines will coordinate all research efforts at the partner HEIs including supervision of research staff based at the UEP.
(2) and (3) Two URAI: the two university research associates that will be identified from the pool of applicants (preferably a graduate student and hailing from nearby towns/provinces of the partner institutions) will assist in project activities from sampling to sample processing, data analysis, manuscript and report writing, and administrative tasks that will be necessary for smooth implementation of the project.

Equipment for UEP:

Item(s)	Cost/unit (PhP)	YR 1		YR 2	
		No. of Units	Sub-Total for YR 1	No. of Units	Sub-Total for YR 2
Inverted microscope and accessories	800,000.00	1	800,000.00		
stereomicroscope and accessories	300,000.00	1	300,000.00		
Oven and accessories	300,000.00	1	300,000.00		
Furnace and accessories	450,000.00	1	450,000.00		
analytical balance and accessories	100,000.00	1	100,000.00		
toploading balance and accessories	50,000.00	1	50,000.00		
UV-Vis spectrophotometer and accessories	700,000.00	1	700,000.00		
Water bath with temperature control and accessories	450,000.00	1	450,000.00		
Centrifuge with different rotors and accessories	1,200,000.00	1	1,200,000.00		
Multiparameter and accessories for water sampling	400,000.00	1	400,000.00		
pH meter, vortex mixer, hotplate, and accessories	150,000.00	1	150,000.00		
freezer and refrigerator for sample storage	100,000.00	1	100,000.00		
compound microscope and accessories	300,000.00	1	300,000.00		
Gopro and underwater camera and accessories	100,000.00	1	100,000.00		
data loggers and accessories	70,000.00	2	140,000.00		
TOTAL (PhP)			**5,540,000.00**		**0.00**

JUSTIFICATION: Describe the need or purpose of the equipment (e.g. how each unit will be used).

Equipment	Purpose	User	Location of the equipment
Inverted microscope and accessories	Will be used for documentation especially looking at details of samples.	For use of the research staff in University of Eastern Philippines during the execution of the grant.	Will be housed and maintained at UEP.
stereomicroscope and accessories	Will be used for documentation especially looking at details of samples.	For use of the research staff in University of Eastern Philippines during the execution of the grant.	Will be housed and maintained at UEP.
Oven and accessories	Will be used for sample drying and pretreatment procedures.	For use of the research staff in University of Eastern Philippines during the execution of the grant.	Will be housed and maintained at UEP.
Furnace and accessories	Will be used for sample processing with focus on total carbon and nitrogen content.	For use of the research staff in University of Eastern Philippines during the execution of the grant.	Will be housed and maintained at UEP.
analytical balance and accessories	Will be used for determination of sample weight.	For use of the research staff in University of Eastern Philippines during the execution of the grant.	Will be housed and maintained at UEP.
toploading balance and accessories	Will be used for determination of sample weight.	For use of the research staff in University of Eastern Philippines during the execution of the grant.	Will be housed and maintained at UEP.
UV-Vis spectrophotometer and accessories	Will be used for determination of target analytes using absorbance spectroscopy.	For use of the research staff in University of Eastern Philippines during the execution of the grant.	Will be housed and maintained at UEP.
Water bath with temperature control and accessories	Will be used for sample processing and pretreatment procedures.	For use of the research staff in University of Eastern Philippines during the execution of the grant.	Will be housed and maintained at UEP.
Centrifuge with different rotors and accessories	Will be used for samples processing for pre-concentration or separation purposes.	For use of the research staff in University of Eastern Philippines during the execution of the grant.	Will be housed and maintained at UEP.
Multiparameter and accessories for water sampling	Will be used for routine measurements of physico-chemical parameters during sampling.	For use of the research staff in University of Eastern Philippines during the execution of the grant.	Will be housed and maintained at UEP.
pH meter, vortex mixer, hotplate, and accessories	Wil be used for pH measurements, sample processing, and temperature-controlled experiments	For use of the research staff in University of Eastern Philippines during the execution of the grant.	Will be housed and maintained at UEP.
freezer and refrigerator and accessories for sample storage	Will be used for sample storage and preservation.	For use of the research staff in University of Eastern Philippines during the execution of the grant.	Will be housed and maintained at UEP.
compound microscope and accessories	Will be used for documentation especially looking at details of samples.	For use of the research staff in University of Eastern Philippines during the execution of the grant.	Will be housed and maintained at UEP.
Gopro and underwater camera and accessories	Will be used for documentation especially during sampling.	For use of the research staff in University of Eastern Philippines during the execution of the grant.	Will be housed and maintained at UEP.
data loggers and accessories	Will be used for data storage/collection.	For use of the research staff in University of Eastern Philippines during the execution of the grant.	Will be housed and maintained at UEP.

Supplies and Materials:

Supplies and Materials for PH side: UEP

Item(s)	Cost/unit (PhP)	YR 1		YR 2	
		No. of units	Sub-Total for YR 1	No. of units	Sub-Total for YR 1
Field and other materials	~1,000.00 to 5,000.00	~25	100,000.00	~25	100,000.00
General laboratory chemicals	~1,500.00 to 6,000.00	~20	100,000.00	~20	100,000.00
Laboratory glassware/plastic ware and other materials	~1,000.00 to 3,000.00	~40	100,000.00	~40	100,000.00
TOTAL (PhP)			**300,000.00**		**300,000.00**

JUSTIFICATION:

Field and other materials: The field and sampling materials, which include various items from ropes, non-slip mats, pails, etc., will be used for sample collection and processing while on the field.

General laboratory chemicals: these chemicals are required in sample pretreatment, clean-up, preconcentration, or cleaning steps in the laboratory.

Laboratory glassware/plastic ware and other materials: these are items needed as sample containers, reaction vessels, transfer and storage containers, and other routine laboratory procedures.

Indirect Cost Rate: UEP

	BUDGET ITEM	Budget Requirement	
		YR 1	YR 2
I. PERSONNEL SERVICES			
1	Personnel Salary	795,120.00	795,120.00
2	Honoraria	90,000.00	90,000.00
	Sub-Total for PS	**885,120.00**	**885,120.00**
II. MOOE			
1	Travel Expenses		
2	Supplies and Materials	300,000.00	300,000.00
3	Communication Expenses		
4	Repairs and Maintenance of Facilities		
5	Transportation and Delivery Expenses		
6	Utilities Expenses		
7	Seaweed Farmers Training/Workshops Expenses	500,000.00	
8	Membership Dues for the Institution		
9	Advertising Expenses		
10	Rent Expenses		
11	Representation Expenses		
12	Subscription Expenses		
13	Survey Expenses		
14	Professional Services		
15	Taxes, Insurance Premiums and Other Fees		
16	Other Maintenance and Operating Expenses		
	Sub-Total for MOOE	**800,000.00**	**300,000.00**

	Indirect for PS	177,024.00	177,024.00
	Indirect for MOOE	160,000.00	60,000.00
	Sub-Total Indirect Cost	**337,024.00**	**237,024.00**
IV. Equipment Outlay		**5,540,000.00**	
TOTAL PROPOSED BUDGET: UEP		**7,562,144.00**	**1,422,144.00**
TOTAL PROPOSED BUDGET FOR 2 YEARS			**8,984,288.00**

Justification for MOOE item #7
Seaweed Farmers Training/ Workshops
Farmer training (20-30 farmers per village in 5 villages per hub) focusing on practices that can minimise losses from disease and epiphytes and reduces marine plastic pollution. Costs will cover community organising/preparation for the workshops, seaweed farmer mobilisation, resource personnel to deliver training, training venue, materials, snacks/refreshments for participants

LINE ITEM BUDGET FOR PARTIDO STATE UNIVERSITY:
TOTAL PROPOSED BUDGET FOR 2 YEARS: 12,054,288.00

Partido State University will participate in research activities including sample collection, preservation for herbarium and chemical analyses, sample pre-treatment and analyses including taxonomic identification, molecular analyses, and morphological characterization, which dictates the equipment being requested for this institution.

Personnel for PH Component: Partido State University

Position	Name	Annual Salary/Rate	Level of Effort	Cost for YR 1	Cost for YR 2
(1) HEI Collaborator	Michael Clores	90,000.00	30%	90,000.00	90,000.00
(2-3) URA I	To be selected	795,120.00	100%	795,120.00	795,120.00
		TOTAL	**PhP**	**885,120.00**	**885,120.00**

JUSTIFICATION: Describe the role and responsibilities of each position.

(1): Dr. Michael Clores of Partido State University will coordinate all research efforts at the partner HEIs including supervision of research staff based at the Partido SU.
(2) and (3) Two URAI: the two university research associates that will be identified from the pool of applicants (preferably a graduate student and hailing from nearby towns/provinces of the partner institutions) will assist in project activities from sampling to sample processing, data analysis, manuscript and report writing, and administrative tasks that will be necessary for smooth implementation of the project.

Equipment for PSU:

Item(s)	Cost/unit (PhP)	YR 1		YR 2	
		No. of Units	Sub-Total for YR 1	No. of Units	Sub-Total for YR 2
Oven drier and accessories	200,000.00	2	400,000.00		
analytical balance and accessories	100,000.00	1	100,000.00		
toploading balance and accessories	50,000.00	2	100,000.00		
furnace and accessories	450,000.00	1	450,000.00		
filtration system (vacuum pump, manifold, and accessories)	200,000.00	1	200,000.00		
UV-Vis spectrophotometer and accessories	700,000.00	1	700,000.00		
Spectrofluorometer and accessories	1,000,000.00	1	1,000,000.00		
Pump, motor with accessories	350,000.00	1	350,000.00		
data loggers and accessories	70,000.00	3	210,000.00		
scuba diving sets and accessories	80,000.00	2	160,000.00		
Compound microscope and accessories	300,000.00	2	600,000.00		
inverted microscope and accessories	800,000.00	1	800,000.00		
Stereomicroscope and accessories	300,000.00	1	300,000.00		
Mechanized rotating drum mixer and accessories	200,000.00	1	200,000.00		
Gopro and underwater camera and accessories	100,000.00	2	200,000.00		
tabletop centrifuge and accessories	150,000.00	1	150,000.00		
pH meter, vortex mixer, hotplate, and accessories	150,000.00	1	150,000.00		
Refractometer and accessories	100,000.00	1	100,000.00		
Dessicator and accessories	250,000.00	1	250,000.00		
electronic image analysis system and accessories	150,000.00	1	150,000.00		

shaker and accessories	500,000.00	1	500,000.00		
Multiparameter and accessories for water sampling	400,000.00	1	400,000.00		
refrigerated centrifuge with different rotors and accessories	1,500,000.00	1	1,500,000.00		
TOTAL (PhP)			**8,970,000.00**		**0.00**

JUSTIFICATION: Describe the need or purpose of the equipment (e.g. how each unit will be used).

Equipment	Purpose	User	Location of the equipment
Oven drier and accessories	Will be used for sample drying and pretreatment procedures.	For use of the research staff in Partido State University during the execution of the grant.	Will be housed and maintained at PSU.
analytical balance and accessories	Will be used for determination of sample weight.	For use of the research staff in Partido State University during the execution of the grant.	Will be housed and maintained at PSU.
toploading balance and accessories	Will be used for determination of sample weight.	For use of the research staff in Partido State University during the execution of the grant.	Will be housed and maintained at PSU.
furnace and accessories	Will be used for drying/carbon content dtermination.	For use of the research staff in Partido State University during the execution of the grant.	Will be housed and maintained at PSU.
filtration system (vacuum pump, manifold, and accessories)	Will be used for sample processing and pretreatment procedures.	For use of the research staff in Partido State University during the execution of the grant.	Will be housed and maintained at PSU.
UV-Vis spectrophotometer and accessories	Will be used for determination of target analytes using absorbance spectroscopy.	For use of the research staff in Partido State University during the execution of the grant.	Will be housed and maintained at PSU.
Spectrofluorometer and accessories	Will be used for determination of target analytes using fluorescence spectroscopy.	For use of the research staff in Partido State University during the execution of the grant.	Will be housed and maintained at PSU.
Pump, motor with accessories	Will be used for sampling.	For use of the research staff in Partido State University during the execution of the grant.	Will be housed and maintained at PSU.
data loggers and accessories	Will be used for data storage/collection during sampling.	For use of the research staff in Partido State University during the execution of the grant.	Will be housed and maintained at PSU.
scuba diving sets and accessories	Will be used for sampling.	For use of the research staff in Partido State University during the execution of the grant.	Will be housed and maintained at PSU.
Compound microscope and accessories	Will be used for documentation especially looking at details of samples.	For use of the research staff in Partido State University during the execution of the grant.	Will be housed and maintained at PSU.
inverted microscope and accessories	Will be used for documentation especially looking at details of samples.	For use of the research staff in Partido State University during the execution of the grant.	Will be housed and maintained at PSU.
Stereomicroscope and accessories	Will be used for documentation especially looking at details of samples.	For use of the research staff in Partido State University during the execution of the grant.	Will be housed and maintained at PSU.
Mechanized rotating drum mixer and accessories	Will be used for sample processing	For use of the research staff in Partido State University during the execution of the grant.	Will be housed and maintained at PSU.
Gopro and underwater camera and accessories	For documentation of samples and sampling processes	For use of the research staff in Partido State University during the execution of the grant.	Will be housed and maintained at PSU.
tabletop centrifuge and accessories	Will be used for sample processing and separation.	For use of the research staff in Partido State University during the execution of the grant.	Will be housed and maintained at PSU.

pH meter, vortex mixer, hotplate, and accessories	Wil be used for pH measurements, sample processing, and temperature-controlled experiments	For use of the research staff in Partido State University during the execution of the grant.	Will be housed and maintained at PSU.
Refractometer and accessories	For measurement of salinity during sampling	For use of the research staff in Partido State University during the execution of the grant.	Will be housed and maintained at PSU.
Dessicator and accessories	For use in drying and sample processing steps	For use of the research staff in Partido State University during the execution of the grant.	Will be housed and maintained at PSU.
electronic image analysis system and accessories	For analysis of images collated during sampling	For use of the research staff in Partido State University during the execution of the grant.	Will be housed and maintained at PSU.
shaker and accessories	Will be used for sample processing and pre-treatment steps	For use of the research staff in Partido State University during the execution of the grant.	Will be housed and maintained at PSU.
Multiparameter and accessories for water sampling	Will be used for routine measurements of physico-chemical parameters during sampling.	For use of the research staff in Partido State University during the execution of the grant.	Will be housed and maintained at PSU.
refrigerated centrifuge with different rotors and accessories	Will be used for samples processing for pre-concentration or separation purposes.	For use of the research staff in Partido State University during the execution of the grant.	Will be housed and maintained at PSU.

Supplies and Materials for PH side: PSU

Item(s)	Cost/unit (PhP)	YR 1		YR 2	
		No. of units	Sub-Total for YR 1	No. of units	Sub-Total for YR 1
Field and sample preservation materials	~1,000.00 to 5,000.00	~30	200,000.00	~30	200,000.00
General laboratory chemicals	~1,000.00 to 4,000.00	~50	100,000.00	~50	100,000.00
Laboratory glassware/plastic ware and other materials	~1,000.00 to 5,000.00	~40	100,000.00	~40	100,000.00
TOTAL (PhP)			**400,000.00**		**400,000.00**

JUSTIFICATION:

Field and sample preservation materials: The field and sampling materials, which include various items from ropes, non-slip mats, pails, etc., will be used for sample collection and processing while on the field.

General laboratory chemicals: these chemicals are required in sample pretreatment, clean-up, preconcentration, or cleaning steps in the laboratory.

Laboratory glassware/plastic ware and other materials: these are items needed as sample containers, reaction vessels, transfer and storage containers, and other routine laboratory procedures.

Indirect Cost Rate: PSU

	BUDGET ITEM	Budget Requirement YR 1	YR 2
I. PERSONNEL SERVICES			
1	Personnel Salary	795,120.00	795,120.00
2	Honoraria	90,000.00	90,000.00
	Sub-Total for PS	**885,120.00**	**885,120.00**
II. MOOE			
1	Travel Expenses		
2	Supplies and Materials	400,000.00	400,000.00

Budget and Justification: PhycoPRO – UCSD

A. Personnel: List employee(s) (including names for each identified position) of the applicant/recipient organization, including in-kind costs for those positions whose work is tied to the grant project.

Position	Name	Annual Salary/Rate	Level of Effort	Cost for Year 1	Cost for Year 2	Total Cost for 2 Years
(1) Collaborator	William Gerwick	105,600.00	12.5%	2,922,536.82	3,030,961.94	5,953,525.76
(2) Collaborator	Lena Gerwick	90,000.00	12.5%	1,026,447.84	1,060,097.90	2,086,545.74
(3) Postdoctoral Fellow	Harpreet Singh	90,000.00	100%	3,667,069.56	3,986,043.60	7,653,113.16
(4) SRA	Evgenia Glukhov	549,360.00	16.67%	877,198.05	917,784.86	1,794,979.92
			TOTAL	8,493,276.27	8,994,888.31	17,488,164.58

JUSTIFICATION: Describe the role and responsibilities of each position.

(1) and (2): Both collaborators, William and Lena Gerwick, will provide general research directions and supervision of research staff based at the Scripps Institution of Oceanography. The expertise of William and Lena in natural products chemistry and bio-assay guided isolation and elucidation of chemical constituents of seaweeds and associated epiphytes will be critical for the success of the project. It is also expected that they will impart valuable strategies in terms of experimental design pertaining to isolation and chemical elucidation of bioactive compounds. Both William and Lena are considered key staff.

(3) : The Postdoctoral fellow will perform most of the research activities pertaining to isolation and chemical elucidation of potential bioactive compounds from seaweed extracts and will be considered key staff.

(4) : The SRA will assist in all research activities and will be considered key staff. In addition to research activities, the SRA will also assist in administrative tasks and coordination of project deliverables with the PH-based project team.

Key staff positions require prior approval after review of credentials of resume and job description.

B. Travel : Explain need for all travel other than that required by this application. Relevant Philippine government policies for travel such as Executive Order 77, series of 2019 on Official Local and Foreign Travels shall apply.

NONE

C. Equipment (for Philippines side only):

NONE

D. Supplies and Materials: UCSD

Item(s)	Cost/unit (PhP)	YR 1		YR 2	
		No. of units	Sub-Total for YR 1	No. of units	Sub-Total for YR 1
Reagents and materials for sample identification and preliminary testing	~13,750.00	~75	825,000.00		
Chemicals for analysis and evaluation of extracts for activity and other parameters	~11,183.33	~75	671,000.00		
Certified reference materials for identification of isolated compounds with activity	~38,500.00	~15	577,500.00		
Reagents for sample analyses of various parameters	~6,600.00			~150	990,000.00
Chemicals and reagents for analysis of seaweed extracts	~5,516.14			~150	827,420.00
TOTAL (PhP)			**2,073,500.00**		**1,817,420.00**

JUSTIFICATION:

Reagents and materials for sample identification and preliminary testing will include various solvents, test kits, and use of sophisticated equipment for sample processing. The costs are projected for assays/analyses of about 75 samples for screening.

Chemicals for analysis and evaluation of extracts for activity and other parameters will include all chemicals (standards, assay kits, chemicals for quality control checks) and consumables for use in instruments for sample processing. The costs are estimated for about 75 purified or pre-concentrated extracts for bioassay guided isolation and further activity screening.

Certified reference materials will include all costs for procurement of compounds with known activity or identified compounds with similar activity for comparison of retention times or mass spectra and will serve as surrogate or internal standards during chromatographic separations and chemical structure elucidation and bioactivity screening (for positive or negative controls).

Reagents for sample analyses of various parameters will include all general laboratory chemicals for routine isolation, separation, and purification or preconcentration of potential bioactive compounds from seaweeds and associated epiphytes following an optimized protocol. The costs are estimated for about 150 samples that will be processed for the screening.

Chemicals and reagents for analysis of seaweed extracts will include all solvents, assay kits, and other consumables needed for routine screening of purified or preconcentrated extracts from seaweed extracts. The costs are estimated for about 150 extracts that will be processed for the screening.

E. Professional Services: These include costs of services such as legal, consultancy, environment/sanitary, general, and other professional services required by the project. A contractual arrangement may be entered into to engage the services of service providers to carry out a portion of the project effort or for the acquisition of routine goods or services under the project. Such arrangements may be in the form of consortium agreements or contracts. A consultant is an individual retained to provide professional advice or services for a fee. All procurement transactions shall be conducted in a manner to provide to the maximum extent practical, open and free competition.

NONE

Budget Justification
Project Title: Lakas 2021-035: PHYTOCHEMICAL CHARACTERIZATION OF MACROALGAE FOR FOOD AND HIGH VALUE PRODUCTS (PHYCOPRO)

F. Construction: NOT ALLOWED

G. Maintenance and Other Operating Expenses (MOOE) may include the following expenses not covered in any of the previous budget categories

Item(s)	Cost for YR 1	Cost for YR 2	TOTAL
Communication expenses			
Communication expenses during project implementation	147,015.00	157,850.00	204,865.00
Repairs and Maintenance of Facilities			
Maintenance of clean room			
Renovation cost			
Transportation and Delivery Expenses			
Freight costs			
Training Expenses			
Membership Dues for the Institution			
		-	
Rent Expenses			
Boats for sampling			
Representation Expenses			
Subscription expenses			
Professional fees			
Taxes, Insurance Premiums, and Other Fees			
Other Maintenance and Operating Expenses			
	487,740.00	715,000.00	1,202,740.00
TOTAL	**634,755.00**	**872,850.00**	**1,507,605.00**

JUSTIFICATION:

Communication expenses will cover cost for availing of internet connectivity and purchasing of SIM and prepaid cards for use of project staff in coordinating sampling activities especially during travel to the Philippines of the UCSD research staff.

Other expenses include fees for access to mass spectral libraries, library/databases of bioactivity and biochemical properties, and inter-calibration purposes that may be required for some analytical methods and a budget for incidentals.

H. Administrative Overhead Cost (AOC):
A maximum fixed AOC of 20% of total PS and MOOE items shall apply.

Budget Justification
Project Title: Lakas 2021-035: PHYTOCHEMICAL CHARACTERIZATION OF MACROALGAE FOR FOOD AND HIGH VALUE PRODUCTS (PHYCOPRO)

BUDGET SUMMARY:

BUDGET ITEM	BUDGET REQUIREMENT		
	YR 1 ___	YR 2	TOTAL
I. PS	8,493,276.27	8,994,888.31	17,488,164.58
II. MOOE	2,708,225.00	2,690,270.00	1,507,605.00
III. EO	0.00	0.00	0.00
IV. AOC	2,240,306.25	2,337,031.66	4,577,337.93
TOTAL PROPOSED BUDGET	13,441,837.53	14,022,189.97	27,464,027.49

Line Item Budget

COMMISSION ON HIGHER EDUCATION

LINE ITEM BUDGET: UP-MSI

Project Title: **Phytochemical characterization of macroalgae for food and high value products (PhycoPRO)**
Project Duration: TWO Years
Implementing Institution/s: UP Marine Science Institute,Mindanao State University-Naawan,Partido State University, Sorsogon State University, and University of Eastern Philippines

	ITEMS		Total
I.	**PERSONNEL SERVICES (PS)**		
	Direct Cost	P	10,119,504.00
	Sub-total for PS	P	
II.	**MAINTENANCE AND OTHER OPERATING EXPENSES (MOOE)**		
	Direct Cost		
	Travel Expenses		6,053,000.00
	Supplies and Materials		2,500,000.00
	Communication Expenses		40,000.00
	Repairs and Maintenance of Facilities		550,000.00
	Transportation and Delivery Expenses		300,000.00
	Utilities Expenses		
	Training Expenses		14,365,000.00
	Membership Dues for the Institution		
	Rent Expenses		250,000.00
	Representation Expenses		160,083.54
	Subscription Expenses		70,000.00
	Survey Expenses		
	Professional Services		800,000.00
	Taxes, Insurance Premiums and Other Fees		250,000.00
	Other Maintenance and Operating Expenses		770,000.00
	Instrument fee for inter-calibration purposes (120,000)		
	Publication expenses (650,000)		
	Sub-total for MOOE	P	**26,108,083.54**
III.	**EQUIPMENT OUTLAY (EO)** - *all in: include customs duties, etc*		
		P	66,750,000.00
	Sub-total for EO	P	**66,750,000.00**
IV.	**ADMINISTRATIVE OVERHEAD COST (AOC)**		
	AOC for PS	P	2,023,900.80
	AOC for MOOE		5,221,616.71
	Sub-total for AOC	P	**7,245,517.51**
	GRAND TOTAL	P	**110,223,105.05**

Prepared and Submitted by:

Michael Y. Roleda, Dr. rer. nat.
(Project Leader)

Irene B. Rodriguez, Dr. rer. nat.
(Co-Project Leader)

Line Item Budget

COMMISSION ON HIGHER EDUCATION

LINE ITEM BUDGET: MSUN

Project Title: **Phytochemical characterization of macroalgae for food and high value products (PhycoPRO)**
Project Duration: **TWO Years**
Implementing Institution/s: UP Marine Science Institute,Mindanao State University-Naawan,Partido State University, Sorsogon State University, and University of Eastern Philippines

	ITEMS		Total
I.	**PERSONNEL SERVICES (PS)**		
	Direct Cost	P	1,770,240.00
	Sub-total for PS	**P**	
II.	**MAINTENANCE AND OTHER OPERATING EXPENSES (MOOE)**		
	Direct Cost		
	Travel Expenses		
	Supplies and Materials		900,000.00
	Communication Expenses		
	Repairs and Maintenance of Facilities		
	Transportation and Delivery Expenses		
	Utilities Expenses		
	Training Expenses		
	Membership Dues for the Institution		
	Rent Expenses		
	Representation Expenses		
	Subscription Expenses		
	Survey Expenses		
	Professional Services		
	Taxes, Insurance Premiums and Other Fees		
	Other Maintenance and Operating Expenses		
	(Provide account)		
	Sub-total for MOOE	**P**	**900,000.00**
III.	**EQUIPMENT OUTLAY (EO)** *- all in: include customs duties, etc*		
		P	8,450,000.00
	Sub-total for EO	**P**	**8,450,000.00**
IV.	**ADMINISTRATIVE OVERHEAD COST (AOC)**		
	AOC for PS	P	354,048.00
	AOC for MOOE		180,000.00
	Sub-total for AOC	**P**	**534,048.00**
	GRAND TOTAL	**P**	**11,654,288.00**

Prepared and Submitted by:

Rey Y. Capangpangan, Ph.D.
(Project Collaborator, MSU-Naawan)

Line Item Budget

COMMISSION ON HIGHER EDUCATION

LINE ITEM BUDGET: ParSU

Project Title: Phytochemical characterization of macroalgae for food and high value products (PhycoPRO)
Project Duration: TWO Years
Implementing Institution/s: UP Marine Science Institute,Mindanao State University-Naawan,Partido State University, Sorsogon State University, and University of Eastern Philippines

	ITEMS		Total
I.	**PERSONNEL SERVICES (PS)**		
	Direct Cost	P	1,770,240.00
	Sub-total for PS	P	
II.	**MAINTENANCE AND OTHER OPERATING EXPENSES (MOOE)**		
	Direct Cost		
	Travel Expenses		
	Supplies and Materials		800,000.00
	Communication Expenses		
	Repairs and Maintenance of Facilities		
	Transportation and Delivery Expenses		
	Utilities Expenses		
	Training Expenses		
	Membership Dues for the Institution		
	Rent Expenses		
	Representation Expenses		
	Subscription Expenses		
	Survey Expenses		
	Professional Services		
	Taxes, Insurance Premiums and Other Fees		
	Other Maintenance and Operating Expenses		
	(Provide account)		
	Sub-total for MOOE	P	**800,000.00**
III.	**EQUIPMENT OUTLAY (EO)** *- all in: include customs duties, etc*		
		P	8,450,000.00
	Sub-total for EO	P	**8,970,000.00**
IV.	**ADMINISTRATIVE OVERHEAD COST (AOC)**		
	AOC for PS	P	354,048.00
	AOC for MOOE		160,000.00
	Sub-total for AOC	P	**514,048.00**
	GRAND TOTAL	P	**12,054,288.00**

Prepared and Submitted by:

Michael A. Clores, Ph.D.
(Project Collaborator, ParSU)

Line Item Budget

COMMISSION ON HIGHER EDUCATION

LINE ITEM BUDGET: SorSU

Project Title: Phytochemical characterization of macroalgae for food and high value products (PhycoPRO)
Project Duration: TWO Years
Implementing Institution/s: UP Marine Science Institute,Mindanao State University-Naawan,Partido State University, Sorsogon State University, and University of Eastern Philippines

	ITEMS		Total
I.	**PERSONNEL SERVICES (PS)**		
	Direct Cost	P	1,770,240.00
	Sub-total for PS	**P**	1,770,240.00
II.	**MAINTENANCE AND OTHER OPERATING EXPENSES (MOOE)**		
	Direct Cost		
	Travel Expenses		
	Supplies and Materials		1,000,000.00
	Communication Expenses		
	Repairs and Maintenance of Facilities		
	Transportation and Delivery Expenses		
	Utilities Expenses		
	Training Expenses		500,000.00
	Membership Dues for the Institution		
	Rent Expenses		
	Representation Expenses		
	Subscription Expenses		
	Survey Expenses		
	Professional Services		
	Taxes, Insurance Premiums and Other Fees		
	Other Maintenance and Operating Expenses		
	(Provide account)		
	Sub-total for MOOE	**P**	**1,500,000.00**
III.	**EQUIPMENT OUTLAY (EO)** *- all in: include customs duties, etc*		
		P	8,450,000.00
	Sub-total for EO	**P**	**8,750,000.00**
IV.	**ADMINISTRATIVE OVERHEAD COST (AOC)**		
	AOC for PS	P	354,048.00
	AOC for MOOE		300,000.00
	Sub-total for AOC	**P**	**654,048.00**
	GRAND TOTAL	**P**	**12,674,288.00**

Prepared and Submitted by:

Richard V. Dumilag, Ph.D.
(Project Collaborator, SorSU)

Line Item Budget

COMMISSION ON HIGHER EDUCATION

LINE ITEM BUDGET: UEP

Project Title: Phytochemical characterization of macroalgae for food and high value products (PhycoPRO)
Project Duration: TWO Years
Implementing Institution/s: UP Marine Science Institute

	ITEMS		Total
I.	**PERSONNEL SERVICES (PS)**		
	Direct Cost	P	1,770,240.00
	Sub-total for PS	**P**	
II.	**MAINTENANCE AND OTHER OPERATING EXPENSES (MOOE)**		
	Direct Cost		
	Travel Expenses		
	Supplies and Materials		600,000.00
	Communication Expenses		
	Repairs and Maintenance of Facilities		
	Transportation and Delivery Expenses		
	Utilities Expenses		
	Training Expenses		500,000.00
	Membership Dues for the Institution		
	Rent Expenses		
	Representation Expenses		
	Subscription Expenses		
	Survey Expenses		
	Professional Services		
	Taxes, Insurance Premiums and Other Fees		
	Other Maintenance and Operating Expenses		
	(Provide account)		
	Sub-total for MOOE	**P**	**1,100,000.00**
III.	**EQUIPMENT OUTLAY (EO)** *- all in: include customs duties, etc*		
		P	8,450,000.00
	Sub-total for EO	**P**	**5,540,000.00**
IV.	**ADMINISTRATIVE OVERHEAD COST (AOC)**		
	AOC for PS	P	354,048.00
	AOC for MOOE		220,000.00
	Sub-total for AOC	**P**	**574,048.00**
	GRAND TOTAL	**P**	**8,984,288.00**

Prepared and Submitted by:

Ronelie C. Salvador

Ronelie C. Salvador, Ph.D.

(Project Collaborator, UEP)

Narrative Proposal
Project Results Workplan/Matrix

Project Title: Phytochemical characterization of macroalgae for food and high value products (PhycoPRO)

Total Funded Duration: TWO Years

CALENDAR YEAR AND QUARTER			YEAR 1				YEAR 2			
			Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Project Duration (in months per quarter; for 24 months)										
OBJECTIVES	**EXPECTED OUTPUT**	**Metrics used to track and measure progress of work leading to output.**	**Quantified relevant % progress**							
Part 1 – Planning										
1.1 Formally initiate project with kickoff meeting	Detailed project work plan	Project plan with details of deliverables.	100%							
1.2 Build project team	Team staffing report	Hiring of research team members	100%							
1.3 Procure equipment	Equipment purchased	Percentage of equipment purchased	10%	50%	75%	100%				
1.4 Procure chemicals, laboratory and office supplies	Chemicals, laboratory and office supplies purchased	Percentage of chemicals, laboratory and office supplies purchased	10%	50%	75%	100%				
1.5 Identify sampling sites and establish local contacts	Sampling sites and local contacts identified	List of sampling sites and local contacts	100%							
1.6 Obtain necessary permits/licenses from pertinent government agencies	Permits/licenses acquired	List of licenses/permits obtained	50%	100%						
1.7 Document progress accomplishment	Annual progress report	Delivery of annual progress report				100%				100%
Part 2 – Seaweed collection for herbarium collection and method optimization										
2.1 Seaweed collection for optimization of methods	Seaweed samples collected	Report on seaweed collection	25%	100%						
2.2 Deposit samples in herbarium	Seaweed samples deposited in herbarium	Report on materials deposited in herbarium		100%						
2.3 Pre-treatment of samples for method optimization	Pre-treatment methods optimized and validated	Report on pre-treatment methods								

Part 3 – Optimization and validation of methods										
3.1 Optimize method for proximal analysis	Optimized method for proximal analysis	Report on validated method		50%	100%					
3.2 Optimize method for total lipid, fucoxanthin, and fucosterol analyses	Optimized method for lipid, fucoxanthin, and fucosterol analyses	Report on validated method		30%	60%	80%	100%			
3.3 Optimize method for fatty acid extraction and analysis	Optimized method for fatty acid extraction and analysis	Report on validated method		30%	60%	80%	100%			
3.4 Optimize method for amino acid analysis and specific component MAAs	Optimized method for amino acid and targeted MAAs analyses	Report on validated method		30%	60%	80%	100%			
3.5 Optimize method for β-carotene extraction and analysis	Optimized method for β-carotene extraction and analysis	Report on validated method		30%	60%	80%	100%			
3.6 Optimize method for iodine analysis and estimation of iodine intake from seaweeds	Optimized method for determination of iodine content and intake from seaweeds	Report on validated method			30%	60%	80%	100%		
3.7 Optimize method for analysis of phenolic content	Optimized method for analysis of phenolic content	Report on validated method			30%	60%	80%	100%		
3.8 Optimize method for determination of antioxidant activity	Optimized method for antioxidant activity determination	Report on validated method			30%	60%	80%	100%		
3.9 Optimize method for total metal analysis	Optimized method for trace metal determination	Report on validated method			30%	60%	80%	100%		
3.10 Optimize method for speciation analysis of specific trace metals of interest	Optimized method for speciation analysis of target trace metals	Report on validated method			30%	60%	80%	100%		
3.11 Optimize in-vitro dialyzability procedure to determine bioavailability and bioavailability of trace metals	Optimized method for in-vitro dialyzability procedure	Report on validated method			30%	60%	80%	100%		
Part 4 – Screening of seaweed and associated epiphytes for bioactivities										
4.1 Training of staff for seaweed handling and preservation for drug discovery related studies	Research staff trained	Report on training conducted	25%	100%						

4.2 Extraction for drug discovery related work, sending of materials to CA	Materials sent to CA collaborator	Report on materials sent to CA collaborator		25%	50%	100%				
4.3 Analysis and evaluation of extracts for activity and other parameters	Extracts evaluated for bioactivity	Results incorporated in manuscript draft				50%			100%	
4.4 Identification of isolated compounds with activity	Compounds identified or characterized	Results incorporated in manuscript draft				25%		50%		100%
4.5 Training of staff for isolation, purification, and identification of potential compounds	Staff trained in relevant processes	Protocols for relevant processes prepared				25%				100%
Part 5 – Seaweed collection and evaluation of phytochemical parameters										
5.1 Sampling for identification and preliminary testing	Seaweed samples collected	Report on Seaweed species identification		50%	100%					
5.2 Deposit samples in herbarium	Seaweed samples deposited in herbarium	Report on materials deposited in herbarium		100%						
5.3 Sampling for analysis of various parameters	Seaweed samples collected	Report on seaweed species collected					20%	50%	100%	
5.4 Analyses of seaweed and seaweed extracts using optimized methods	Various parameters analyzed in seaweed samples	Report on data acquired						20%	80%	100%
5.5 Data processing and analyses	Data collected	Report on data acquired						10%	70%	100%
Part 6 – Training/Workshop on seaweed farming										
6.1 Preparation of seaweed farming information materials	Information materials prepared	Prepared information materials for dissemination	25%	75%	100%					
6.2 Conduct of training/workshop for seaweed farmers	Training/workshops carried out with partner HEIs for seaweed farmers	Report on the training/workshop conducted		10%	20%	40%	60%	80%	100%	
Part 7 – Post-implementation										
7.1 Write research papers	Research papers	Number of research papers							50%	100%
7.2 Project evaluation	Final project report	Final project report								100%

2021-0621-1111-3115
UNIVERSITY OF THE PHILIPPINES
3F, Quezon Hall, Diliman, Quezon City
Tele/Fax: (632) 8962-6345; 89818722; 89818500 local 2528
Email: ovpaa@up.edu.ph

Office of the Vice President for Academic Affairs

9 August 2021

HON. J. PROSPERO E. DE VERA III, DPA
Chairman
Commission on Higher Education
C.P. Garcia Ave., Quezon City

Subject: Endorsement of CHED-PCARI proposal of IHITM 2018-035: "Phytochemical Characterization of Macroalgae for Food and High Value Products (PhycoPRO)" headed by Dr. Michael Y. Roleda (UP Diliman) and Dr. William Gerwick (UC San Diego)

Dear Chairman De Vera,

May we respectfully inform you that the UP Board of Regents, in its 1362nd meeting last 29 July 2021, approved the CHED-PCARI project proposal titled, *"Phytochemical Characterization of Macroalgae for Food and High Value Products (PhycoPRO)"* to be implemented by the University of the Philippines Diliman (Dr. Michael Roleda) in collaboration with the University of California San Diego (Dr. William Gerwick).

The PhycoPRO project is envisioned to enhance the status of the seaweed industry in the Philippines by providing critical knowledge through a comprehensive phytochemical characterization of seaweed species with high economic potential. This study will identify and prioritize species that have sufficient natural standing stock and species that are culturable or a cultivation technology can be developed and capitalize on their unique chemistry for commercial applications.

Following your instructions, we subjected the proposal to an expert reviewer who agrees with the CHED-PCARI technical reviewers that the proposal is 'good to go'! It is very well written, of profound significance to the livelihood of seaweed farmers and is backed by the impressive credentials, experience, and expertise of the Co-Project Investigators. AVPAA for Research Carla Dimalanta and her team also reviewed the line-item-budget (LIB) of the project and, except for two minor comments, which were relayed to the proponents found the LIB in order.

Attached herewith are the copies of the BOR certification, revised project proposal, LIB, and budget justification.

Thank you.

Truly yours,

Maria Cynthia Rose Banzon Bautista
Vice-President for Academic AffairsUniversity of the Philippines

Republic of the Philippines
Mindanao State University at Naawan
9023 Naawan, Misamis Oriental
http://www: msunaawan.edu.ph

Office of the Chancellor

March 28, 2022

PCARI Advisory Group and
J. PROSPERO E. DE VERA III, DPA
Chairperson
Commission on Higher Education
Higher Education Development Center (HEDC) Building
C. P. Garcia Avenue, Diliman, Quezon City

Dear Sir:

Greetings of peace and good will!

This letter is in reference to the proposed project of **Dr. Irene B. Rodriguez and Dr. Michael Y. Roleda** from UP MSI entitled, ***"Phytochemical characterization of macroalgae for food and high value products (PhycoPRO)"*** for possible funding under the Commission on Higher Education Philippine-California Advanced Research Institutes (PCARI) Program. The overall goal of the project is to contribute to knowledge generation on the potential uses of seaweeds that will have societal and economic impacts to coastal communities in particular, and mankind in general.

We would like to express our great enthusiasm for being chosen by the UP- Marine Science Institute (UP MSI) as partner university for this project. We believe that this will enhance our capability in implementing collaborative research endeavors.

After series of discussions, MSU at Naawan signifies our commitment and support for the implementation of the project. Likewise, we commit to assist the project team in delivering the target deliverables and participate in any research activities called by the project leader.

Thank you very much and more power!

Sincerely yours,

ELNOR C. ROA, Ph.D.
Chancellor II

Republic of the Philippines
PARTIDO STATE UNIVERSITY
Camarines Sur

Office of the University President

March 28, 2022

HON. J. PROSPERO E. DE VERA III, DPA
Chairperson
Commission on Higher Education
Higher Education Development Center Building
C.P. Garcia Ave, Diliman, Quezon City

Through: **HON. ALDRIN A. DARILAG, Ph.D., CHRP, RMT, RN**
Commissioner, Commission on Higher Education

Sir:

This is to respectfully and formally express the commitment of Partido State University to participate as collaborating SUC in the CHED-PCARI program: **Phytochemical characterization of macroalgae for food and high value products (PhycoPRO).** The proponents of this program are Dr. Michael Y. Roleda and **Dr. Irene B. Rodriguez** (Balik Scientist in ParSU) from the Marine Science Institute University of the Philippines; and, Dr. William H. Gerwick and Dr. Lena S. Gerwick from the Scripps Institution of Oceanography University of California San Diego California, USA.

As such, believing in the overall goal of this program to contribute to knowledge generation on the potential uses of seaweeds that will have societal and economic impacts to coastal communities in particular, and mankind in general, **Partido State University will participate in all research activities and delivering outputs of the project.**

Thank you very much for this worthwhile opportunity.

Sincerely,

RAUL G. BRADECINA, Ph.D.
SUC President III

San Juan Bautista Street, Goa, Camarines Sur 4422
Telefax No: +639772785045; +639499872405; Email: op@parsu.edu.ph; Website: www.parsu.edu.ph

Republic of the Philippines
Sorsogon State University
Office of the President
Sorsogon City
Tel No. (056) 211-0200, email address: ssc@sorsogonstatecollege.edu.ph

23 February 2022

SORSOGON STATE UNIVERSITY
RELEASED
DATE: 02-24-22
TIME: 3:54PM
BY:
RECORDS SECTION

J. Prospero E. De Vera III, PhD
Chairman
Commission on Higher Education

Dear Chairman De Vera,

Greetings of peace!

This is to express our commitment as Collaborating Agency of the proposed project, **Phytochemical characterization of macroalgae for food and high value products (PhycoPRO)** lead by Dr Michael Y. Roleda and Irene B. Rodriguez of UP Diliman, The Marine Science Institute. Rest assured that we will exert full effort for its consummation.

Thank you very much.

Very truly yours,

HELEN R. LARA, Ed.D.
SUC President III
Sorsogon State University

Republic of the Philippines
UNIVERSITY OF EASTERN PHILIPPINES
University Town, Northern Samar, Philippines
Web: http://uep.edu.ph; Email: uepnsofficial@gmail.com

12 May 2022

J. PROSPERO E. DE VERA, III, DPA
Chairperson
Commission on Higher Education
Diliman 1101, Quezon City

Sir:

Greetings!

This is in connection to the proposed research project on **PHYTOCHEMICAL CHARACTERIZATION OF MACROALGAE FOR FOOD AND HIGH VALUE PRODUCTS (PhycoPRO)** to be implemented by the University of the Philippines - Marine Science Institute and University of California San Diego – Scripps Institution of Oceanography.

The University of Eastern Philippines through the Office of the University Board Secretary has submitted the said proposed research project to the Board of Regents for endorsement/clearance on the long-term commitment of this university to support basic and applied seaweed research.

This study will be beneficial to the university and its component communities as it will enhance the R&D capacity of UEP after the completion of the project, consider the feasibility of establishing a "seaweed nursery", and development of practical uses/applications and/or sustainability of the research beyond the duration.

We look forward to cooperating with your institution in the conduct of this research work. Thank you very much.

Very truly yours,

KARINA MILAGROS C. LIM, PhD
Vice President for
Research, Development,
and Extension
University of Eastern Philippines

OFFICE OF THE SECRETARY OF THE UNIVERSITY
UNIVERSITY OF THE PHILIPPINES

CERTIFICATION

This is to certify that during the **1362nd Meeting of the Board of Regents held on 29 July 2021**, the Board APPROVED the following:

"CHED-PCARI Project "Phytochemical Characterization of Macroalgae for Food and High Value Products (PhycoPRO)" by Dr. Michael Roleda and Dr. William Gerwick "

WITNESS MY HAND AND SEAL this 5th day of August 2021 at Diliman, Quezon City.

ROBERTO M.J. LARA
Secretary of the University
and of the Board of Regents

Quezon Hall, University of the Philippines, Diliman, Quezon City 1101, Philippines
Tel. # (632) 927 67 80, Telefax # (632) 920 68 97, 920 68 85, e-mail osu@up.edu.ph

Republic of the Philippines
MINDANAO STATE UNIVERSITY
Marawi City

OFFICE OF THE SECRETARY OF THE UNIVERSITY
AND OF THE BOARD OF REGENTS

EXCERPTS FROM THE MINUTES OF THE 259th MEETING OF THE BOARD OF REGENTS HELD ON JUNE 13, 2022 VIA ZOOM TELECONFERENCING AND FACE-TO-FACE AT HOTEL KIMBERLY MANILA, 770 PEDRO GIL ST., MALATE, MANILA

==========

x-x-x x-x-x x-x-x

"RESOLUTION NO. 239, S. 2022"

RESOLVED, that upon the recommendation of the President of the University and the Executive Committee of the Board, MSU at Naawan Naawan's project proposal entitled Phytochemical Characterization of Macroalgae to be submitted to the Commission on Higher Education, is hereby **APPROVED**.

APPROVED"

x-x-x x-x-x x-x-x

CERTIFIED TRUE AND CORRECT:

ATTY. SHIDIK T. ABANTAS
Secretary of the University
and of the Board of Regents

STA:rlf
24 June 2022
(Not valid without seal)

Republic of the Philippines
PARTIDO STATE UNIVERSITY
Goa, Camarines Sur

RESOLUTION PASSED DURING THE 84th REGULAR BOARD OF REGENTS MEETING HELD ON APRIL 4, 2022 AT CATANDUANES STATE UNIVERSITY

Resolution No. 34, s. 2022

WHEREAS, the Partido State University is mandated to undertake advanced studies, research, production and extension services, and provide leadership in these areas pursuant to Section 2 of RA 9029;

WHEREAS, to adhere to the mandate, Partido State University is currently hosting a Balik Scientist with specialization in marine biogeochemistry, metallomics, environmental biotechnology, and environmental analytical chemistry;

WHEREAS, the Balik Scientist is a Co-Project Leader in the project entitled, "Phytochemical characterization of macroalgae for food and high value products (PhycoPRO)", which was submitted to CHED-PCARI for possible funding;

WHEREAS, the PhycoPRO project is envisioned to enhance the status of the seaweed industry in the Philippines by providing critical knowledge through a comprehensive phytochemical characterization of seaweeds to identify species that have higher economic potential and pinpoint locations where their growth will be maximal;

WHEREAS, the Partido State University have faculty researchers working on seagrass ecosystems and are interested in pursuing seaweeds research to facilitate introduction of seaweed farming to coastal communities in Camarines Sur;

WHEREAS, the Partido State University is willing to commit on this project and to sustain the quest for knowledge, even after the duration of the CHED-PCARI funding;

NOW THEREFORE, BE IT RESOLVED AS IT HEREBY RESOLVED TO APPROVE THE COMMITMENT OF PARTIDO STATE UNIVERSITY TO PARTICIPATE AS COLLABORATING SUC IN THE CHED-PCARI PROJECT: PHYTOCHEMICAL CHARACTERIZATION OF MACROALGAE FOR FOOD AND HIGH VALUE PRODUCTS (PHYCOPRO), SUBJECT TO THE FINAL APPROVAL OF THE COMMISSION ON HIGHER EDUCATION.

ADOPTED this 4th day of April, 2022 during the 84th Regular Board of Regents Meeting at Catanduanes State University, Virac, Catanduanes.

I hereby certify to the correctness of the afore-going approved resolution:

LEIH ANNE R. ODOÑO
Board Secretary V

ATTESTED:

RAUL G. BRADECINA, Ph.D.
SUC President III

cRepublic of the Philippines
SORSOGON STATE UNIVERSITY
Sorsogon City

May 24, 2022

THE HONORABLE MEMBERS
BOARD OF REGENTS
SORSOGON STATE UNIVERSITY

THRU: **DR. ALDRIN A. DARILAG**
CHED Commissioner
Chairman, SorSU-BOR

SIR:

Greetings!

The Sorsogon State University in its commitment to realize its vision as a research university would like to pursue international research engagements which will significantly impact both in local and global scale. With this, SorSU is collaborating with the University of the Philippines, University of California San Diego and other State University and Colleges (SUCs) under the Philippine-California Advanced Research Institutes (PCARI) Project in the research titled Phytochemical characterization of macroalgae for food and higher value products (PhycoPRO).

This research aims to contribute to knowledge generation on the potential uses of seaweeds that will have societal and economic impacts on coastal communities and beyond. Its main objectives are: *(1) performing a phytochemical characterization of at least 50 seaweed species from the Philippines; (2) assess the potential anti-cancer activity of extracts and purified compounds from seaweed associated epiphytes;* and *(3) conduct training workshops and seminars on seaweed farming techniques, collection preservation methods, and propagation strategies for faculty and researchers of partner HEIs, farming communities, and other stakeholders.* As a collaborating SUC, Sorsogon State University will assist in conducting the workshops and training for seaweed farmers, which will aid them in properly harvesting and preservation procedures to maximize financial earning capacity.

With this, the University hereby respectfully submit by way of this referendum **the request for endorsement of the proposed long term commitment of SorSU in the collaborative initiative for the CHED-PCARI project: Phytochemical characterization of macroalgae for food and higher value products (PhycoPRO) for approval of the CHED Commission En Banc.**

May this request merit your consideration. Thank you very much.

Very truly yours,

HELEN R. LARA, EdD
SUC President III

Comments:

Approved ☑
Disapproved ☐

ALDRIN A. DARILAG, PhD
CHED Commissioner
Chairman, SorSU-BOR

Republic of the Philippines
SORSOGON STATE UNIVERSITY
Sorsogon City

REFERENDUM NO. __ s. 2022
May 24, 2022

SUBJECT MATTER:

Request for endorsement of the proposed long term commitment of SorSU in the collaborative initiative for the CHED-PCARI project: Phytochemical characterization of macroalgae for food and higher value products (PhycoPRO) for approval of the CHED Commission En Banc.

LEGAL BASES:

- Republic Act 11088 (Sorsogon State University Charter)
- Republic Act 8292 (Higher Education Modernization Act of 1997)
- Intellectual Property Code (RA No. 8293)
- Republic Act No. 9147 (Wildlife Act)

SorSU BOR Members	ACTION	SIGNATURE
DR. ALDRIN A. DARILAG CHED Commissioner and Chair of the Board		
DR. HELEN R. LARA SorSU President and Vice-Chair of the Board	Approved	
CONG. EVELINA G. ESCUDERO House of Representatives Committee on Higher and Technical Education	Approved	
HON. MA. BERNADETTE E. QUIRANTE Senate, Committee on Education, Arts and Culture	Approved	
DIR. AGNES E. TOLENTINO NEDA ROV Director		
DIR. ROMMEL SERRANO DOST ROV Director		
DIR. RODEL TORNILLA DA ROV Executive Director		
DR. JHONNER RICAFORT Alumni Regent	Approved	
HON. VINCE P. JANORAS Student Regent	Approved	
(Vacant) Faculty Regent		

FOR APPROVAL OF THE BOARD

Request for endorsement of the proposed long term commitment of SorSU in the collaborative initiative for the CHED-PCARI project: Phytochemical characterization of macroalgae for food and higher value products (PhycoPRO) for approval of the CHED Commission En Banc.

CSW FORM

A. Proponent: Office of the President

B. Action Requested:

- For Approval of the Board

C. Rationale/ Background Information

The Sorsogon State University in its commitment to realize its vision as a research university would like to pursue international research engagements which will significantly impact both in local and global scale. With this, SorSU is collaborating with the University of the Philippines, University of California San Diego and other State University and Colleges (SUCs) under the Philippine-California Advanced Research Institutes (PCARI) Project in the research titled Phytochemical characterization of macroalgae for food and higher value products (PhycoPRO).

This research aims to contribute to knowledge generation on the potential uses of seaweeds that will have societal and economic impacts on coastal communities and beyond. Its main objectives are: *(1) performing a phytochemical characterization of at least 50 seaweed species from the Philippines; (2) assess the potential anti-cancer activity of extracts and purified compounds from seaweed associated epiphytes;* and *(3) conduct training workshops and seminars on seaweed farming techniques, collection preservation methods, and propagation strategies for faculty and researchers of partner HEIs, farming communities, and other stakeholders.* As a collaborating SUC, Sorsogon State University will assist in conducting the workshops and training for seaweed farmers, which will aid them in properly harvesting and preservation procedures to maximize financial earning capacity.

With this, the University hereby respectfully submit by way of this referendum the request for endorsement of the proposed long term commitment of SorSU in the collaborative initiative for the CHED-PCARI project: Phytochemical characterization of macroalgae for food and higher value products (PhycoPRO) for approval of the CHED Commission En Banc.

D. Supporting Documents

- SorSU Letter of Commitment as Collaborating Agency
- PCARI Project Form No. 3b: Narrative Proposal
- Summary of the involvement of the collaborating SUC in the implementation of the activities under the PhycoPRO
- Highlights of the Minutes of the Meeting with CHED Comm. Darilag & Phycopro Proponents *(February 18, 2022)*

E. Legal Bases

- Republic Act 11088 (Sorsogon State University Charter)
- Republic Act 8292 (Higher Education Modernization Act of 1997)
- Intellectual Property Code (RA No. 8293)
- Republic Act No. 9147 (Wildlife Act)

F. Proposed BOR Resolution

Resolution No. __ s, 2022

Request for endorsement of the proposed long term commitment of SorSU in the collaborative initiative for the CHED-PCARI project: Phytochemical characterization of macroalgae for food and higher value products (PhycoPRO) for approval of the CHED Commission En Banc.

G. Certification

I hereby certify to the accurateness and completeness of all the attached documents and that all pertinent University and Government rules and regulations regarding the matter are fully complied with.

HELEN R. LARA, EdD
President

Republic of the Philippines
Sorsogon State University
Office of the President
Sorsogon City
Tel No. (056) 211-0200, email address: ssc@sorsogonstatecollege.edu.ph

23 February 2022

SORSOGON STATE UNIVERSITY
RELEASED
DATE: 02-24-22
TIME: 3:54PM
BY:
RECORDS SECTION

J. Prospero E. De Vera III, PhD
Chairman
Commission on Higher Education

Dear Chairman De Vera,

Greetings of peace!

This is to express our commitment as Collaborating Agency of the proposed project, **Phytochemical characterization of macroalgae for food and high value products (PhycoPRO)** lead by Dr Michael Y. Roleda and Irene B. Rodriguez of UP Diliman, The Marine Science Institute. Rest assured that we will exert full effort for its consummation.

Thank you very much.

Very truly yours,

HELEN R. LARA, Ed.D.
SUC President III
Sorsogon State University

Republic of the Philippines
UNIVERSITY OF EASTERN PHILIPPINES
University Town, Northern Samar
Web: http://uep.edu.ph Email: uepnsofficial@gmail.com

BOARD OF REGENTS
REFERENDUM NO. 04, S. 2022

APPROVAL OF THE REQUEST OF THE UNIVERSITY FOR FAVORABLE SUPPORT/ENDORSEMENT PERTAINING TO THE IMPLEMENTATION OF THE PHYTOCHEMICAL CHARACTERIZATION OF MACROALGAE FOR FOOD AND HIGH VALUE PRODUCTS WITH A PROPOSED Budget OF P 79, 959, 718.14

WHEREAS, the request for **favorable support/endorsement pertaining to the implementation of the Phytochemical Characterization of Macroalgae for Food and high value products with a proposed budget of p 79, 959, 718.14** has been presented to the Honorable Board of Regents;

WHEREAS, the university commits to support the government's goal towards the enhancement of the status of the seaweed industry in the Philippines by providing critical knowledge through a comprehensive phytochemical characterization of seaweeds to identify species that have higher economic potential and pinpoint location where their growth will be maximal;

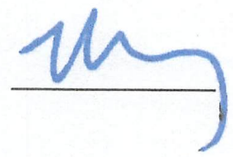
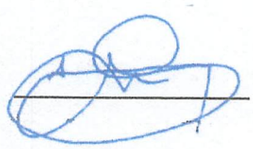
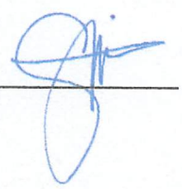
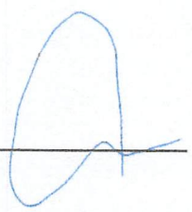
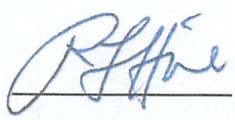
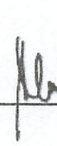
WHEREAS, that as a collaborating agency of the University of the Philippines-Diliman in the implementation of the Phytochemical Characterization Project of Macroalgae for Food and High Value Products amounting to **P 79, 959, 718.14** , the University of Eastern Philippines hereby expresses its favorable support to the said endeavor;

WHEREAS, that Honorable Board of Regents has reviewed the subject request and has found the same to be in order;

WHEREAS, the Administrative Committee endorsed the same for approval of the UEP Board of Regents through Finance Committee Referendum No. 01 Series of 2022;

NOW THEREFORE, BE IT RESOLVED AS IT IS HEREBY RESOLVED, THE APPROVAL VIA REFERENDUM OF THE UEP'S FAVORABLE SUPPORT TO THE IMPLEMENTATION OF THE PHYTOCHEMICAL CHARACTERIZATION PROJECT OF MACROALGAE FOR FOOD AND HIGH VALUE PRODUCTS AMOUNTING TO P 79, 959, 718.14.

DATE: **May 30, 2022**

	APPROVED	DISAPPROVED
HON. BONIFACIO G. UY Regional Director NEDA, Regional Office VIII		
HON. ERNESTO M. GRANADA Regional Director DOST, Regional Office VIII		
HON. SEARY ANNE C. BALLICUD President UEP Federation of Student Councils		
HON. ALBERT P. YRUMA President UEP Alumni Association Federation		
HON. ROMUALDO L. HIPE President UEP Federation of Faculty Unions		
HON. ANGEL C. ENRIQUEZ Regional Executive Director DA, Regional Field Office VIII		
HON. EMIL L. ONG (+) Representing: Hon. Mark O. GO Chairperson, Committee on Higher and Technical Education House of Representatives		

	APPROVED	DISAPPROVED
HON. DON L. ABALON Representing: Hon. Emmanuel Joel J. Villanueva Chairperson, Committee on Higher and Technical Education House of Representatives Senate of the Philippines		
HON. CHERRY I. ULTRA President, UEP Vice Chair, Board of Regents		
HON. ALDRIN A. DARILAG Commissioner, CHED Chairman, UEP Board of Regents		

Attested:

DR. NELIA M. ADORA
Board Secretary

Personal information:

Name:	Michael Y. Roleda
Contact Address:	Marine Science Institute, University of the Philippines
Email:	myroleda@up.edu.ph; michael.roleda@daad-alumni.de

Academic qualifications:

2005	Dr. rer. nat. (Biology), University of Bremen, Germany
1997	MSc (Marine Science), Marine Science Institute-University of the Philippines
1988	BSc (Biology), University of San Jose-Recoletos, Cebu, Philippines

Professional positions held:

2018–present	Professor, Marine Science Institute, University of the Philippines
2013–2018	Researcher, Norwegian Institute of Bioeconomy Research (NIBIO) [*previously Bioforsk Norwegian Institute for Agricultural and Environmental Research*]
2010–2013	Research Fellow/Lecturer, Botany, University of Otago, New Zealand
2009–2010	Algal Physiologist, Scottish Association for Marine Science, Scotland
2006–2009	Research Associate/Lecturer, Institute for Polar Ecology, Kiel, Germany
2005–2006	Postdoctoral Scientist, Alfred Wegener Institute, Bremerhaven, Germany
2001–2005	Researcher, Biologische Anstalt Helgoland, Helgoland, Germany
2000	Director, Marine Biological Station-DLSU, Batangas, Philippines
1997–2000	Associate Professor, De La Salle University (DLSU), Manila, Philippines
1992–1997	Research Associate, Marine Science Institute, University of the Philippines

Other positions:

- Associate Editor, Journal of Phycology (Wiley)
- Associate Editor, Marine Biology (Springer)
- Associate Editor, Frontiers in Marine Science
- Guest Editor, Frontiers in Marine Science; Frontiers in Microbiology

Research Interest:

Algal stress physiology; applied phycology; biodiversity; cultivation, selective breeding and hybridization; ecophysiology; ecotoxicology; photobiology; reproductive biology

Publication summary:

I have **91** ISI journal articles (33 first author, 2 single author, 15 senior/last author and 41 co-author). My H-index is 33 (source: WoS, https://apps.webofknowledge.com accessed 20.02.2022). I am a co-author of 6 book chapters and first author on one book chapter.

Other publications include eight peer-reviewed articles in non-ISI-journal papers, conference proceedings and scientific reports, and 4 scientific reports and extended abstracts. Participation in international and national conferences consist of 45 and 23 oral presentations, respectively, and 32 posters. I have been invited to present 20 public lectures, seminars, and workshop.

Teaching experience:

2018–present	University of the Philippines, Philippines
2010–2012	University of Otago, New Zealand
2006–2009	Christian Albrecht Universität zu Kiel, Germany
1997–2000	De La Salle University, Philippines

Graduate student supervision:

1. Bienson Ceasar V. Narvarte (MS Marine Science, 2020) Inorganic carbon utilization of tropical calcifying macroalgae and the survival of rhodolith *Sporolithon* sp. (Sporolithales: Rhodophyta) under long-term low pH conditions. Marine Science Institute, University of the Philippines.
2. Pablo Leal (PhD Botany, 2016) Effects of ocean acidification, temperature and copper on the development of early life stages of the native kelp *Macrocystis pyrifera* and the invasive kelp *Undaria pinnatifida* from southern New Zealand. Department of Botany, University of Otago, New Zealand. https://ourarchive.otago.ac.nz/handle/10523/6791
3. Pamela Fernandez (PhD Botany, 2015) The effects of ocean acidification on photosynthesis, growth and carbon and nitrogen metabolism of *Macrocystis pyrifera*. Department of Botany, University of Otago, New Zealand. https://ourarchive.otago.ac.nz/handle/10523/6227

4. Yuanyuan Feng (PhD Botany, 2014) Environmental controls on the physiology of the marine coccolithophore *Emiliania huxleyi*. Department of Botany, University of Otago, New Zealand. https://ourarchive.otago.ac.nz/handle/10523/5589

Current Research projects:

2022–2023 Selection and curation of safe and healthy eucheumatoid seedlings for the future (SecureFuture). Safe Seaweed Coalition

2021–2022 Selection of new strains of *Kappaphycus alvarezii* from wild individuals and their progeny. Sea6 Energy

2018-2022 Environmental controls on the physiology of different strains of Philippines eucheumatoids. *UP-MSI inhouse project.*

2020-2021 Growth performance of newly established *Kappaphycus alvarezii* seedstock from wild population. *Outright Research Grant, UP-Diiliman.*

2020–2021 Selection and distribution of new strains of Kappaphycus alvarezii developed from spores of local wild populations to restore seaweed farming productivity in the Philippines GSS/RF/047 GCRF Global Seaweed Star, UK Research and Innovation

2020-2022 Nutrient physiology of commercially important eucheumatoids: two-phase cultivation strategy to optimize biomass and carrageenan yield and quality. *Balik PhD, UP-System.*

2019-2021 Diversity of wild and farmed eucheumatoids in the Philippines at the intra- and inter-specific scale: population genetics of *K. alvarezii* and molecular taxonomy of eucheumatoids. *UK Research and Innovation, Global Challenges Research Fund.*

2019-2021 Diversity of and population genetics of wild *Kappaphycus alvarezii* in Guiuan, Samar, Philippines. *ECWRG, UP-System*

2019-2020 Predicting Responses between Ocean Transport and Ecological Connectivity of Threatened ecosystems in the WPS (PROTECT WPS): Seaweed Component. *DENR-BMB.*

2018-2021 Coastal Acidification: How it affects the Marine Environment and Resources in the Philippines. *DOST.*
Project 2: Impacts of ocean acidification on the base of the marine food web and their effects on marine production
Project 3: Possible influence of ocean acidification on specific reef resources

Service to the scientific community:

i. Reviewing journals:
2008-2020: 20-50 annual invitations to review manuscripts for top journals such as *Oikos*; *Global Change Biology*; *Limnology and Oceanography*; *Planta*; *Plant Cell and Environment*; *Marine Ecology-Progress Series*; *Oecologia*; *Marine Biology*; *New Phytologist*; *Marine Biotechnology*; *PloS One; Proceedings of the Royal Society B; Journal of Experimental Botany*; *Photochemistry and Photobiology*; *Scientific Reports*, among others.

ii. Reviewing international research grants:
Fonds de la Recherche Scientifique, Belgium; Estonian Research Council, Estonia; COST Action, European Cooperation in Science and Technology, EU; National Science Foundation, U.S.A. (NSF); Commission for Scientific and Technological Development, Chile (CONICYT); Deutsche Forschungsgemeinschaft, Germany (DFG); National Science Centre (Narodowe Centrum Nauki), Poland; Alaska Coastal Marine Institute, U.S.A. (CMI); Czech Science Foundation, Czech Republic (GAČR); The Netherlands Organization for Scientific Research (NOW)

PUBLICATION LIST (Top 10 cited publications)

1. Bartsch I, Wiencke C, Bischof K, Buchholz CM, Buck BH, Eggert A, Feuerpfeil P, Hanelt D, Jacobson S, Karez R, Karsten U, Molis M, **Roleda MY**, Schubert H, Schumann R, Valentin K, Weinberger F, Wiese J (2008) The genus *Laminaria* sensu lato: recent insights and developments. European Journal of Phycology 43: 1-86.
2. **Roleda MY**, Slocombe SP, Leakey RJG, Day JG, Bell EM, Stanley MS (2013) Effects of temperature and nutrient regimes on biomass and lipid production by six oleaginous microalgae in batch culture employing a two-phase cultivation strategy. Bioresource Technology 129: 439-449.
3. Boyd PW, Dillingham PW, McGraw CM, Armstrong EA, Cornwall CE, Feng Y, Hurd CL, Ringold-Gault M, **Roleda MY**, Timmins-Schiffman E, Nunn BL (2016) Physiological responses of a Southern Ocean diatom to complex future ocean conditions. Nature Climate Change 6: 207-213.

4. **Roleda MY**, Morris JN, McGraw CM, Hurd CL (2012) Ocean acidification and seaweed reproduction: increased CO_2 ameliorates the negative effect of lowered pH on meiospore germination in the giant kelp *Macrocystis pyrifera* (Laminariales, Phaeophyceae). Global Change Biology 18: 854-864.
5. Gómez I, Wulff A, **Roleda MY**, Huovinen P, Karsten U, Quartino ML, Dunton K, Wiencke C (2009) Light and temperature demands of benthic marine micro- and macroalgae in polar regions. Botanica Marina 52:593-608
6. **Roleda MY**, Boyd PW, Hurd CL (2012) Before ocean acidification: calcifier chemistry lessons. Journal of Phycology, 48: 840-843.
7. Holzinger A, **Roleda MY**, Lütz C (2009) The vegetative arctic freshwater green alga *Zygnema* is insensitive to experimental UV exposure. Micron 40:831-838.
8. Wiencke C, **Roleda MY**, Gruber A, Clayton MN, Bischof K (2006) Susceptibility of zoospores to UV radiation determines upper depth distribution limit of Arctic kelps: evidence through field experiments. Journal of Ecology 94: 455-463.
9. Duarte B, Martins I, Rosa R, Matos AR, **Roleda MY**, Reusch TBH, Engelen AH, Serrão EA, Pearson GA, Marques JC, Caçador I, Duarte CM, Jueterbock A (2018) Climate Change Impacts on Seagrass Meadows and Macroalgal Forests: An Integrative Perspective on Acclimation and Adaptation Potential. Frontiers in Marine Science 5:190.
10. Tatters AO, **Roleda MY**, Schnetzer A, Fu F, Hurd CL, Boyd PW, Caron DA, Lie AAY, Hoffmann LJ, Hutchins DA (2013) Short- and long-term conditioning of a temperate marine diatom community to acidification and warming. Philosophical Transactions of the Royal Society B: Biological Sciences 368: art. No. 20120437

Dr. Irene B. Rodriguez
The Marine Science Institute
University of the Philippines Diliman
Quezon City, Philippines
Email: ibrodriguez@msi.upd.edu.ph
ORCID ID: https://orcid.org/0000-0001-9848-9526
Lab website: https://msi-ibrlab.com/

Education

Dr. *rer. nat.* Chemistry	Karl-Franzens University Graz, Austria
MS Chemistry	University of the Philippines Diliman, Quezon City, Philippines
BS Chemistry	University of the Philippines Los Baños, Laguna, Philippines

Research and Teaching Experience

01/2019 – Present	**Associate Professor**, The Marine Science Institute, University of the Philippines Diliman (UP Diliman), Quezon City, Philippines
04/2012 – 12/2018	**Postdoctoral Research Fellow**, Research Center for Environmental Changes, Academia Sinica, Taipei, Taiwan
11/2008 – 5/2015	**Assistant Professor**, Institute of Chemistry, UP Diliman, Quezon City

Awarded Research Grants (2019 to present)

1. UK-RI: Selection and distribution of new strains of *Kappaphycus alvarezii* developed from spores of local wild populations to restore seaweed farming productivity in the Philippines (New Kappa Star); 2020-2021; Co-Project Leader; Budget: £ 49,995.17.
2. DOST-PCAARRD: Trace metals and nutrients from aerosol deposition: Impacts on community structure and occurrence of harmful algal blooms (MEDIATE-HABs); 2020-2023; Project Leader; Budget: PhP 4,783,784.00
3. DOST-PCAARRD: Macronutrient, carbon cycling, and trace metals: Impacts on phytoplankton community structure and toxin production of harmful algal blooms (Trace-HABs); 2020-2023; Project Leader; Budget: PhP 12,508,077.00
4. DOST-PCAARRD: Metallomics and Inorganic Biogeochemistry Facility for Research and Development; 2020-2022; Project Leader; Budget: PhP 19,850,000.00
5. UP-MSI: Investigating the role of nickel availability in the hydrogen production by the nitrogen-fixing cyanobacterium *Trichodesmium erythraeum*; 2020; Project Leader
6. UP-MSI: Influence of Fe availability and temperature variability on superoxide dismutase production of Symbiodiniaceae; 2019; Project Leader

Scholarly Development: Publications (representative)

1. Chen CC, Rodriguez IB, Chen YL, Zehr JP, Chen YR, Hsu SD, Yang SC, and Ho T-Y (2022). Nickel superoxide dismutase protects nitrogen fixation in *Trichodesmium*. Accepted paper in *Limnology & Oceanography Letters*.
2. Reich HG, Tu W-C, **Rodriguez IB**, Chou Y, LaJeunesse TC and Ho T-Y (2021). The importance of iron availability for coral symbionts' response to heat stress. *Journal of Phycology*, 57, 3-13, 10.1111/jpy.13078-20-150.
3. Reich HG, **Rodriguez IB**, LaJeunesse TC and Ho T-Y (2020). Endosymbiotic dinoflagellates pump iron: differences in iron and trace metal needs among the Symbiodiniaceae. *Coral Reefs*, https://doi.org/10.1007/s00338-020-01911-z.
4. Tuo S-H, **Rodriguez IB** and Ho T-Y (2020) H_2 accumulation and N_2 fixation reduction by Ni limitation in *Cyanothece*. *Limnology and Oceanography*, https://doi.org/10.1002/lno.11305
5. Li T, Lin X, Yu L, Lin S, **Rodriguez IB** and Ho T-Y (2019). RNA-seq profiling of *Fugacium kawagutii* reveals strong metabolic response and impacts on symbiosis potential of deficiencies of iron and other trace metals. *Science of the Total Environment*, https://doi.org/10.1016/j.scitotenv.2019.135767
6. **Rodriguez IB** and Ho T-Y (2018) Trace metal requirements and interactions in *Symbiodinium kawagutii*. *Frontiers in Microbiology*, 9:142. doi: 10.3389/fmicb.2018.00142
7. **Rodriguez IB** and Ho T-Y (2017) Interactive effects of spectral quality and trace metal availability on the growth of *Trichodesmium* and *Symbiodinium*. *Plos ONE*, 12:e0188777. doi: https://doi.org/10.1371/journal.pone.0188777

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PRC: 0008670

rey.capangpagan@msunaawan.edu.ph
09778203227
ORCID: 0000-0002-7078-9903
WoS Researcher ID: AAX-7641-2021
Citations: 132; h-index: 6

BIONOTE

Rey Y. Capangpangan is a Professor of Chemistry at MSU-Naawan, who specializes in the wide applications of nanotechnology and Material Science. His research interests are in areas of synthesis, characterization, and applications (biomedical and environmental) of various polymers and nanomaterials. Dr. Capangpangan is able to publish several scientific papers in ISI/Scopus indexed journals. He also received numerous awards, including the Presidential Award for Excellence in Research, Presidential Award for Excellence in Resource Generation, Outstanding Faculty, National Awardee (PCHRD-NHRS), among others. He also received a Research Fellowship in the Hebrew University of Jerusalem, Israel. He is actively involved in mentoring young researchers.

EDUCATION

2011 – 2015
Ph.D. in Chemistry
National Tsing Hua University
Taiwan

2005 – 2008
M.Sc. in Chemistry
University of the Philippines
Diliman

1997-2001
B.Sc. in Chemistry
MSU-Iligan Institute of Tech.

EXPERTISE

Sensor Fabrication
Nanoscience and Material Science
Polymer Chemistry
Environmental Monitoring
Green Synthesis
Proteomics
Nanomaterial Characterization

REY YONSON CAPANGPANGAN, Ph.D., RCh

Professor VI/ Chemist

PROFESSIONAL EXPERIENCE

Professor VI (Aug. 2021 – Present)
Mindanao State University – Naawan, Misamis Oriental, 9023, Philippines

Professor I (June 2004 – July 2021)
Caraga State University – Ampayon, Butuan City, 8600, Philippines

Laboratory Chemist (October 2001 – June 2004)
TECHNOLAB Laboratory Services – Cabancalan, Mandaue City, Cebu 6014, Philippines

RECENT JOURNAL PUBLICATIONS

Latest Journal Publications

1. Photocatalytic Activity of Cellulose Nanocrystals/Zinc Oxide Nanocomposite Against Thiazine Dye under UV and Visible Light Irradiation, *ASEAN Journal of Chemical Engineering* (JUST ACCEPTED MANUSCRIPT)
2. Single-Dip Colorimetric Detection of Cyanide using Paper-based Analytic Device based on Immobilized Silver Nanoparticles, *International Nano Letters* (JUST ACCEPTED MANUSCRIPT)
3. Variability and Source Characterization of Regional PM of Two Urban Areas Dominated by Biomass Burning and Anthropogenic Emission, *Aerosol and Air Quality Research*, 2022, **22**(6), 220026 (DOI: 10.4209/aaqr.220026)
4. Assessment of Impact of Meteorology and Precursor in Long-Term Trends of PM and Ozone in a Tropical City, *Aerosol and Air Quality Research*, 2022, **22**(1), 210269
5. Preparation of spin-coated poly(vinyl alcohol)/chitosan/gold nanoparticles composite and its potential application for colorimetric detection of cyanide in water, *Polish Journal of Environmental Studies*, 2022, **31**(2), 1-8 (DOI: 10.15244/pjoes/140276)
6. Improvised centrifugal spinning for the production of polystyrene microfibers from waste expanded polystyrene foam and its potential application for oil adsorption, *Journal of Engineering and Applied Science* (Springer), 2021, **68**, 25
7. Functionalized silver nanoparticle-decorated paper sensor for rapid colorimetric detection of copper ions in water, *Funct. Compos. Struct.*, 2021, **3**, 035007 (DOI: 10.1088/2631-6331/ac25e9)

RESEARCH EXPERIENCE

Projects handled/implemented

1. Project Staff, L3 – DOST-GIA CRADLE Program (2021 – 2022)
2. Project Leader – DOST-PCIEERD Project No. 08905 (2020-2021)
3. Project Leader – DOST-PCIEERD Project No. 09021 (2020-2021)
4. Project Staff (Collaborator) – eAsia Project with MSU-IIT (2020-2022)
5. Program Leader – CHED Funded Program in Response to COVID-19 (2020)
6. Project Leader – CHED Funded Project under the DARE TO K12 Program (2018-2020)
7. Project Staff (Collaborator) – CHED Funded under the DARE TO K12 (2018-2020)
8. Principal Investigator – Industry-supported Research Project (PHILSAGA) (2018-2019)
9. Principal Investigator - Industry-supported Research Project (SRMI) (2017-2018)
10. Project Staff – NGO-funded project (Haribon Foundation) (2018)

For more information: Please visit the website: https://rycapangpangan.wixsite.com/reycaps

Capsule Curriculum Vitae

CURRICULUM VITAE
MICHAEL A. CLORES

Educational Background:

Doctoral degree — 2011 – 2015
De La Salle University – Manila
PhD in Biology

2002 – 2005
De La Salle University – Manila
PhD in Science Education, Major in Biology

Master's degree — 2000 – 2002
Ateneo de Naga University
Master of Arts in Education, Major in Curriculum and Instruction

College degree — 1993 – 1998
Universidad de Santa Isabel
Bachelor of Science in Physical Therapy

Special Training:

J-1 Exchange Scholarship: Visiting Professor/Scholar
Focus: SPECIFY Databasing & Natural History Collections Management Work, August to September 2018
Herpetology Division, Kansas University Biodiversity Institute and Department of Ecology and Evolution, Natural History Museum, University of Kansas, USA.

Employment Background:

July 2021 to Present
Campus Dean/Associate Professor
Partido State University – Caramoan Campus

July 2020 to March 2021
Professor
Graduate School
Bicol State College of Applied Sciences and Technology

June 2000 to June 2020
Professor
Ateneo de Naga University

Other Positions Held:

September 2020 to March 2020
Director
Research, Extension, Production, and Enterprise Development,
Bicol State College of Applied Sciences and Technology

Recently published/ prepared/ presented papers:

1. Bernstein, J.M., Bautista, J., Diesmos, A.C., **Clores, M.A.** Cuesta, M.A., Sanguila, M.B. Brown, R.M. (2021) [In Press]. Multivariate Characterization of Phenotypic Variation from throughout the Geographic Ranges

of Philippine False Coral Snakes: Two Species or Four? ***Ichthyology & Herpetology*** 109, No. 4, 2021, 000–000. DOI: 10.1643/h2021035

2. Brown, R.M., Bautista, J.B., ***Clores, M.A.*** Cuesta, M.A. (2021) *DENDRELAPHIS MARENAE* (Gauke's Bronze-back Tree Snake). DIET. ***Herpetological Review***, 52(1), 153-154.
3. **Clores, M.A.**, Bautista, J.B., Fernandez, J.B., Cuesta, M.A., Brown, R.M. (2020). Diversity and distribution of amphibians and reptiles in the Caramoan Island Group (CIG), Maqueda Channel, Southern Luzon, Philippines. ***Journal of Asia-Pacific Biodiversity***. 14 (2020)1-14. https://doi.org/10.1016/j.japb.2020.11.005
4. Brown, R. M., Meneses, C.G., Wood, P.L.J., Fernandez, J.B., Cuesta, M.A., **Clores, M.A.**, Tracy, C., Buehler, M.D. Siler, C.D. (2020). Unexpected Discovery of another New Species of Philippine False Gecko (Gekkonidae; *Pseudogekko*) from the Bicol Peninsula of Luzon Island. ***Herpetologica*** 76(3), 315-329, (9 September 2020). https://doi.org/10.1655/Herpetologica-D-19-00029.1
5. **Clores, M.A.**, Conde, M.A.A., Perez, J.D. (2020). Biomass Leaching and Dynamics of Nutrients, Microbial Abundance and Activity during the Decomposition of Seagrass *Cymodocea rotundata* Necromass. ***Applied Environmental Research***, 42(1): 1-13. https://doi.org/10.35762/AER.2020.42.1.1
6. **Clores, M.**, & Cuesta, M. (2019). Trophic Models of Seagrass Ecosystems in Maqueda Channel, Caramoan Peninsula, Philippines. ***Applied Environmental Research***, 41(3), 14-31. https://doi.org/10.35762/AER.2019.41.3.2
7. Sarcia, S.M., Ang, M. P., **Clores, M.A.,** Oropesa L.J.B. (2016). Grazing Activity of Collector Urchin (Tripneustes gratilla, Linnaeus 1958) on the Leaf Stratum of Seagrasses in Sabitang-Laya Ilsand, Caramoan Peninsula, Philippines. ***Bicol Science Journal*** (3) 1, 8-20.

Note: cor complete list of papers published, please see https://scholar.google.nl/citations?user=X9RLwPYAAAAJ&hl=de or https://www.researchgate.net/profile/Michael-Clores-2

Awards received:

- First Recipient of the **Fr. Frank X. Lynch SJ. Award for Most Outstanding Researcher.** Ateneo de Naga University. February 20, 2020.
- Recipient: **Fr. Raul Bonoan, S.J, and Jose Quiros Professorial Research Chair Award.** Ateneo de Naga University
- Recipient of the **Best Paper Award.** 2nd International Congress on Action Research, Action Learning (ARAL2017). De La Salle University, Manila. April 1, 2017.
- Recipient of **Outstanding Research Paper Award**. First Place (Faculty Research Category). CHED Council of Deans for Graduate Education (CODGE V), March 2016

Organizations:

- Associate Member, National Research Council of the Philippines (Approved: April 2022)
- Biology Teachers Association, Inc. (BIOTA, Inc.)-National Chapter. Position: Member, Board of Directors (2021-2023)
- CHED Regional Quality Assessor Team (RQAT) Member
- Asian Association for Biology Education (AABE) – Regular Member
- Philippine Association of Marine Sciences (PAMS) – Regular Member
- Philippine Society for Microbiology, Inc. (PSM) ~Bicol Chapter. Position: President (2007-2008) – Ex-Officio

RICHARD V. DUMILAG, PhD

Associate Professor V
Fisheries Department, Sorsogon State University – Magallanes Campus
Magallanes, Sorsogon 4705 Philippines
richard.dumilag@sorsu.edu.ph +639361585522

DEGREES

PhD Biology, Institute of Biology, University of the Philippines, Diliman, 2017
MSc Marine Science, The Marine Science Institute, University of the Philippines, Diliman, 2012
BSc, Pamantasan ng Lungsod ng Maynila, Intramuros, Manila, 2003

EMPLOYMENT HISTORY

Mindanao State University – Tawi-Tawi, Professor VI, 2019-21
Far Eastern University, Assistant Professor, 2013-19
UP-MSI, University Research Assistant, 2010-16

GRANTS

Project Leader, DOST-PCAARRD, Molecular characterization, selection, and production of high quality eucheumatoid cultivars in the Bangsamoro Autonomous Region in Muslim Mindanao (BARMM) under a NICER Program: Establishment of Seaweed Research and Development Center (SeaRDeC) to Support the Seaweed Industry in BARRM, 2020-2021

Co-Project Leader, United Kingdom Research and Innovation–Global Challenges Research Fund (UKRI- GCRF) 'GlobalSeaweedSTAR' Programme (Grant No.GSS/RF/015). Diversity of wild and farmed eucheumatoids in the Philippines at the intra- and inter-specific scale: population genetics of *K. alvarezii* and molecular taxonomy of eucheumatoids, 2019-2020

Project Leader, Far Eastern University Office of the Research. Establishment of the Far Eastern University Herbarium (FEUH) Phase 2: Molecular Analysis, Project Title 3: Molecular-assisted alpha taxonomy of Philippine economically important seaweeds, 2018-2019

Project Leader, Far Eastern University Office of the Research. Establishment of the Far Eastern University Herbarium (FEUH) Phase 1: Specimen Collection, Project 2: Building strengths of the FEUH: Rhodophyte diversity of Sorsogon, 2017-2018

Consultant, National Fisheries Research Development Institute, Plant Division (NFRDI). Diversity of *Gracilaria* in the Philippines, 2017

Systematics of Philippine Bangiales. International Association of Plant Taxonomy (IAPT) Research Grants Program in Plant Systematics, Bratislava, Slovakia, Travel Grant, 2016

Accelerated Science and Technology Human Resource Development Program, Student Research Support Fund, 2016

Project Leader, Far Eastern University Office of the Research. DNA barcoding, genetic variation and biogeography in northern Philippine species of *Porphyra* (Bangiales, Rhodophyta), 2015-2016

PUBLICATIONS

Dumilag RV, Crisostomo BA, Aguinaldo ZA, Hinaloc LAR, Liao LM, Roa-Quiaoit HA, Dangan-Galon F, Zuccarello GC, Guillemin M-L, Brodie J, Cottier-Cook EJ, Roleda MY. The diversity of eucheumatoid seaweed cultivars in the Philippines. Reviews in Fisheries Science and Aquaculture. In Press.

Dumilag RV, Javier RF. 2022. Ethnobotany of medicinal seaweeds of Ilocos Norte, Philippines. Philippine Journal of Science 151(3): 1135–1156.

Gamus GCV, **Dumilag RV**, Yap SL. 2021. Use and Distribution of *Bangia* cf. *fuscopurpurea* (Bangiales, Rhodophyta) in the Northern Philippines. Philippine Journal of Science 151 (S1) 73–77.

Brakel J., Sibonga RC, **Dumilag, RV**, Montalescot V... Gachon CMM. 2021. Exploring, harnessing and conserving marine genetic resources towards a sustainable seaweed aquaculture. Plants People Planet 3(4): 337–349.

Dumilag RV, Samuyag AMP, Robles RJF. 2021. First report of *Chamaebotrys proliferus* (Rhodymeniaceae, Rhodophyta) from the Philippines. Philippine Journal of Science 150(1):263–266.

Roleda M.Y., Aguinaldo ZA, Crisostomo BA, Hinaloc LAR, Projimo VZ, **Dumilag RV**, Lluisma AO. 2021. Discovery of novel haplotypes from wild populations of *Kappaphycus* (Gigartinales, Rodohyta) in the Philippines. Algae 36(1): 1–12.

Dumilag RV, Lin S-M, Zuccarello GC, Kraft GT. 2020. The identity of *Eucheuma perplexum* (Solieriaceae, Gigartinales) and its distinction from *Euchuema serra* as exemplified by a proposed new epitype. Phycologia 59(6): 497–505.

Ferrer MSR, Gomez RN, **Dumilag RV.** 2020. COI-5P gene sequence variation in Philippine *Hydropuntia edulis* (Gracilariaceae, Rhodophyta). Philippine Journal of Science 149 (3-b) 1135–1140.

Dumilag RV et al. 2020. The Ulvophyceae (Chlorophyta) of eastern Sorsogon, Philippines, including *Halimeda magnicuneata* sp. nov. (Bryopsidales). Botanica Marina 63(5): 439-45.

Dumilag, R.V. 2019. Edible seaweeds sold in the local public markets in Tawi-Tawi, Philippines. Philippine Journal of Science 148 (4): 803–811.

Dumilag RV, Nelson W, Kraft GT. 2019. Validation and phylogenetic placement of the Placentophoraceae fam. nov. (Gigartinales, Rhodophyta). Phycologia 58 (3): 269–275.

Dumilag RV, Aguinaldo ZA, Alcoriza VAM, Balucanag MPSB, Dulalia ART, Sayasa AR. 2019. DNA barcodes of *Caulerpa* species (Caulerpaceae, Chlorophyta) from the northern Philippines. Philippine Journal of Science 148 (2): 343–353.

Ferrer MSR, Gomez RN, Nacido CB, Moises MT, **Dumilag RV.** 2019. Genetic diversity of Philippine *Gracilaria salicornia (*Gracilariaceae, Rhodophyta) based on mitochondrial COI-5P sequences. Biologia 74 (6): 599–607.

Dumilag RV. 2019. Detection of *Pythium porphyrae* infecting Philippine *Pyropia acanthophora* based on morphology and nuclear rRNA internal transcribed spacer sequences. Journal of General Plant Pathology 85(1):72–78.

Dumilag RV. 2018. *Betaphycus gelatinus* and B. *philippinensis* (Gigartinales, Rhodophyta) are conspecific. Phytotaxa 372 (1): 22–34.

Dumilag RV & Yap SL. 2018. *Pyropia lunae* sp. nov. and *Pyropia islae* sp. nov. (Bangiales, Rhodohpyta) from the Philippines. Botanica Marina 61(5): 467–480.

Dumilag RV. 2018. Unmasking a cryptic ethnotaxon: a case study on the identity of *Dermonema virens* (Nemaliales, Rhodophyta) in the Philippines. Webbia: Journal of Plant Taxonomy and Geography 73(1):89– 96.

Dumilag RV, Gallardo WGM, Garcia CPC, You Y, Chavez AKG & Agahan L. 2018. Phenotypic and mtDNA variation in Philippine *Kappaphycus cottonii* (Gigartinales, Rhodophyta). Mitochondrial DNA Part A 29(6):951–963.

Dumilag RV & Monotilla WD. 2018. Molecular systematics and biogeography of Philippine foliose Bangiales (Rhodophyta). Journal of Applied Phycology 30(1):173–186.

Dumilag RV, Aguinaldo ZAA, Mintu CB, Quinto MP, Ame EC, Andres AC, Monotilla WD, Yap SL, Cao EP, Vital PG, Fontanilla IKC. 2017. A review of the current taxonomic status of foliose Bangiales (Rhodophyta) in the Philippines. Phytotaxa 312(1):47–59.

Dumilag RV & Aguinaldo ZAA. 2017. Genetic differentiation and distribution of *Pyropia acanthophora* in the Philippines. European Journal of Phycology 52(1):104–115.

Dumilag RV, Aguinaldo ZAA, Mintu CB, Quinto MP, Ame EC, Andres AC, Monotilla WD & Yap SL. 2016. Morphological and molecular confirmation of the occurrence of *Pyropia tanegashimensis* (Bangiales, Rhodophyta) from Palaui Is., Sta. Ana, Cagayan, Philippines. Phytotaxa 255(1): 83 –90.

Dumilag RV, Salvador RC & Halling C. 2016. Genotype introduction affects population composition of native Philippine *Kappaphycus* (Gigartinales, Rhodophyta). Conservation Genetics Resources 8(4): 439- 441.

Dumilag RV, Orosco FL & Lluisma AO.2016. Genetic diversity of *Kappaphycus* species (Gigartinales, Rhodophyta) in the Philippines. Systematics and Biodiversity 14(5):441–451.

Dumilag RV & Lluisma AO. 2014. Resolving the phylogenetic affinities of *Kappaphycus inermis* within the genus *Kappaphycus* (Gigartinales, Solieriaceae) using mitochondrial and plastid markers. Phytotaxa 162(4):223 –231.

Dumilag RV, Liao LM & Lluisma AO. 2014. Phylogeny of *Betaphycus* (Gigartinales, Rhodophyta) as inferred from COI sequences and morphological observations on *B. philippinensis*. Journal of Applied Phycology 26(1):587 –595.

Dangan-Galon FD, **Dumilag RV** & Ganzon-Fortes E. 2013. Photosynthesis-irradiance curves of four marine macroalgae from Bolinao, Pangasinan and Calatagan, Batangas, Philippines. Scholarly Journal of Scientific Research and Essay 2(8): 134-138.

CURRICULUM VITAE

RONELIE CHATO SALVADOR
College of Agriculture, Fisheries and Natural Resources
UNIVERSITY OF EASTERN PHILIPPINES
Catarman 6400, Northern Samar

Phone : 09054852744
Email address : roneliesalvador@yahoo.com

Date and Place of Birth : Sept. 24, 1957, Surigao City
Citizenship : Filipino
Civil Status : Single
Sex : Female

Degrees :

Ph.D. Fisheries (Aquaculture) – UP in the Visayas, Miag-o, Iloilo, Nov. 2004
MSc. In Fisheries (Aquaculture) – UP in the Visayas, Miag-ao, Iloilo, 1993
BSc. Fisheries - Univ. of Eastern Philippines (UEP), Catarman, N. Samar , 1981

Work Experiences :

- Prof. 6 - UEP (Jan. 1, 2017 to Present)
- Prof. III – UEP (Sept. 1, 2014 to Dec. 31, 2016)
- Asso. Prof. V- (UEP) (Oct. 1, 2012 to Aug. 30, 2014)
- Assoc. Prof. III – UEP (July 2009 to Sept. 30, 2012)
- Postdoctoral researcher on seaweed biotechnology- Marine Science Institute, Univ of the Philippines, Diliman, Quezon City (June 2010 to Jan. 2011)
- Assoc. Prof. II - UEP (Oct. 2005 to June 2009)
- Assoc. Prof. III - UEP (July 1995 to Sept 2005)
- Visiting Aquaculture Research Specialist - SEAFDEC-AQD, Tigbauan, Iloilo (Oct. 2005 to Dec. 2007)
- Asst. Prof. II - UEP (Jan. 1995 to June 1995)
- Instructor III - UEP (Mar. 1993 to Dec. 1994)
- Instructor II - UEP (Aug. 1991 to Feb. 1993)
- Instructor I - UEP (Nov. 1988 to July 1991)
- Technical Research Asst. - BFAR-NSIRDP (July 1988 to Sept.1988)
- Instructor - UEP (June 1986 toJune 1987)
- Sr. Fishery Biologist - BFAR, Northern Samar (Nov. 1984 to June 1986)
- Project Officer - Ministry of Human Settlements (Feb. 1983 to Feb.1984)
- Technical Staff Assistant - Ministry of Human Settlements (April 1982 to Feb.1983)

Present designations :

- Dean, College of Agriculture, Fisheries and Natural Resources, UEP
- Coordinator for Agronot studies, UEP

Current research involvement:

Program Leader – Saganang Buhay sa Ilog (SALOG) research program (2020-2022)
Funding agency: National Research Council of the Philippines
Study Leader - Production and profitability of Parasol method of *Kappaphycus* farming

Major Researches completed:

- Development, validation and adoption of culture-based fisheries in Catubig and Pambujan Rivers, Northern Samar, Philippines
- Assessment of culture performance of *Kappaphycus* spp. In Northern Samar
- Histochemical profiling of 'ice-ice' disease infected *Kappaphycus alvarezii* (Doty) Doty (Solieriaceae, Rhodophyta)
- Development of aquatic plant-based diets for *Kuhlia rupestris*
- Assessment of culture performance of indigenous freshwater clam, *Batissa* sp. in Catubig River
- Stock assessment and culture of freshwater prawn, *Macrobrachium rosenbergii*, in Northern Samar
- Inventory/assessment of seaweed resources of Northern Samar, Philippines
- Culture of tilapia hybrid in marine waters
- Growth, survival and bacterial load of sex-reversed Tilapia hybrid (*Oreochromis niloticus* x *Oreochromis mossambicus*) in seawater netcages
- Micropropagation of *Kappaphycus alvarezii* (Doty) Doty (Solieriaceae, Rhodophyta)
- Testplanting of selected *Kappaphycus* strains in Northern Samar
- Generation of novel *Kappaphycus* varieties or strains from fused protoplasts

Research work experiences in other institutions:

June 2009-Jan.2010	-	Postdoctoral researcher on seaweed biotechnology, Marine Science Institute, Univ of the Philippines, Diliman, Quezon City
Oct. 2005- Dec. 2007	-	Visiting Aquaculture Research Specialist (SEAFDEC-AQD, Tigbauan, Iloilo); conducted research on strain improvement of farmed carrageenan-producing marine macroalgae

List of Technical Papers Published:

A. **Full Papers**

Tan KC, LB Mante, AD Mante and **RC Salvador** (2020). Yields from rice-fish farming at different water depths and fish stocking densities in wetland rice ecosystem. Pacific Journal of Science and Technology 4 (1): 28-34.

Dumilag, RV, RC **Salvador** and C. Halling (2016). Genotype introduction affects population composition of native Philippine *Kappaphycus* (Gigartinales, Rhodophyta). Conservation Genet Resour. DOI 10.1007/S12686-016-0591-2.

***Chato-Salvador* RC**, Tablizo FA and LLuisma AO (2014) Non-enzymatic isolation of somatic cells from *Kappaphycus* spp. and *Eucheuma denticulatum* (Solieriaceae, Rhodophyta). European Journal of Phycology, 49:4,486-492, DOI: 10.1080/09670262.2014.977963.

Salvador RC and Serrano AE (2014). Germination and growth of somatic cells of Philippine strains of *Kappaphycus alvarezii* (Doty) Doty (Solieriaceae, Rhodophyta). ELBA Bioflux 6(1):36-45.

Vista, EB and **Salvador RC** (2014). Socio-economic status of artisanal fishermen in two rivers in Northern Samar , Philippines. Proceedings of the Second International in Fisheries Sciences. Colombo, Sri Lanka, July 30-31, 2014.

Salvador RC and Serrano AE (2005) Isolation of protoplasts from tissue fragments of Philippine cultivars of *Kappaphycus alvarezii* (Doty) Doty (Solieriaceae, Rhodophyta). J. Appl. Phyco. 17(1): 15-22.

Salvador RC and Salvador AA (1997). The standing stock of Freshwater Prawn, *Macrobrachium rosenbergii* (De Mann) in selected rivers of Northern Samar. Proceedings of the 27th FIMFS Convention. San Carlos Univ., Cebu City.

Dureza LA, Gempis CC, **Salvador RC**, Sombero RP and Belmonte BD (1994). Effects of selected factors on the growth and survival of Tilapia hybrids raised in cages in saline waters. In: LG Villacorta and LA Dureza (eds) Tilapia Farming: Genetic improvements and advances on culture technology. Proceedings of the Third National Symposium and workshop on Tilapia Farming. UPVisayas, Miag-ao, Iloilo.

Salvador RC, Gabona SF and Salvador AA (1994). Grouper species in the waters off the Northern Coast of Samar Island, Philippines. Pacific J. of Science and Technology 8: 29-33.

Salvador RC ,Salvador AA and Bacalando NY (1992). Seacage culture of sex-reversed Tilapia hybrid in Northern Samar. In: RD Guerrero III and DDG Fernandez (eds) Tilapia Farming in the Philippines: Recent developments. Proceedings of the Second National Symposium and Workshop in Tilapia Farming. CLSU, Muñoz, Nueva Ecija.

Salvador RC and Salvador AA (1992). Culture trial of red Tilapia (*Oreochromis* sp) in polyethylene tanks. Pacific J. of Science and Technology 7 (1&2) : 40-46.

B. Published Abstracts

Salvador RC, De la Peña, MR and Santos FF (2007) Farmed carrageenan-producing seaweeds of the Philippines. Programs and Abstracts 39th Annual FIMFS Convention. Visayas State University, VISCA, Baybay City, Leyte.

Lim MM, Porlaje RS and Chato-Salvador RC (2006) Rheological properties of carragenophyte farmed in Eastern Visayas, Philippines. 38th Annual FIMFS convention, University of San Carlos, Cebu City.

Chato-Salvador RC, Santos FF, Ceniza PB and Consumo CA (2006) Growth performance and carrageenan properties of Philippines *Kappaphycus* and *Eucheuma* cultivars cultured in land-based facilities. 2006 Visayas Zonal Network Annual R & D Review and Planning Workshop.

Salvador RC and Salvador AA (2002). Inventory and assessment of seaweeds in Northern Samar. Proceedings of 30th FIMFS Pearl Anniversary convention. UEP, Catarman, Northern Samar.

Salvador RC, Salvador AA, Azamar MSL, Orane NL and Sorio MZ (1994). Growth response of *Macrobrachium rosenbergii* juveniles to various indigenous feedstuffs. Proceedings of the National Seminar-workshop on Fish Nutrition and Feeds. SEAFDEC AQD, Tigbauan, Iloilo.

Salvador RC, Atencio MD and Honrade MB (1994). Effect of feeding rates on growth and survival of Tilapia (*Oreochromis mossambicus*) cultured in netcages in seawater. Proceedings of the National Seminar-workshop on Fish Nutrition and Feeds. SEAFDEC AQD, Tigbauan, Iloilo.

C. Published and copyrighted books and manuals

Chato-Salvador, RC. 2016. Laboratory Manual in Fish Health Management. University of Eastern Philippines, Catarman, Northern Samar, Philippines. 58 p.

Chato-Salvador, RC. 2016. Laboratory Manual in Aquatic Botany. University of Eastern Philippines, Catarman, Northern Samar, Philippines. 51 p.

Chato-Salvador RC, MR De la Peña and FF Santos. 2017. Farmed Carrageenan-producing Seaweeds of the Philippines. SPPI, ICCO and CFC. 82 p.

Chato-Salvador RC. 2017. Farming of Carrageenan-producing Seaweeds (A Practical Guide). SPPI, ICCO and CFC. 27 p.

Chato-Salvador, RC. 2017. Simplified Methods for Assessment of Mangrove Ecosystems. SPPI and UEP. 23 p.

Intellectual Property Products with IPR

- Method of extracting crude enzymes from Golden Apple Snail (***Pomacea canaliculata)*** **– Utility Model # 2/2021/050056**
- Crude enzyme extracts from Golden Apple Snail (Pomaceae canaliculata) - Utility Model (submitted to IPOPhil and still in publication in IPOPHL Official Gazette)
- A method to dissociate tissues and isolate somatic cells from tissues of the red seaweeds *Kappaphycus* spp. and *Eucheuma* spp. (still in publication in the IPOPHL Official Gazette)

Awards Received :

A. Non-academic distinction/recognition

Title of Award	Area	Awarding Body/Agency	Level
Award of distinction	Public Service	UEP	Institutional
Loyalty Award			
Best Paper in Resource Assessment and Management (1995)	Research	FIMFS	National
Outstanding Applied Research Award (Bronze) (1992)	Research	VICARP	Regional
Outstanding Basic Research Award (Bronze) (2000)	Research	VICARP	Regional
Best Poster Paper Award (2000)	Research	VICARP	Regional
AFMA R & D Paper Award (2004)	Research	DA-BAR	National
2005 CHED Republica Award	Research	CHED	Zonal
Best poster paper award (Oct. 2007)	Research	PCAMRD	Zonal
Best poster paper award (Nov. 2007)	Research	FIMFS	National
Best Research Paper -1st Prize (Nov. 2010)	Research	DOST-PCAMRD	Zonal
Best Research Paper- 1st Prize (Jan. 2011)	Research	NARRDS/DOST-PCAMRD	National
Best Paper (Capture Fisheries Category) -1st Place (Oct. 2012)	Research	NRFDI	National
Best Paper (Aquacuture Category)-2nd place (2013)	Research	NFRDI	National
Best Paper (1st Place) (July 2015)	Research	VICARP-RRDEN	Regional

Best Paper Award (Research Category) Dec. 2015	Research	VICARP-RRDEN	Regional
2017 Dr. Elvira O. Tan Award (Outstanding Published Paper- Aquatic Sciences Category) (June 2017)	**Research**	**PCAARRD-DOST**	**National**
2017 CSC Dangal ng Bayan Awardee	**Public Service**	**Civil Service Commission**	**National**
Best Scientific Poster in Agricultural Sciences (July 2018)	**Research**	**National Academy of Science and Technology**	**National**

B. Academic Distinctions:

College Scholar Award - 1st semester AY 1999-2000, UP in the Visayas, Miag-ao, Iloilo

University Scholar Award - 2nd semester AY 1999-2000 and 1st semester AY 2000-2001, UP in the Visayas, Miag-ao, Iloilo.

Study/Scholarship Grants received :

MSU Full scholarship grant - AY 1973-1978
PCAMRD Thesis Grant - AY 1992-1993
PCAMRD Doctoral Study Grant - AY 1999- 2003
Netherlands Fellowship Program - May – June 2009

Membership in Organizations :

Member - Network of Women in Fisheries (Winfish)
Member - U.P.Visayas Graduate Students in Fisheries Association (UPV-GRASFA)
Member - Federation of Institutions of Marine and Freshwater Sciences (FIMFS)
Member - Philippine Fisheries Society
Member- Philippine Fisheries Association
Member- Society of Fisheries technologists in Eastern Visayas
Member- Netherlands Fellows Foundation of the Philippines, Inc.
Founding Member - Asian Society of Icthyologists
Member - Biodiversity Conservation Society of the Philippines
Associate Member - National Research Council of the Philippines

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18 SEP 1963

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Kasarian/Sex
F

Lugar ng kapanganakan/Place of birth
KIBAWE BUKIDNON

Petsa ng pagkakaloob/Date of issue
02 DEC 2019

Petsa ng pagkawalang bisa/Valid until
01 DEC 2024

Maykapangyarihang nagkaloob/Issuing authority
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RYAN L.

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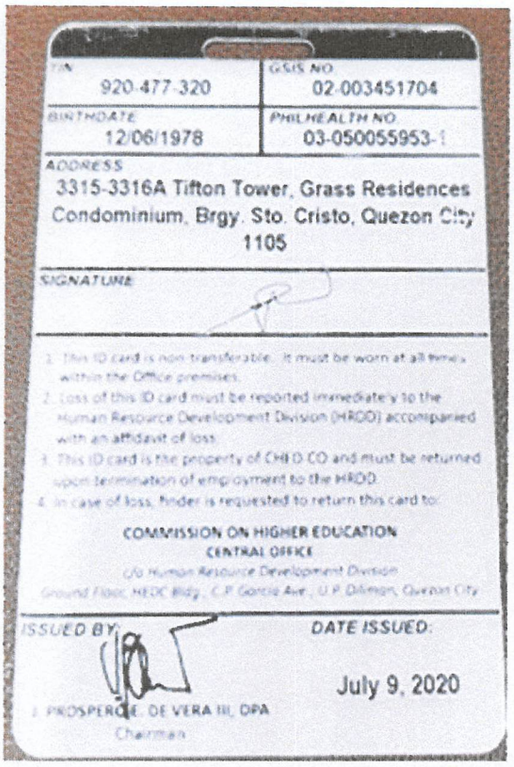
OIC, Executive Director, UniFAST



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ISSUED BY:

J. PROSPERO E. DE VERA III, DPA
Chairman

DATE ISSUED:

July 9, 2020

Maka-Diyos,
Maka-Tao,
Makakalikasan
at Makabansa.

3

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CHERRY I. ULTRA, PhD
SUC President III

DR. ALDRIN A. DARILAG, CHRP, RMT, RN
CHED Commissioner

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TIN: 142-377-197
Birthdate: October 09, 1965

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